

Econ 401 Game Theory

Numbered Transcript

ECON 401 • Dr. Daniel Bonevac

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Lecture 1: Games and Strategies

At a glance — Bonevac introduces game theory — games and strategies.

- 1.1 Well, welcome. This is going to be a course on game theory. Now, game theory is a topic that has been explored by mathematicians, political scientists, economists, philosophers, and so there are many different ways of approaching the subject. Some of them are highly technical, highly mathematical. I'm going to try to keep the math to a minimum here and talk about this from the point of view of applications, applications in fields like economics and philosophy, but more generally in life.
- 1.2 Because I think game theory is a really valuable thing to know, quite apart from any academic endeavor; it has to do with how to live in a lot of contexts where our actions depend in part on the actions of other people. So in this first lecture, I'm going to talk in general about kinds of games and what a strategy is, and lay out some of the basic concepts, but also give you a sense of what this is good for.
- 1.3 Then we'll start getting into the actual details of how it helps us with all sorts of problems as we go to subsequent lectures. But for now, I've got to say, this is a relatively young subject. Professionally, I'm a philosopher, which means I deal with topics that have been under discussion for thousands of years. Game theory isn't like that. It stemmed from a 1928 paper written in German by John von Neumann. And so, this subject is just under a century old. And it was developed later by a number of other people. In fact, 11 economists have ended up winning Nobel Prizes for working game theory. So, it's been a huge area of research.
- 1.4 But what is it about? It's about decision-making, but also specifically about strategic thinking. And so, what I mean by that is that it has to do with thinking, where the results of our actions aren't things we can evaluate independently. It depends on what other people are doing. When you start thinking about it, you realize that's most of life. It's really widely applicable, and it's kind of amazing that nobody really investigated it very seriously until 1928.
- 1.5 But in any case, what is it about? Well, it's about coordination. It's about competition, conflict. It's about cooperation, and how, in general, people interact with each other. So, we're going to be thinking about what it is to be an intelligent, rational decision maker, and then how such decision makers interact.
- 1.6 Well, let me go back, being a philosopher, all the way to the ancient world. Aristotle said that we are essentially rational animals. The function of man, and he uses the Greek term *ergon* here, which means "the work," the function, what we're for, is "an active life of the element that has a rational principle." Rational activity, acting according to a rational plan, is really what we're all about.

- 1.7 Now, you can study that in isolation, and for centuries people did, thinking about decisions and what it is to make a good decision. That, in a sense, goes at least all the way back to Aristotle. But when we start thinking about the interactions between ourselves and other people, it becomes more complicated. A rational decision maker is called a player in game theory, and we're going to try to think of players as having well-defined objectives. They know what they want, and they understand the game being played.
- 1.8 Now, in real life, that's not always accurate, and so we're going to be dealing mostly with games of perfect information, where we actually know what we want, we know what other people want, and we understand the nature of the relationship. In real life, it turns out, well, imperfect information is more the norm. On the other hand, I'm not going to really talk much about that, partly because it gets complicated, but partly because it's analyzed typically in terms of reducing those games to multiple games of perfect information.
- 1.9 If we understand those, we'll be able to figure out, OK, I don't know whether I'm playing this game or that game, but I can analyze them each individually and figure out something on that basis. So we're going to think of ourselves as agents, as players, who understand what we want. We're going to know roughly what we think is desirable anyway. We're also going to know what's possible. Now, there's a difference here because knowing what's best, that's a matter of my own subjective point of view. I might prefer things that other people don't.
- 1.10 I was recently being interviewed, for example, and the last question was kind of a funny question. "The smell of what food would make you think of home?" I was apparently the first person this person had ever interviewed that answered "sauerkraut." Most people, it's maybe a spaghetti sauce or baking bread or something like that, muffins. Well, for me, it was sauerkraut. But anyway, that was something that I give a very high degree of preference to. Most people wouldn't.
- 1.11 On the other hand, we're going to assume that people know what's possible, and that's more of an objective matter. Is this a possible course of action or not? It's not just my opinion about that. Some things really are possible; some things really aren't. So what are we going to be talking about? Strategic thinking. I've already mentioned that sometimes the results of my actions don't just depend on the laws of probability or something; they depend on what other people do.
- 1.12 So, in some contexts, I can just look at the probabilities and say, "How might this work out?" And think in terms of what has the highest expected utility for me, what would be best for me. For example, if I'm in Las Vegas and playing slot machines or roulette or something, it doesn't really depend what other people are doing around me. It's just a matter of the probabilities of what I'm doing. Now, they're always stacked against you, so I do not advise doing any of those things, but nevertheless, it's something that I don't have to worry about, other people.

- 1.13** On the other hand, think about sitting at a poker table. It matters what the other people are doing. In fact, I have a friend who goes to Las Vegas and plays poker late into the night. He gets up around 10 p.m. And decides to go down and start playing poker because he figures, by 3 a.m., everyone else is going to be exhausted, and he'll still be fresh. And so, he essentially, I can't say cheats, but takes advantage of exhausted people, OK, and he does very well. Is he a great poker player? I don't know. I think it's really just a question of, well, it depends what the other people are doing, and if the other people are exhausted, you have an advantage.
- 1.14** So there are lots of cases like that, where the actions of somebody else help to determine the outcome, and that's what we're going to be looking at. Those kinds of cases that aren't like roulette, but are more like poker, where I have to think, "What is the other person thinking? What do they want? And what are their options?" And so, I have to anticipate what they might do and think about the best course of action for me, given what the other people are doing.
- 1.15** So, the way von Neumann puts it is this. "The fate of each player depends not only on his own actions, but on those of the other players." That's what I mean by strategic thinking. We have to put ourselves in somebody else's shoes and think, "How are they seeing it, and what are they likely to do?" I can't just think, "What's the best thing for me to do in isolation?" I have to think, "What's the best thing for me to do, if you do that or this other thing?" I have to ask what you're going to do.
- 1.16** And so, there's something really important in game theory, a basic, basic principle, which is, I have to put myself in somebody else's shoes. I have to think, "How does this look from your perspective? What do you want? What are you actually likely to do? What's the best thing for me to do? What's my best response if you do this or that?" So I have to see things from your point of view to really get anywhere at all.
- 1.17** So, in some parts of life, we don't really have to worry about that, but in some we do. That's because it's not just a question of me figuring out how to solve this problem. It's a matter of, What do I do if that person does this, if that person does that? Often, it's a matter of interpersonal relationships, or sometimes it's just realizing I'm in a competition or I'm in a situation where what I want conflicts with what somebody else wants, and so I have to think, "How do I beat them?" But in any case, it's not just a simple matter of problem-solving or thinking about probabilities; it's a matter of anticipating other people's actions.
- 1.18** Well, as I mentioned, von Neumann thought about all of this and published a paper, a relatively short paper, in 1928. It took it a while to get noticed, partly because it was published in 1928. You might think, yeah, a lot of stuff started happening around the following year and so on. It took some time for people to really realize what he had done. But the real theory was developed by von Neumann and an economist, Oskar Morgenstern in a book called Theory of Games and Economic Behavior, in 1944. John Nash then followed that up with the beginning of a series of papers in 1951.

- 1.19** And so, the main development of the theory really happened toward the tail end of World War II and in the beginnings of the Cold War. That was actually a major motivation for the development of the theory. Partly, we're in a world war. How do we win? Talk about situations of competition and conflict. Here's one that's life-and-death, existential. But then the Cold War—how do we begin to think about, well, all sorts of questions involving nuclear weapons development, placements of troops in Europe and other parts of the world, establishing military bases and so on?
- 1.20** Much of this work for a while was funded by the government, specifically by the army, by the navy, having research units that said, "We've got to figure this out, because we need a theory of competition and conflict." But it quickly evolved into something much broader. It also involves cooperation, it involves coordination, it involves lots of peacetime things as well as wartime things, though that was some of the initial development.
- 1.21** Nash, in 1951, published this paper on non-cooperative games. And so, the difference between a cooperative game and a non-cooperative game is really just, can we form coalitions or are we in it on our own? And so, here we're primarily going to be talking about games that are non-cooperative, where we're just thinking about individual players having their own preferences, their own interests.
- 1.22** If we have groups, we can easily generalize to just say, well, think of a group of players then, not an individual player. But sometimes the formation of coalitions is its own separate matter, so in the last lecture here, I'll talk a little bit about social choice theory and about the theory of coalitions, parties voting, things like that that extend more into political science, where these ideas become very useful as well.
- 1.23** OK, so how are we going to model a player, a rational agent? We're going to think about a game as, first of all, involving a set of players, and then each of those players has a set of actions from which they can choose. So, they might be the same. It might be that we're all playing cards, and it's a question of call or fold or raise, etc. Or it might be that we each have our own individual actions.
- 1.24** And so, think about a student and a professor. The professor may have actions involving assigning grades. The student may have, as a course of action, submitting various things, or now that my semester is over, complaining to the professor about the grades they received. But whatever the strategy is, it's a different set of actions for different players, so we can't assume they're always all the same.
- 1.25** There are going to be a set of possible consequences of these actions. So if I do this, what might happen? And we have to think, well, here's what happens if the other players do this, if the other players do that, and so on. We'll think about a function that links these consequences to actions that might have those as consequences. They're not just free-floating consequences. They're things that are tied to various actions. Then we'll think about a preference relation or a utility function. In other words, I'll

think of myself as a player, as preferring this to that and that to this and so on. Or I'll just assign numbers. I'll think, well, this is worth so much to me; this is worth something else to me.

1.26 Now, sometimes, money—and hence the utility of this in economics— money serves as a great example of such a utility function. In other cases, though, all we really have is information about what I like best and what I like second best and so on. OK, so let's talk a little bit about these concepts of preferences and of utility, since the whole theory, in a sense, depends on that. We're going to talk about payoffs, which are these consequences of actions, possible consequences of possible actions, really. Then we have some preference relation for a player on those—what do I like best and so on—and we're going to think then about utilities; here's what that's worth to me.

1.27 So let's start with the idea of preferences. This is a concept that's not only really fundamental in game theory, but in all of economic thinking. Economics is generally the subject that devotes itself to how people in societies allocate limited resources to satisfy their wants. So, we have to think, what do people want? There's always a problem of scarcity. We never have enough to give everybody everything they want, and so how do we solve that problem? How do we allocate things within a society?

1.28 The key concepts of economics all depend on the idea of preferences. Think about some of the basic ones: trade, for example, an exchange of goods or services. Trade creates value. I have something you want. You have something I want. We make a trade. We're both better off as a result, typically, as long as we know what we're doing. And so, it typically creates value. Buyers and sellers exchange items to make themselves better off; at least, they think they're going to be better off.

1.29 So, I am thinking about something I prefer less, and then something I prefer more. Last night, for example, I went to a baseball game. I bought a cap. I decided that cap was worth more to me than the money it cost. It was a questionable decision, I've got to say. They charge a lot for those caps at a baseball stadium. But nevertheless, I thought, "Well, this is worth it to me. I want this cap." So, I ended up thinking, at least, that I'd be better off.

1.30 Markets are things that bring buyers and sellers together. I went into this shop where they have various sporting paraphernalia, Arizona Diamondbacks caps and shirts and so on. That was, in effect, a market saying, "You're interested in this? We're interested in selling it," and it brings buyers and sellers together. But notice, it all depends on the concept of preferences. I have to think, "Do I prefer to keep the money, or do I prefer the baseball cap?" The other people in the shop are making the same decision.

1.31 It's a matter there, I can't really say what that cap is worth to me, what's the maximum I would pay for it. I mean, an economist would think in those terms. I don't know. I was willing to pay that, though it's sort of like, hmm, but was I willing to pay more? Well, there was a cap that was a lot more. It was cool. I didn't buy it because I decided it's not

worth \$70 to me. Incredible. Anyway, but the point is, yeah, I'm thinking in terms of what I prefer, and the whole idea of economic exchange depends on that.

- 1.32** So does the idea of incentives. I respond to positive incentives. I get rewards for certain things, in other words, things I prefer, and then I may suffer certain penalties. There are negative incentives as well, and so, those disincentivize me. They're things I prefer not to have happen. And so, that, too, depends on the idea of preferences. Giving you an incentive is giving you something you would prefer, and giving you a penalty or disincentive, that's a matter of giving you something you would prefer not to have, like a fine for speeding or something like that.
- 1.33** Economics depends on the idea of marginal analysis. That is to say, what would be the cost of one more of these? How much would we have to charge to justify producing one more? When we think of it that way, those questions of more and less depend on preferences, too. A benefit and a cost of one more unit? Well, I have to think about the benefit. In other words, how much I would prefer that, but also the cost—how much I'd prefer not to bear that cost.
- 1.34** There's also the idea of trade-offs. I then think, well, OK, it's going to cost me this much to have that. Milton Friedman wrote a book titled *There's No Such Thing as a Free Lunch*. And Thomas Sowell has said, in a related vein, "There are no solutions, there are only trade-offs." But what's a trade-off? I think, well, getting this benefit, this thing I prefer, also entails costs, things I would prefer not to have to pay. And I can think in terms of opportunity cost. After all, I could be doing this or I could be doing other things.
- 1.35** And so, I think about the highest-value alternative, which, in my case, at the moment, since my semester is done, is not very much. But anyway, I would think about things I would prefer to have and then things I'm sacrificing to obtain something else, things I would prefer not to have to give up. So, all of this depends on the idea of preferences. So, modeling preferences is key to economics, but it's also key to game theory, and really, in a sense, just to rational activity in general. Even apart from what other people want, I have to think about what I want, what I prefer to what.
- 1.36** A simple example is, I walk into an ice cream shop, and there are all these flavors, all of which sound enticing. What should I choose? I've got to think, "Do I prefer mint chocolate chip to Belgian chocolate?" et cetera. Those can be hard decisions sometimes. OK, so all of this depends on the idea of preferences. Now, we're not going to assume very much about preferences. We're just going to assume, first of all, that we've got certain options.
- 1.37** If I've got two things I'm choosing between X and Y, either I prefer X to Y, or I prefer Y to X, or I regard them as on a par, as equally desirable. We're going to assume those are the only options. I like this better, I like that better, or I think they're about the same. Now, I want to say that's already a theoretical commitment. We're leaving certain

things out. For example, in assuming that this is better or this is better or they're the same, from my point of view, I'm assuming I can compare them. So I'm pushing aside here the possibility that these are just incomparable.

- 1.38** Bernard Williams talks about somebody who's contemplating a career as a musician, but also as a medical doctor. They're good at science. They think, "I could go to medical school, but on the other hand, I'm terrific playing the violin. Maybe I should try to be a member of an orchestra." And they face a real conflict like that. I've known some students who are like that, who were very musically talented, but also were terrific in science; or in one case, I'm thinking of computer science, and they had to make decisions, "Which direction do I go?"
- 1.39** You might say, it's not a simple thing of, "Oh, I prefer a career as a doctor to that of a musician," or they're about the same. It's more like, they're really different lives. I don't know how to compare them. And so, that's something where we could say, yeah, maybe there are cases like that, maybe, where I don't know how to compare them. Maybe there are even cases where, actually, I kind of prefer this to that, but that to this, too. There are different aspects of them.
- 1.40** Like the mint chocolate chip, I do love the mint. On the other hand, the Belgian chocolate is so rich and creamy. I might think, it's not just that they're about the same. In fact, we're going to assume that means they're just about the same for you. But you might say, "No, not really. I kind of prefer this one, but also that one." Well, we're going to say, then forget that. We'll just assume that means they're about the same for you.
- 1.41** We're also not going to think about dependent preferences. So, I say to a student, "What do you want to major in?" The student says, "Well, whatever my parents tell me I should major in, or whatever my friends are doing." We're going to forget about that sort of thing. We're going to assume you have preferences of your own. They're not dependent on the preferences of other people. Or "happy wife, happy life. I don't know, it's whatever my wife wants." Then we'll just drop you out as a player and put your wife in as the player because she apparently has preferences.
- 1.42** There are also going to be cases where, well, it depends on the circumstances. How often have I had Belgian chocolate in the last month? We're going to neglect that and just say, well, take the snapshot now. Forget about that dependency, too. Then there's something else that we can't capture just thinking about preferences, or at least not directly. We can kind of do it indirectly, as we'll see. But intensity is something that really requires something a little bit more.
- 1.43** So, I say, "I prefer this to that." But how much? Is it like that or is it like that? OK, maybe I prefer it a little bit. I somewhat prefer it. Or maybe it's, "Oh, I love this. I really don't like that." Those are important differences that, for the moment, we're ignoring. Now, we're going to come back to that. That's why it might be to our advantage to talk about

utilities or the value of something to me, as opposed to just what I prefer, because then we can measure intensities.

- 1.44** So what are we really then assuming about preferences? I've mentioned we are already leaving certain possibilities out. We're assuming that we've got something that can be referred to as "completeness," or "connectedness," or a "trichotomy principle," which is, this is better than that, that's better than this, or they're about the same. But in general, we're going to treat preference as a binary relation, and so we'll think of a relationship of something being at least as desirable as something else, which you can symbolize as a greater than or equal to symbol.
- 1.45** You can put a little squiggle to indicate they're about the same value, they're equally valuable to you. You could also say X is strictly preferred to Y, like a greater-than symbol. And so, we're going to assume that, well, these are all interdefinable. To say this is better is to say it's at least as good, and moreover, not equally good; it's actually better.
- 1.46** We could say, what is it for something to be at least as good? Well, either it's better, or it's equally valuable. What does it mean for it to be equally valuable? Well, it's at least as good as this, but that's also at least as good as the other thing. So we can really start with any one of those concepts, or at least any one of the preference relations—the strict one, the greater-than, or the greater-than-or-equal-to, the at least as good as—and define the rest.
- 1.47** Now, such a relationship is reflexive if everything bears it to itself, so everything is at least as good as itself. We'll assume that. But everything is as good as itself. Nothing is better than itself. We'll say there's a completeness if we've got either this one or that one. In other words, either it holds in this direction or it holds in that direction. So either X is at least as good as Y, or Y is at least as good as X.
- 1.48** We'll also assume transitivity. That is to say, if I think X is at least as good as Y, and Y is at least as good as Z, then X is at least as good as Z. The same thing for strictly better than. If X is better than Y, and Y is better than Z, X is better than Z. So I like mint chocolate chip, let's say, better than Belgian chocolate. I like Belgian chocolate better than I like vanilla. Then I like mint chocolate chip better than I like vanilla. It's an intuitive principle; it makes sense. We write it down in math, and it frightens half my students. But on the other hand, when you think about the basic concept, it's pretty simple.
- 1.49** So, that is going to imply another principle that sometimes will be useful, though we won't talk about it a whole lot until late in this course. And it has an ugly name, acyclicity. But the idea is, these don't hold in a circle. In other words, I don't say, well, I like X better than Y, Y better than Z, but Z better than X. Think about what that would mean. I like mint chocolate chip better than Belgian chocolate, and Belgian chocolate better than vanilla. But actually, I like vanilla better than mint chocolate chip.

- 1.50** If I really said that, you'd think you're deeply confused. Either you really mean they're all kind of equally good, or you mean maybe sometimes you like that better, sometimes you like this better. But if I just said that, you'd think, this person's crazy. And so, we're going to assume that doesn't happen. That's going to seem like a basic principle of rationality.
- 1.51** Now, there are people who have challenged all of these. They've said, "Well, maybe things aren't really transitive. Maybe there are cycles." We'll see later, especially in voting systems, the real possibility of cycles, which is frightening, because it turns out majority voting, for example, can lead to cycles like that. So, maybe each individual voter is rational, but the group of voters look completely irrational.
- 1.52** Unfortunately, in smaller groups or when there are lots of options, that happens a lot, so it's a bit frightening. We simplify it in our political system by having lots of voters and relatively few options. But if you think about a primary where you've got dozens of candidates, for example, and not many people showing up to vote in the primary, this sort of thing can happen a lot.
- 1.53** So, transitivity. Just think about that as a basic principle of rationality. I like mint chocolate chip better than Belgian chocolate, that better than vanilla. So I like mint chocolate chip better than vanilla. Seems reasonable. But now, Rubinstein has given people in general, starting with his students, but then a more general group of people who have read his books or watched his videos, a questionnaire that says, "What do you like better?" and he catches people in these intransitivities and in these cycles.
- 1.54** Now, only 15% of the people who respond to these surveys show no intransitivity at all. So, we could say 15% of those who responded here are actually completely rational agents in our sense. Other people show some sign of irrationality, having intransitive preferences, having these cycles. In fact, one person actually had 66 of these cycles out of 84 possible comparisons. That's incredible. But what's the general pattern here? I mean, how can you tell whether you're liable to fall into these cycles?
- 1.55** Well, in a way, it comes down to something pretty straightforward. Are you deciding on the basis of simple rules? For example, with the ice creams, maybe I have a simple rule, which is, if chocolate is involved, then I prefer it to anything where chocolate is not involved. Simple rule, and then I've still got to decide about the mint chocolate chip versus the Belgian chocolate, and so on. But at least that gives me a general heuristic. If you've got some simple rules like that, you tend to be highly rational. You don't get involved in these cycles. But the more holistic your judgment is, well, yes, it has chocolate like that, but you know, I don't know if you can mix it with nuts, but that doesn't mix as well with nuts. If I've got many criteria, it's easy to fall into these cycles.
- 1.56** In fact, a little aside, increasingly, admissions offices to colleges and universities say, "Oh, we do admissions holistically. We look at the whole person." Now, how they do that when they get 100,000 applicants, I have no idea. But in any case, they claim to

look at the whole person. And one school actually says, "There are 26 factors we consider." That is a recipe for this kind of intransitivity, for having cycles for, in short, irrational decisions. So, simple heuristics are way better if you want to actually produce something that makes sense in terms of an ordering of preferences that's coherent.

1.57 When you have lots of things going on, in a way it's very intuitive. The more criteria you have, the more considerations you're thinking about, the harder it is to keep all that straight in your head and to actually produce something that's coherent as a result. It's not too surprising that this results—it's also not too surprising that a few schools have said, "You know what? We're going to go back to the old way. We're just going to look at your GPAs and test scores. Forget about all the rest." That does simplify the situation completely. It also ends up meaning that the decisions are, from that point of view, much more rational. There aren't going to be these intransitivities.

1.58 Well, this has the result of partitioning your options into various groups. You might say, "These are roughly equal to me, but I like those better than these. They're roughly equal, and then I like those better than these." So, in short, it partitions the space of all the various options into certain categories. Within the category, things are about even. They're equally preferable for you. But this category, you'd like better than things in this category, and so on. Those things in general are called equivalence classes, and in game theory, we call them indifference sets, so I'm indifferent among these various options.

1.59 But for example, you say, "Hey, I'll give you \$20." OK, that's nice. How will you do it? Well, twenty \$1 bills, one \$20 bill, and so on. There are lots of options. You could Venmo it to me, et cetera. OK, that's nice. I prefer that to receiving \$10 in any of those forms. But within the form, I don't really care how you give it to me. It doesn't matter. And so, that would be a simple example of, yeah, I prefer \$20 to \$10, but the form that that transfer takes, I don't really care. I'm indifferent between all of those.

1.60 That brings us to this notion of utility. Sometimes it matters to us, not just what you prefer, but how much you prefer it. Then utility matters. The idea comes most famously, I guess, from Jeremy Bentham, who presents a theory he calls utilitarianism, something based on utilities. And so, it's a question of actually having a measure, a value function, that tells you how much you value something. Instead of just saying, "I like this better than that," I say, "Oh, this is worth \$20 to me. This is only worth \$10." I'm thinking in terms of numerical values I could place on this, cardinal numbers instead of just ordinal numbers, like, I choose that one first, this one second, and so on.

1.61 Now, you may not derive your preferences from a utility function. In many cases, I don't go into the ice cream shop and think, "How much more would I pay for the mint chocolate chip than vanilla?" I don't think of it in terms of utilities. But on the other hand, sometimes we do. In certain contexts, it's really important that we do. You can't just go to Las Vegas and hope to prosper by thinking, "I kind of like having money to not

having money." That's not enough. You need to think about it more seriously and more quantitatively.

- 1.62** So, you're going to, in a lot of cases, need something more. You're going to need some value function, some utility function, that says, "Here's how much this is worth to me." We can think of a utility function as representing your preferences when the things you prefer more get higher numbers. We'll say, if I like this one at least as much as this one, then the utility I assign it is at least as high as the utility of that one. The numbers have to match in order, in other words, the order of the preferences.
- 1.63** Now, it might be that, for example, if we're talking about money, it's just the money. I say, "Yeah, \$20 is worth \$20 to me, \$10 is worth \$10," and so on. But if you start thinking about large sums, it may be different. So, \$10 million and \$20 million, I don't think \$20 million is actually twice as valuable to me as \$10 million. I don't frankly know what I'd do with the extra \$10 million, and probably I'd just worry about it. Instead, you might say, "Yeah, it's going to tail off as I have more and more of this thing."
- 1.64** Initially, what it's worth to me would track the numerical value of the dollars pretty well, but at a certain point, it's going to tail off. I mean, if I'm Elon Musk, and somebody says, "Hey, I'll give you a dollar if you do something," what's a dollar to Elon Musk? On the other hand, the poor guy at the street corner who says "anything helps," a dollar presumably makes much more difference to his actual welfare.
- 1.65** So, this is not going to be a simple tracking. It's going to depend on all sorts of things, and it may depend on individuals. Some people might care a lot about money, for example. Some might care much less. But the idea is, we're going to have a utility function that represents your preferences, if higher numbers go along with preferring things more.
- 1.66** Now, for any given preference ordering, I could have infinitely many ways of assigning numbers to those because it all depends on intensities. I can say this is better than that, that's better than this. But how much? Are they equally far apart? Or is this one really close to that one? Or is this one way up here? Or this one way down there? It's different.
- 1.67** So, in a card game, for example, having an ace might be more valuable than having a king, a king more valuable than a queen, the queen more valuable than a jack, that more valuable than any of the number cards. On the other hand, how much more valuable? Well, it depends on the game you're playing. And so, the preference ordering you can define, but then how much more valuable is an ace than a king? Well, that depends on the game. And so, it might depend on all sorts of things. We have to understand the game we're playing.
- 1.68** The same thing happens in other kinds of games. Here's one of my favorites, Ticket to Ride, a sort of railroad game. You build the railroads and you get points according to the length of the track between any two cities. When you play that game, the length of

the route is pretty simple: 1, 2, 3, 4, etc. But the point values don't just go that way. They go 1, 2, 4, 7, 10, 15, 21, and so on. The longer it is, it's not just, "Oh, you get an extra point for that. Oh, you get six extra points for that," because it's really hard to get something that long. So that's a utility function, as well.

1.69 It turns out, every preference ordering can be represented by a utility function, at least on a finite set, because I have these preferences, then I define the utility function by just assigning numbers, so that's not even very difficult. Now, let's think about choices. After all, given my preferences or given my utilities, I have to actually choose something. It won't be enough for me to stand at the ice cream shop and say, "Well, here's my preference ordering, or here are my utilities." I actually have to tell them what I want, and so I have to make a choice in the end.

1.70 I'm going to assume that people are rational in the sense that, not only do they have a preference ordering or a utility function, but when you make a choice, you're aware of your alternatives and you pick the one with the highest value for you, so among the highest on the utility function or the thing you prefer most. It would be very strange to say, "I like mint chocolate chip better than Belgian chocolate, so I'll take the Belgian chocolate." What? I'm assuming that if you're rational, you'd choose the thing you prefer.

1.71 Or, "Wow, I'm looking for a new car, and Mazdas are highly reliable, so I prefer a Mazda. But on the other hand, my friend is selling a BMW, so I'll buy the BMW." That wouldn't make much sense if I prefer the Mazda. I've got to think in terms of what I prefer and then choose from the things I most prefer. Well, some choices then are rationalizable in the sense that they are explainable by a preference function and, oh, that's what you prefer most. But others aren't. So we're going to think of a choice function as making—of a choice, really—as rationalizable, if there is an underlying preference relation where this is among the things at the top.

1.72 It might be the top, or it might be just one of a bunch of things that are tied at the top. But in any case, it's one of those best things. But there are lots of things that aren't rationalizable. To give you the idea, this isn't a trivial thought. Here is Nassim Nicholas Taleb's idea of what to order in an unfamiliar restaurant, when you're with a group of people. You don't know what's on the menu, you don't know what's good. So he says, "Just order whatever the fattest guy at the table orders."

1.73 OK, now, why is that not—I mean, it's quite reasonable from one point of view. There's a guy who cares about the taste of food, so I'm going to order what that guy orders. That makes sense. On the other hand, it's not rationalizable because there's no stable preference ordering that is indicated there. It depends who the fattest guy at the table is. You're thinking, "Oh, that guy's ordering the beef. I'm going to order the beef." But then a fatter guy comes to the table and says, "Well, I'm having—" and they say, "Oh, well, then." So, it's dependent on something else. It's not really rationalizable on its own terms.

- 1.74** The same thing is true with the wisdom of crowds. Suppose you say, "I'm going to do what everybody else around me is doing." People have conducted experiments. They have people just go to a street corner, start looking at the sky, and then others start coming and looking to see what they're looking at. OK, you think, "They must be looking at something."
- 1.75** I used to be very good at fantasy football, but I started getting too busy to pay much attention. So I said, "I will follow the wisdom of crowds. I will just see who other people are starting and start those players." And you can do that. They actually give you these statistics and so on. Well, I've gone from being a contender for the Super Bowl every year in my fantasy league to being among the worst teams, because it turns out, the other people in the crowd are also people who are saying, "I'm going to just follow the wisdom of the crowd," so it turns out, this is not a good strategy. But anyway, it's clearly not a rationalizable strategy. It doesn't depend on a preference ordering of the players or a value function of how much that player is worth. It's just like, what are other people deciding to do this week?
- 1.76** Pandering. Politicians telling people what they want to hear. Clearly not rationalizable. If somebody just says, "Well, I represent the voters of Texas, so I'm going to just advocate whatever they advocate," maybe that's their job, actually. I'm not opposed to that thought. But on the other hand, it's not a rationalizable thing. It depends then what other people are doing.
- 1.77** The same thing is true with a strategy of follow the leader. There are lots of contexts in life where actually my strategy has been, "Just do what the person ahead of you is doing." It's gotten me through many complicated situations. But one time, to indicate the non-rationalizable aspect of this, at the last minute, the person who was supposed to be the commencement marshal for the University of Texas got sick. I was there as a faculty member to be on stage, and they suddenly said, "You, you're going to be the commencement marshal."
- 1.78** Well, I've never done this before. I don't know how it works. I was the guy in the front, holding the big banner, and everybody had to follow me to find their seats and then to go up onto the stage. Well, I had no idea what I was doing. My strategy had always been, "Follow that guy." And so suddenly, I'm that guy. I don't know what to do. And so, the history majors graduated under the anthropology banner, and the anthropologists graduated under the philosophy banner, and so on. It was a total disaster. I've never been asked to do it again. I'm a very good follower in that context. I'm a terrible leader. The fact that you were guy number two does not mean you know how to be guy number one.
- 1.79** There is another silly example of that, where I was trying to drive from Pittsburgh to Binghamton, and there was a huge rainstorm in northeastern Pennsylvania. A lot of the creeks and rivers were flooded. So I got to about a little town called Monroe, and all the bridges were washed out. I had no idea where to go. But there was a car that also was

there at the washed-out bridge that seemed to understand. So it backed up and started going off in this direction. So I thought—and this is long before the days of GPS or anything like that—I thought, I'll follow that guy. He seems to know what he's doing. But I don't know what bridges are still open, what roads are flooded.

1.80 So I follow this person, and I followed him for several miles till finally he stops. And by the way, it's pouring rain, so I can't see more than a few feet in front of me. And then it turns out, this is the guy's driveway. He just went home. So, at that point, I got out of the car. I was like, "Help, I'm trying to get to the other side of the river in Twanda." And so, he gave me a way of doing it, so I got there eventually. But yeah, follow the leader—in that case, you have to know where the leader is going, and in this case, I had no idea. That was not a rationalizable choice.

1.81 Now, there's another more complicated situation, more interesting, I think, philosophically. Aristotle holds that virtue is an intermediate between extremes. Honesty, for example, is a mean. It's a mean between dishonesty, but then bluntness or rudeness. You should tell the truth to the extent that you want to tell the truth, in the right way, in the right circumstances, to the right people, and so on. But it requires discretion, and if you don't have that discretion, you can, in effect, be too honest.

1.82 You might think, "Oh, I prefer the middle ground. I seek the golden mean, so I just want whatever's in the middle." That means it depends what the options are on both sides of you, so that can change, making this not a rationalizable method. Here's an example. If you were looking at the Democratic primaries in 2020, initially, you might look and say, "I ordered them in this order, so this candidate seems to be the one in the middle." But then, after March 1, when a lot of people have dropped out, you might think, "Now it's a different person in the middle."

1.83 It's not unique to the Democratic Party. Four years earlier in the Republican Party, it was the same thing. You looked after the Iowa caucuses, and maybe it looked like Donald Trump was the one in the middle. Then later, it might look as if Marco Rubio is the one in the middle. It all depends on who's dropped out and how many are on which side. Just thinking "I want the middle ground" is not going to lead you to a stable situation.

1.84 One more example of dependence. Suppose you've got these three coworkers, and they all schedule a party on the same evening. Carmen, Bob, and Ed, let's call them. You like all these people. You like Bob and Ed, especially, but they hate each other. You're caught in the middle all the time between them. If just Carmen and Bob had invited you to a party, you'd definitely go to Bob's party, because you like Bob a lot.

1.85 On the other hand, if Carmen, Bob, and Ed invite you to a party, if you go to Bob's party, Ed's going to be mad at you. Go to Ed's party, Bob's going to be mad at you. So you go to Carmen's party. That doesn't reflect any underlying preference ordering. You really prefer Ed and Bob to Carmen. But on the other hand, given that these two hate

each other, it's dangerous to choose either one of them, so you end up doing the thing you less prefer, just because it keeps you out of trouble. That, too, would be a case where we don't have rationalizability.

- 1.86** Now, there's actually a pretty simple test for whether or not your preferences are rationalizable. It's something developed by Paul Samuelson in 1938. If you choose X when Y is also available, then don't choose Y when X is also available. Are there situations where you'd prefer this to that? Good. There shouldn't be other situations or other mixes of options where you prefer that one to this one.
- 1.87** Now, things can depend on circumstances. Like, wait, did Ed invite me to a party, too, the same night? And then you have to choose. Then that's a change in circumstances that does affect your preferences in a way. But here, I'm thinking of just the addition of certain options, and in that situation, your preferences depend on what's there. We're really ruling that out. Here's my point. To say that a choice is rationalizable in the sense we're using it, doesn't mean it's reasonable or that deviating from that is unreasonable. You could have all sorts of reasons for doing something that doesn't reflect an underlying preference ordering. But we're generally going to simplify life by putting those aside.
- 1.88** So, there's a general idea here that if you have a preference relation on a finite set, you can always come up with some utility function, some value function, that tells you how much that's worth to you, that would represent that preference ordering. It doesn't always work for infinite sets, which is kind of interesting, and it's very easy to see why. If you have multiple criteria, and you say, "First I go to this criterion, and if something's better on that one, I choose it. But if they're tied, I go to this other criterion," that is really a two-dimensional thing. It can't be reflected in one dimension if you've got an infinite set of possibilities.
- 1.89** But for the moment, let's put all those complications aside and just go back to thinking about what it is to be a rational agent. I've got a set of actions from which I'm choosing, and they lead to certain consequences. I've got preferences among them, or they're worth a certain amount to me, reflected by a utility function. Now I'm playing a game. I have to think about this in the context of other rational agents who also have actions, and they lead to consequences for them. And so now I'm in a more complicated situation. I have to develop strategies.
- 1.90** What's a strategy? Well, it's going to be a way of playing the game. I'm also going to have payoffs. That is to say, I choose this, other people choose that, here's my payoff. That can be represented in terms of preferences like, "Oh, that's first choice. That's the thing I like best." Or it can be represented by utility functions or just more numerical values, saying, "This is worth 10, this one's worth five, and so on."
- 1.91** So, what is a strategy? It's a rule for saying how I'm going to play the game. Maybe it will be, "I always raise in poker." Terrible rule, but nevertheless, that would be a

strategy. It's going to specify an action. A mixed strategy is going to specify probabilities over actions. So, for example, in poker, if I always do the same thing, that's a terrible strategy. That would be a pure strategy. Instead, I've got to mix it up a bit. I've got to think, "Well, I have a bad hand here. Should I fold? Maybe I should bluff." And I've got to combine those to be very effective and combine them in the right way.

- 1.92** So we'll later on take mixed strategies very seriously. For a while, we're going to put them aside and mostly think about pure strategies. A pure strategy does give me a complete definition of how I'm going to play the game. I give the same response to every move by the other player, and it determines, in that way, the move the player is going to make, no matter what the situation is. A strategy set is the set of these available pure strategies that I have.
- 1.93** I might think, for example, I can raise, I can call, I can fold in poker, or I might think, playing tennis, let's say, I can hit it forehand, or I can hit it backhand. Those are my options. I might have a pure strategy that says, "My backhand's terrible. Always use forehand." Not a great strategy again for tennis, but nevertheless, something that would be a pure strategy. A mixed strategy mixes those up, basically says, "Sometimes I'm going to hit forehand, sometimes I'm going to hit backhand. Sometimes I'm going to call, sometimes I'm going to fold, sometimes I'm going to raise." And so, it assigns probabilities to these pure strategies. It allows me to respond in different ways to the same set of moves by other players.
- 1.94** Now, sometimes mixed strategies are extremely useful, basically, when you don't want the other player to know what you're doing. Now, if you're coordinating with the other player, if you are cooperating with them, you want them to know what you're doing, so a pure strategy is generally much better. But if I'm competing with you or in conflict with you, I want to keep you guessing, and so it's really important.
- 1.95** As Sun Tzu said, "The essence of warfare is deception." I've got to trick you into thinking, "Hey, I don't want you to know." So, I don't want you to know whether I'm going to hit this forehand or backhand. I don't want you to know what play I'm going to call as quarterback. I don't want the batter to know what pitch I'm going to throw. In fact, last night, Paul Skenes, by the eighth inning, he struck out the side, but it was clear he didn't have much on the ball anymore, at least in terms of speed. He wasn't throwing those 98-mile-an-hour fastballs anymore.
- 1.96** In fact, I don't think he threw one the entire inning, but he struck everybody out anyway, by just varying the pitches, making it clear that the batter had no idea what was coming. When you've got, say, five really good pitches, even if one of them is no longer there because your arm is tired, you still have four and you can deceive the batter. So, deception turns out to be critical.
- 1.97** At other times, a mixed strategy would be terrible. Let's say, in traffic, should I drive on the right side of the road? Should I drive on the left side? If the light turns yellow,

should I stop or should I speed up? If I think, "Well, I think it'll be 50-50. Sometimes I'll pretend I'm in England and drive over there," this is a terrible thing. That's a coordination problem, and if I'm trying to coordinate with you, or like, let's say, "All right, we're going to meet for dinner," "Great. Where do you want to go?", I said, "Well, I might be at this place or this place or this place," that's not a good strategy for coordinating.

- 1.98** I have to know what kind of game I'm playing. Here I'm going to talk primarily about four different kinds of games. Some are coordination games, like in traffic. We have to coordinate. I have to stay in my lane, you have to stay in your lane. We have to decide what side of the road we're driving on and so on. We have to all play by the same rules or we're going to be in trouble.
- 1.99** I don't know if any of you have ever driven in Latin America, but there are traffic laws, there are traffic signs, but as far as I can tell, they're just vague suggestions in people's minds, not actual rules. And so, the coordination problem gets much more difficult. It turns into a competition. Cooperation is the same way. If we're going to cooperate, I need to know what you're doing, and you need to know what I'm doing.
- 1.100** But if we're competing, then I do want you not to be sure. I don't want you to know whether I'm going to throw a changeup or a fastball. If we're in a situation of conflict especially, I don't want you to know whether we're going to attack over there or over here. And so, there are going to be times where I have to know what kind of game I'm playing. In fact, a lot of interpersonal conflicts can come down to this sort of problem. I think I'm trying to cooperate with you. You think you're competing with me.
- 1.101** And so, if you think, "How could that happen?", imagine that you are, let's say, junior partners in a law firm, all competing to get a partnership. I should say, junior associates in a law firm, trying to get a partnership. Well, one of you thinks, "Look, we're going to be better off if we all cooperate and coordinate our activities, so we look good as a group." Somebody else says, "I want to look like I'm the best. I'm actually in competition with you." All of a sudden, there's a danger here, because I think I'm cooperating, you think you're competing.
- 1.102** Now, coordination problems, they're all around us. Think about air traffic control. That's a great example of a coordination game. Traffic is another one, but there are many of these. At the university, we have to decide who's teaching what in what classrooms. That's a huge coordination problem. Students might have the nightmare of suddenly walking into an exam and finding out, "I didn't know there was an exam today."
- 1.103** The professor's version is, you suddenly find out you're supposed to be teaching some class, and you haven't been going all semester. You didn't know you were to be teaching it. If it's like the dreams I have, it's in something you don't know. What? You're supposed to be teaching 17th-century French poetry. Oh my gosh, and it's November.

Everybody's been sitting there waiting for two months, and I don't know French, like, oh, no. So that's the professor's version. But in a way, it's a coordination problem, like, oh, the coordination failed.

- 1.104** There are times when it's important to cooperate. In fact, there's this game that I have called Pandemic. Actually, during the year of the pandemic, we did not play this game. I kept saying, "We should play this game." And everybody else thought, "No, we're living it. We are playing this game in real life. We don't need to get out the board game." But it's important for the players to cooperate in order to defeat the pandemic.
- 1.105** Other games are competitive. We're trying to get to the end first, for example. Or there are games, actually, this is one of my favorite childhood games, Conflict, where actually we are opposing military powers and we're trying to use our naval power and air power and our armies in such a way as to defeat the other player.
- 1.106** There are other distinctions among games that are really important that I want to mention here initially. So one of them is the distinction between simultaneous and sequential games. They're also called extensive games. The idea here is that, in a simultaneous game, we all play, in effect, at the same time. We don't know what the other players are doing, because we have to commit before we find out. But in a sequential game, it's more like we take turns. I make this move, then you respond and make that move, then I move again and so on. Those are importantly different kinds of games.
- 1.107** There are also some games that are zero- or constant-sum games, where there's a fixed amount, and if I get more, you get less. The amount totally for the players remains the same. But other games are mutually beneficial. We can cooperate and we both benefit, like economic exchange. We both end up better off if we know what we're doing and we actually successfully complete the trade. Some of them are destructive, like war. We actually have a negative outcome for both of us, perhaps.
- 1.108** So, in a simultaneous game, we choose actions; we don't know what the other players are choosing. We have to commit before we find out. There are lots of things like that. Somebody has a request for proposals, and we have to write a proposal for a grant, let's say, or for a contract, and we don't know what the other people are proposing, so we don't know how we compare. We have to commit before we find out.
- 1.109** Applying for a job or for admission to a college is like this. I apply, I don't know what other people are doing and applying, and so I've got to commit before I find out anything about the competition. A closed, sealed bid auction is like this. I bid a certain amount. I don't know what other people are bidding. Football play calling is like this. The defense calls a play, the offense calls a play, and we go out there and play that. Only in some cases, let's say the quarterback calls an audible, do we actually have a response to what the other team is doing; typically, we have to commit before we know what the other team is doing.

- 1.110** Voting is like that. I go into the voting booth and I cast my ballot. I don't know what other people are doing behind their screens, and what ballot they are casting, who they're voting for. I might vote differently if I knew what everybody else was doing. In battle, that's often the case. I have to send my troops here before I find out where the opposing general is sending his troops.
- 1.111** Invitations are like that. I invite a bunch of people to my party, let's say, and I don't know what other events maybe are happening on the same evening. I don't know whether this person is going to accept before I invite that person. Do I invite both Bob and Ed? Well, gee, I'd like to know, if Ed accepts, then I don't want to invite Bob, or some complicated thing like that. And exams—a student writes an exam paper, let's say, and they don't know what other students are writing. At least you hope they don't know what other students are writing, so that they have to commit before they actually know what's going on with the competition.
- 1.112** Now, a lot of the games we talk about are going to be simultaneous games like that. In fact, many of the most famous examples from game theory—prisoner's dilemma, stag hunts, games of chicken, the battle of the sexes, mutual hostage games, pure competition—we're going to talk about each of those, and they're big things that are discussed for the most part in the literature. The mutual hostage game really isn't, but the others are.
- 1.113** A lot of these were already, in 1951, developed in Nash's original paper on the topic. It's kind of amazing. He has examples there, gives us six examples of games, and they turn out to be ones that have just been discussed mostly in the literature since. Competitive games, prisoner's dilemma, stag hunts, the battle of the sexes—they're all there already in his 1951 paper.
- 1.114** But in a sequential game, it's quite different. Most board games are like this. I know what move you made. I just saw you make it. Now it's my turn. I make my move. Chess is like this. Checkers is like this, and so on.
- 1.115** So, negotiations are like this. You bid that, I think, "Well, am I going to accept that, or am I going to counter-propose something?" A trial is like that. This attorney presents this side of the case, the other attorney responds. Admissions officers can do this, because they can say, "We admit these people. We put these people on a waiting list, so it's going to depend." Open bid auctions are like that. How much am I going to bid? I don't know. I have a maximum, let's say, and then other people start bidding, and I may change my mind, depending on other players' moves.
- 1.116** Dating is like this. You respond to what the other person is doing. Marriage is like that. All human relationships are really like that. They are in part sequential, where you just look at how people behave and you behave a certain way in response. As I mentioned, chess is like this, poker, blackjack, debate, conversation. So lots of things. In fact,

increasingly in linguistics, people think about conversations as games and use the concepts of game theory to analyze linguistic communication.

- 1.117** Some games are zero-sum. That is to say, any gain for me, you lose. Or sometimes it's just a constant sum. Hey, this is the budget, and if your group gets more, somebody else gets less. A gain for one player is balanced by a loss for the others. Those games are strictly competitive. Lots of things are like that—oil and gas licensing, promotions, budgeting, real estate negotiations, football. You might see the yards gained by the offense are yards given up by the defense. A lot of bargaining really ends up being like this.
- 1.118** We're going to write games in a certain form, which I'll call strategic form or normal form. We'll put the players on opposite sides in different dimensions, and then we will essentially say, OK, here are their moves. We'll list the basic moves. For example, you can drive on the right or drive on the left, and then we'll give preferences. Those things will be the payoffs, and we might just say, in ordinal terms, this is that player's first choice; this is their second choice.
- 1.119** Or we might actually have numerical values, cardinal values, to say it's worth this much to that player. So these outcomes can be represented by utilities or just by payoffs, for example, monetary payoffs, or they can be lists, preferences. We'll typically write the outcome for player A before the outcome for player B. So, for example, here would be a form where we say, A can play right or left. Let's say they play right. B can play right or left. They play right. So they each get their first choice. That would be the optimum for both of them.
- 1.120** Here's a very simple zero-sum game. To give you the feel for this, two-finger morra. It's very simple. There are two kids playing this game. You can either put out one finger or two fingers. It's simultaneous. You have to commit before you know what the other player is doing, and you call even or odd. Let's say I call even. Well, then I put out one finger. I win if you also put out one finger. But if you put out two, then I lose.
- 1.121** Now, notice the sequential aspect of this game would be utterly destructive. For example, I say, "I call even," and I play one. Now it's your turn. Then I always lose. It only matters if it's going to be simultaneous. In that case, OK, great. We've put out one, so even, I win. I get the penny, you lose the penny. It's a zero-sum game. If, on the other hand, you put out two, now it's odd, so you win, now I lose a penny, you gain a penny, and so on.
- 1.122** So that would be an example of a very simple case, where I think, "What's my best response?" Well, if you're putting out one, and I'm even, I want to put out one. If you're going to put out two, I want to put out two. What's weird is, you would think, well, obviously, the best thing for me to do is just let you have no idea of what I'm going to do, mix them up 50-50. It turns out that people are bad at randomizing. We'll talk much

more about that when we get to mixed strategies. It's rational here to play each 50% of the time.

- 1.123** There are also mutually beneficial games, and we're going to talk a lot about those, things where we can both benefit. And so, examples of that are, as I mentioned, economic exchange. One caveman wishes he had a rock; the other wishes he had a stick. If they trade, they're both better off, mutual advantage. A lot of friendship is like this. Romantic relationships are like this. You hope both parties benefit. It's not a zero-sum interaction.
- 1.124** Or it might just be that you decide, "What are we going to do this evening? We both prefer to read. We each do our own thing." And so, yay, we both get what we most wanted: to have an evening just quietly reading at home. Now, in a case like that, where I think, "Yeah, I'd rather read, no matter what you're going to do," and the other player thinks that, too, reading is called our dominant strategy. That is to say, it's better for me, no matter what you do. That's going to be one of the most important concepts we think about.
- 1.125** Is this something that makes me at least as well-off as anything else, given that you're doing this? That's an important question. I think, is this among my best responses to what you're doing? But if it's a best response, no matter what you're doing, that makes it a dominant strategy. Only sometimes will I have a dominant strategy where I can actually say, "You know what? It doesn't matter what you do. I'm better off doing this in any case." Sometimes that will happen, but most of the time it will not.
- 1.126** I'll close with a famous example where that happens. This is Pascal's wager. Pascal wrote, "Let's weigh the gain and the loss in wagering that God is. Let's estimate the two chances. If you gain, you gain all. If you lose, you lose nothing. Wager then without hesitation that He is." Now, notice this is not an argument that God exists. This is an argument for belief, saying, "Look, you're better off believing whether God exists or not." And that's in effect saying belief is your dominant strategy.
- 1.127** So we could represent this in normal form, almost as a game against the universe. So here you're playing, and either God exists or God does not exist. But you're thinking, suppose God does exist, and I believe, then I gain the glories of heaven. Suppose I don't believe. Oh, I go to hell. A terrible outcome for me. Clearly, my best response, if there's a God, is to believe.
- 1.128** Now, suppose there's no God. He says, well, if I believe, at least I will live as if there's a God. That is to say, my life will have meaning for me, and I will lead a life of virtue to the extent that it's possible for me. Suppose, on the other hand, I don't believe. Then I'm going to fall into vice. I will lead a life that even I consider meaningless. I'm better off believing even if there is no God. In short, he says, "I'm better off believing whether there's a God or not, so belief is my dominant strategy." Now, to say that doesn't mean

God does exist. He's basically saying, you should believe, whether God exists or not, because you're better off believing. That's to say, belief is your dominant strategy.

- 1.129** There are going to be lots of games where we see that people do have a dominant strategy. A general rule for us is going to be: to be rational, play your dominant strategy, but not always. So next time, we'll look at some examples where, yes, it makes perfect sense to play your dominant strategy, and other cases where, if we all play our dominant strategy, we're all in deep trouble.
- 1.130** So, in a sense, individual rationality and group rationality are going to turn out to be rather different things. Being rational as a member of a group is not the same as having the group be rational. Sometimes, in fact, if we all play our dominant strategy, we all suffer. But more on that next time.

Lecture 2: Coordination and Convention

At a glance — Coordination games and convention.

- 2.1** Welcome to the second lecture. I'm going to be talking about coordination games and about social conventions, which are really examples of solutions to large-scale coordination games. Remember, we talked about different kinds of games involving coordination, cooperation, competition, and conflict. This is the first kind, and I think it's worth discussing some of these in detail. These are typically games where we want pure strategies, so they're often relatively simple.
- 2.2** I don't want to keep you guessing if I want to coordinate with you. In general, we have to find ways of getting along together in society, so that we don't get in each other's ways, so we can communicate, so we can actually achieve our own goals, both together and then individually without interference. That's what this sort of game is all about, really. How do I solve a certain kind of problem, where either I want to do my own thing without interference from you or I actually want your cooperation, I want your coordination, so that we can accomplish a task together? So we want to establish some kind of agreement between the two of us.
- 2.3** In general, a "coordination game" is one where players are rewarded if their strategies, in some sense, coordinate—if they match. Now, that doesn't mean we have to do exactly the same thing, but we have to do things that actually work together with whatever the other players are doing, in such a way that they make a coherent whole. We need to do our thing. It might be, we do the same thing. "Hey, let's all meet at that restaurant at that time." Or it might be that we simply have to do our parts. "Here's our joint project. So you're going to do this part, I'm going to do that part, you're going to do this part," and so on. And so, coordination is a matter of us all arranging our actions in concert with one another to achieve our overall goals.
- 2.4** The person who was really vital for understanding coordination games is John Nash. He was the subject of the movie *A Beautiful Mind*. He won the 1994 Nobel Prize in economics, along with John Harsanyi and Reinhard Selten. He's somebody who was born the very year that game theory was first developed, in that 1928 paper by John von Neumann. He died in 2015, and he had a really remarkable life. He studied at Carnegie Institute of Technology—Carnegie Tech, now Carnegie Mellon University—and then at Princeton. He spent most of his life at MIT and then at Princeton, though he also battled mental illness for much of his life.
- 2.5** Unfortunately, many of the superstars of mathematical logic, game theory, and other fields in the 20th century were like this. They had unbelievably great technical talent, and they also just suffered from mental illness. I've wondered whether there's a connection. Emil Post was the same way. Post was a brilliant logician, and yet his doctors would tell him, "Stop working on mathematics, stop working on logic. It's

making you sicker." His friends would try to observe this, and then they'd encounter a problem they couldn't solve, and they would say, "Post could solve this in a second." They would ask him about it, so he would solve it, publish a brilliant paper, and then go back into the asylum.

- 2.6 Nash was a bit the same way. He haunted the Princeton campus for many, many years, and students knew who he was and occasionally could go over and talk to him. He was very nice if people talked to him. But on the other hand, he mostly kept to himself, as the weird guy sitting over there.
- 2.7 In any case, he came up with one of the most foundational concepts of game theory: the Nash equilibrium. We've talked about what it is to make a best response to what another player does. So, in two-finger morra, if I've called even and you play one, my best response is also to have played one. Then together, we'll form an even number. One plus one is two, and so I win.
- 2.8 A dominant strategy is one where it turns out the same thing is my best response, no matter what you do. That's relatively rare. But a Nash equilibrium is a much more pervasive concept. The idea is that each player is playing their best response to what the other players are doing. So, I do this, and that's the best thing I can do, given what other players are doing. On the other hand, they are also doing the best thing they can do, given what I'm doing. Now, in a zero-sum game, you might think, "Well, that doesn't work out very well." Indeed, only then are mixed strategies Nash equilibria. But in a lot of games, the Nash equilibrium is something that does involve pure strategies and is a really important notion.
- 2.9 Here's how Nash himself expressed it. "Each player's strategy is optimal against those of the others." Even if I lose the game, in the end, I say, "Well, I couldn't have done any better. I made my best possible move in every single case."
- 2.10 Now, why are these important? Because they're uniquely stable. No player can do better individually. That is to say, no player can actually unilaterally achieve a better outcome. If all I can do is change my own strategy, I have nothing to gain by doing that. If we're in a Nash equilibrium, I am already making my best response to what you're doing. That's not to say that we as a group couldn't do better; maybe we could, but we'd have to coordinate. We would have to get together and all change our strategies, or at least many of us would. I can't do it alone. So, nobody can improve his outcome unilaterally. That does make these highly stable.
- 2.11 If I'm in an outcome that is not a Nash equilibrium, then there's some player that could improve their outcome unilaterally and has an incentive, therefore, to do it. Those outcomes tend to be unstable in the game. Somebody's got to realize, "Wait, I could make a better move and push us in that direction." On the other hand, if I'm in an equilibrium, I can't do anything myself to improve my situation.

- 2.12** Here is Nash's existence theorem, which is an important result in game theory. Every game with a finite number of players, in which each player can choose from finitely many options, has at least one Nash equilibrium. In fact, [Robert Wilson] proved that each such game has an odd number of Nash equilibria. So coordination problems are solvable. That's the good news.
- 2.13** We can come up with equilibria that actually solve coordination problems. We can come up with equilibria that solve other kinds of problems. In some cases, not in all cases—competition problems with pure strategy—sometimes we can. Sometimes it's a mixed-strategy solution. So these won't always be pure-strategy solutions. Sometimes they're going to involve probabilities and keeping the other player guessing. If I've got cooperation problems, very frequently, as we'll see, there are Nash equilibria, and those are important things to identify.
- 2.14** So among those six games that Nash talked about in his 1951 paper, there is one that is a coordination game, a classic sort of coordination game. He uses an actually easily understandable notation, but it's not the standard notation now.
- 2.15** A coordination game might look like this. I, as player A, have a choice between going to the right and going to the left. You, as player B, have a choice of going to the right or going to the left. If we both go right, we both get a payoff of one. If we both go left, we each get a payoff of one. Now, I am, as player A, thinking, "I don't know what you're going to do. If you go right, then I get a payoff of one if I also go right—on the other hand, minus 10 if I go left." So that's a case of us not coordinating. I really suffer from not coordinating with you. And so, my best response is to coordinate with you and also go right.
- 2.16** If you go left, on the other hand—well, if I go right, I get minus 10. On the other hand, if I go left also, I get one. And so my best response is to go left, given that you're going left. It's the same from your point of view. If I'm going right, you're better off going right, and if I'm going left, you're better off going left. There are two outcomes in this case, in which each player is playing a best response to what the other is doing. I am thinking, "If you're going right, it's best for me to go right." But the same thing is true of you. If I go right, it's best for you to go right, so that's a Nash equilibrium.
- 2.17** Now, it's very easy to see in this case that there are two of these. It doesn't matter whether we both go right or both go left. What's important is that we coordinate and do the same thing. This is a classic driving game. We think about driving, and we think, "Am I going to drive on the right or on the left?" I think this is an apocryphal story, but I've seen it in a number of places. In 1895, there were only two cars in Ohio. Guess what happened? They collided. I've seen the same story about St. Louis and other things, so I'm not sure this really happened. But if you don't have any rules about coordinating, the odds are you collide and get a bad outcome.

- 2.18** One of my favorite TV shows of all time was Buffy the Vampire Slayer. Even though it was filmed in the late '90s, nobody had a cell phone in that series. If they had had cell phones, half the plots would have been destroyed because half the plot is, "Where's Buffy? What's Buffy doing?" If we could have called Buffy, it would have simplified life immensely because then the team could have coordinated. Half of what's going on there is a coordination problem that nobody has enough information to solve, and there are no conventions to solve.
- 2.19** In any case, the driving game is one of these classic coordination problems. In fact, it's very much exactly what Nash talked about in 1951. If we both drive on the right, we can get to our destination safely, let's say. If we both drive on the left, we can get to the destination safely. On the other hand, if you're driving on the left and I'm driving on the right or vice versa, it's risky. We could very well end up colliding. And so, it's very important that we operate under the same set of rules.
- 2.20** If I think of this in ordinal terms instead, I get my first outcome, driving safely, if we both go right, and the same is true for you. If we both drive on the left, we also get our first outcome, getting there safely. Risky drive—that's if one of us is on one side, one of us driving on the other. That becomes dangerous. That's clearly a second choice.
- 2.21** And so, if I think about this from my point of view as player A and think, "Okay, you're driving on the right. It's definitely better for me to drive on the right. If you're driving on the left, definitely better for me to drive on the left," but it's the same for you. I drive on the right, better for you to drive on the right. I drive on the left, better for you to drive on the left. We end up with two Nash equilibria, one where we both drive on the right or we both drive on the left. Both of those are good.
- 2.22** Now, I mentioned that Nash's existence theorem says there's not only a Nash equilibrium in all such cases, but there's an odd number. We only have two here. There's got to be a third. Well, there is a third where we each flip a coin and drive on the left 50% of the time or the right 50% of the time. In this case, that's a disastrous one. We don't want to do that. "Okay, what are we going to do today, people? Let's see, we'll drive on the left, pretend we're in England." That's not going to work out very well, unless everybody else is doing the same thing.
- 2.23** We've got three Nash equilibria here: two pure-strategy ones that allow us to coordinate, one that would be insane given the practical circumstances of this case. This is typical of coordination problems. I've got to adjust my actions to the actions of others to promote good outcomes for everybody—good outcomes for me, in particular, but also good outcomes for everyone else. That's what's nice about coordination games. A solution for me is typically a solution for everybody else as well. I'm not just winning; we're all winning by obeying the same rules and adopting the same conventions.

- 2.24** It is going to be typical in these kinds of cases that there's more than one way of doing that. You could all agree that you're going to drive on the right. You could all agree that you're going to drive on the left. It doesn't really make any difference. They both work. Coordination problems are often going to be like that. We want to coordinate and meet at the same restaurant for dinner. Pick a restaurant. It doesn't really matter. There might be a number of ones that are successful coordination solutions in this case. We just want to be at the same one.
- 2.25** So, we need to coordinate, because often the problem is, which one? Hence, the difficulties of the plots of Buffy the Vampire Slayer. Which possible solution is Buffy pursuing now? And so, with driving, maybe we're in the UK and Australia, all driving on the left. Maybe we're in the United States or most of Europe, driving on the right.
- 2.26** Now, here I've represented this as if we have exactly the same preferences. But actually, it doesn't matter if we don't. For example, let's say I'm an American, used to driving on the right, and you're someone from Britain, let's say, who's used to driving on the left. Then actually, we both have preferences about these two solutions. I would really rather drive on the right and have you driving on the right, too. I know that's not your first choice. If you're going to drive on the left, on the other hand, then I want to drive on the left, too. But that's clearly my second choice.
- 2.27** Now, each of us has as a third choice that the person from Britain does what he wants. He drives on the left; I drive on the right. That's not great, but at least we're used to doing that. The worst outcome is where I'm driving on the left and he's driving on the right. This is really awkward for both of us. That makes it even more dangerous. In this case, my first choice is, we both drive on the right. My second choice is, we both drive on the left. Third is, we each do our own thing. The worst is, we each do the opposite of what we'd really like to do.
- 2.28** You do the same. You say, "I'd rather drive on the left and have you drive on the left." Second choice is, we both drive on the right. Third, we each do our own thing. Then the worst-case scenario is, we're both awkward. It doesn't actually matter in this case because, look, if you're going to drive on the right, I want to drive on the right. Yay, first choice. If you're going to drive on the left, it's my second choice, but it's still best for me to drive on the left.
- 2.29** And so, we get the same Nash equilibria, whether that's really my first choice or not. As long as I prefer one or the other, and you also prefer one or the other, it doesn't make a difference to the outcome. Now, it might make a slight difference in the mixed-strategy one, but on the other hand, we don't want that one anyway. We want to coordinate and do the same thing. And so, as a result of that, we could say, look, having a preference among the Nash equilibria won't actually change the form of the game solution at all.

- 2.30 However, it will mean that one of us is going to actually get our preference and the other isn't. I might feel awkward. I'm in England, I have to drive on the left. I've done that in a stick shift. It's weird having the stick in your left hand and changing gears like that. But on the other hand, after a few hours of terrified driving, you can get used to it and do it.
- 2.31 What surprised me when I was at Oxford is that it's not just that people drive on the left, people walk on the left. So I was constantly on the wrong side of the sidewalk or on the wrong side of the staircase. I'm going up on the right, and the people coming down are, oh, okay. So, that was actually sort of harder to coordinate. Oddly, when I got back to the United States, then it was kind of hard to shift back to walking on the right, because I was now confused, what do I do? And actually, on the Oxford campus, it's also confusing because there's so many people from other countries at Oxford that it's like, on campus, it's all chaos. But in the rest of the town, everybody keeps to the left.
- 2.32 Now, if you think about the form we've just talked about, where we do have preferences among the equilibria and different preferences, but there's still two good Nash equilibria there, this is actually the form of a game that is known in the literature as the Battle of the Sexes. Now, I don't think that's—in a sense, it's just, look, a coordination game where we care, and care differently, about the different coordination solutions. But it's also in Nash's paper; he has an example like this, which has this structure.
- 2.33 Here's the idea. One of the partners prefers opera; the other prefers football. And so they're deciding what to do. A, let's say, prefers opera. That's first choice. We both go to the opera. Second choice is, we both go to the football match. Third choice is, we do our own thing. You go to the football match; I'll go to the opera. Worst choice is, you go to the opera, and I go to the football match.
- 2.34 And so, in this case, we're looking at this and saying, "It has just the same structure." Here are two people who care about doing something together. They'd rather do something together than go off and do things individually. On the other hand, they have preferences about what they do together. This one wants to go to the opera. This one wants to go to see the football game. Or it could be, we're going to the movies. This person wants to see a comedy. This person wants to see a drama. This person likes sci-fi. This person likes horror, etc. So these things come up all the time. We want to go to the same movie. On the other hand, we have to try to figure out what. We have preferences about that.
- 2.35 So this is a really common lifetime scenario, and again, we get two Nash equilibria. It doesn't actually matter that one of us prefers opera and the other prefers football. Still, the two equilibria solutions are, if you're going to be at the opera, my best choice is to be at the opera, too, even if that's not what I prefer, because I'd rather be with you. And similarly, you're going to the football match, then I want to be at the football

match, because being with you is more valuable than the particulars of what we're doing together.

- 2.36** And so, in those cases, there is an element of competition here, an element of taking turns maybe or coordination in some way, cooperation, compromise. But the most important thing is, we coordinate and end up at the same place. Even though we have different preferences about that, they're both Nash equilibria. I cannot unilaterally do better. Let's say I really prefer going to the opera, but you want to go to the football game, so I go to the football game. Can I unilaterally do better? Well, I could go to the opera instead, but that's actually a worse choice because then I'm going alone. We would have to together make a choice that I would prefer, but you wouldn't prefer it.
- 2.37** So there is an element of competition here. Is it your Nash equilibrium or mine that we're going to choose as a coordination? That's an element of competition, yes, but there's also an element of cooperation. We've got to actually yield to the preferences of the other person in some cases. Now, sometimes it might be systematic. I'm in England, I have to go along with driving on the left or I'm going to be in big trouble. If it's a matter of walking, it's not that I'm going to violate some official rule; I won't be ticketed. On the other hand, it'll be awkward because I'm going up the side of the staircase that other people are coming down on.
- 2.38** If I'm on a road in Scotland, which is a narrow one-lane road, I have to know who's supposed to yield to whom and what side I'm supposed to be on. If it looks like, oh, good grief, there's not room for both of us, and I instinctively go to the right, and that person driving on the left goes to the left, this is bad. This has happened to me. Luckily, the person was understanding. I was obviously an American, and it was just like, "Silly American." There was no actual collision. But in any case, we have to coordinate in such cases and agree about this.
- 2.39** Now, this brings me to another kind of game where paying attention to the needs of the other person and the preferences of the other person is really important. I've called it a mutual hostage game. This is one that's in Nash's paper, but is really not discussed much in the literature, although I think it's really quite important.
- 2.40** Now, the idea of mutual hostages, like us being hostage to each other, sounds bizarre. What do you mean? What kind of thing would this be? But it's a matter of me being unable to really affect the outcomes for me very much, but I can affect them for you. And similarly, you can affect the outcome for me, but not really for yourself. In other words, we each depend on the other. Now, that's an odd sort of thing. At first glance, you might think, "When does that happen?" But I'm actually convinced it happens a lot. Here is an example of this.
- 2.41** Let's say what I do makes a difference only for you. It doesn't make any difference for me. And similarly, what you do doesn't matter for you, but it matters for me. This sort of case—and here, let's just say, imagine a scenario where, let's say, one of us is

injured, and the other has the opportunity to help, but could help or could just pass on by, as in the Good Samaritan scenario. Well, then you might say, "Look, it's not a huge cost for me." In fact, let's, just to take the pure case, imagine that it's no cost to me to help you.

- 2.42 On the other hand, it matters to you. And similarly, then you could do something that matters for me, although it doesn't matter for you. Let's say you need help, and I'm thinking, "Well, I could help." That's a good outcome. On the other hand, if I also pass on by, it's the same for me. In other words, I'm in the weird situation where, if I'm thinking, "What's my best response to what you're doing?" it doesn't matter. Everything's a best response, because nothing I do affects me very much. It affects you.
- 2.43 And so, if I think of this in Nash's terms, think, "What's the best response from my point of view?" it doesn't make any difference. Here's what Nash says. All the pure strategies turn out to be best responses. All the mixed strategies turn out to be best responses from my point of view, because nothing I do matters to me. The same thing is true for you. If I start analyzing this in terms of Nash equilibria, I realize everything's as good as everything else.
- 2.44 But from one point of view, that's clearly not true. Let's say we both help, then we both, as I've drawn it here, get a unit of one. We get a value of one. If we both just pass on by, we get a value of zero. And so, it doesn't matter to me whether I pass on by, but on the other hand, what you do matters to me. What I do matters to you. All of a sudden, everything looks as good as everything else. And yet, from one point of view, it's clearly not as good as everything else. And so, this is a weird case where thinking about Nash equilibria doesn't really help us. Everything turns out to be a stable solution. But I can't unilaterally improve my situation because nothing I can do improves my situation, or makes it worse.
- 2.45 So, what are cases like this? I think the reason people haven't thought about it much is, they've thought, "Nothing's like this." But I think there are cases. So, for example, Alexander Solzhenitsyn in *The Gulag Archipelago* talks about this. He says, look, there was a feeling in the Soviet Union that nothing an individual could do would make a difference. But he said that sense, that nothing you could do could make a difference, even for you, it didn't matter, because you were as likely to be arrested if you were guilty as if you were innocent. I mean, it just didn't matter. And so, if nothing you do makes a difference, well, then you become apathetic. Why care what you're doing, because it doesn't matter to you?
- 2.46 And he says, the thing is, once you become apathetic like that, you become actually unable to act. So he writes, "Universal innocence also gave rise to a universal failure to act." Why? "For several decades, political arrests were distinguished in our country precisely by the fact that people were arrested who were guilty of nothing and were therefore unprepared to put up any resistance whatsoever. There was a general feeling of being destined for destruction, a sense of having nowhere to escape from the GPU-

NKVD, which, incidentally, given our internal passport system, was quite accurate. A submissive sheep is a find for a wolf."

- 2.47** If nothing you do is going to matter to the secret police and will not affect the outcome for you at all because it won't matter whether you're innocent or guilty—it's largely chance whether you get arrested or not—well, then, you become apathetic. There's no point in acting. There's no point in doing anything.
- 2.48** Here's another thing that I think is an example of this. It's an example of other things, too. But the tragedy of the commons is, in a sense, a mutual hostage game, because anything I do in a particular situation is going to have an imperceptible effect. It might have some effect, but it's very small. It's going to look as if, to me, nothing I do makes a difference. And yet, what we do collectively may make a large difference.
- 2.49** In the classic example, a series of sheep farmers may each have their own pasture, but there's a common pasture where they also may graze their sheep. Each one has an incentive to graze the sheep on the common pasture. If other people are doing it, well, you'd better do it, too, or your sheep are going to look pretty bad compared to theirs. If, on the other hand, they're not doing it, you can get your sheep in better shape than theirs by doing this. Everybody has an incentive to use the common pasture, so everybody does. As a result, the pasture is destroyed, and everybody is worse off.
- 2.50** That's an example of the tragedy of the commons, but there are many of these types of examples that we'll talk about later involving environmental pollution, for example, involving various other kinds of difficulties, where each one of us is such a small part of the whole that it feels like what I'm doing makes no difference. Does it matter whether I toss this in the landfill trash or recycle it? Maybe it doesn't, or at least it doesn't matter much. It looks to me as if the effects on me are zero.
- 2.51** But on the other hand, it may be that, if we all collectively do this, then we're better off than if we don't. So, my own actions seem insignificant. Consequently, I become apathetic. I don't really think, "Oh, look, it's not going to make any difference what I do." The result of that is that we together start acting in ways that are destructive collectively.
- 2.52** The same thing can happen. Lately, it seems as if something that a number of people have been writing about in The New York Times and so on is what they call "micro-shoplifting" or something like that, where basically, they figure, "It's a big company, and I go in and I steal something from the drugstore, what does it matter? It doesn't have any effect." Well, it does have an effect. It's just that your particular contribution to the welfare of that company is admittedly very [small]. It doesn't matter a lot to them whether you steal the tube of toothpaste.
- 2.53** On the other hand, clearly, if a lot of people are stealing lots of tubes of toothpaste, all of a sudden you don't have a store or a company anymore, and everybody's worse off. And so, these kinds of situations are not exactly mutual hostage games, because what

I'm doing does make a difference; it's just an imperceptible difference. We'll come back to those kinds of problems.

- 2.54** But in a sense, the problem of social capital, which we'll come back to a number of times in this course, is also like this. What do I mean by "social capital"? Well, it's a matter of our ability to solve coordination problems. Basically, do we have the right networks of communication, of trust, of coordination, of cooperation, to let us actually find these solutions together? In a situation which is a high-trust situation, we do, and so we're able to coordinate successfully. We can cooperate successfully. We can agree on a certain Nash equilibrium, or on taking turns on Nash equilibria or something that allows us to have a successful solution.
- 2.55** But if you don't have that level of trust, you can't do it. If I think, let's say I have a friend, and actually I've had friends like this, where you say, "Yes, I'll meet you at such and such a place, at such and such a time," and maybe they show up and maybe they don't. Okay, highly unreliable. I have found that, if you do this with a group of people and they all agree, "Yes, we'll be there," so you get there first and order the pitcher of beer, and then none of them show up, you end up just getting really drunk without anticipating this, because you can't let that beer go to waste, and nobody showed up.
- 2.56** In any case, the problem is, you've got to have these networks of trust so that you can say, "Look, we're going to do this, right?" And you trust the other people to do it. We'll come back to this problem again and again involving group projects, involving just simple things like, this time we'll go to the opera, next time we'll go to the football match, and so on. We have to have ways of solving these problems.
- 2.57** How, in society, on the large scale, do we solve these coordination problems? Let's say, Battle of the Sexes, it's just a question of taking turns about what we do. "All right, we went to see that kind of movie last week. Let's watch this kind of movie this week." But on a larger scale, it's not like that. It can't be like, "Well, honey, which side of the road should we drive on today?" That's not going to be something that's going to work; we need a society-wide solution.
- 2.58** How do people do it? One way is to enact rules or laws. You might have a law that says, look, if you're driving on the road, you have to drive on this particular side. You can't drive on the other side. That's a violation of the traffic law. Or you have to go, at most, this speed. You can't go over that. We're trying to coordinate our speeds in such a way as to maximize safety. It might be a question of more informal rules, however, like walking on this side of the sidewalk or that side of the sidewalk.
- 2.59** Some of these might be written into laws with real penalties; some of them might be social conventions of kinds where it's just like, you get at most a frown or something like that. Some of them might be policies within a corporation or some other kind of group. You might have policies about how you deal with situations. It's not a violation

of the law if you do something else. On the other hand, it's a violation of policy. We do things this way here, so that we know what each other is doing.

- 2.60** Or it might be a matter of social conventions, where, just informally, we all adopt a certain pattern of behaving, and so we all just generally do this. Well, we'll talk more about what those conventions involve, but they're really ways of solving these coordination problems. Maybe we do take turns, as often happens in a relationship, where people are saying, "Okay, you got to do that last time. This time, we'll take my preference," and so on.
- 2.61** It might be that we simply communicate. We just say, "Where are we meeting?" And we just need to coordinate, hence the Buffy the Vampire Slayer problem of, if we can't coordinate, we have problems. And we'll come back to that next time with the prisoner's dilemma, where a huge part of this is, we separate the prisoners and don't let them talk. Then there might be penalties for violation. That would go along with laws, but there might be more informal penalties of ways in which we just say, "Look, you're not doing things the right way. You're violating the convention, you're violating the policies. You're doing things that actually lead to negative outcomes for you."
- 2.62** I think one of the things that makes this interesting—and here I'm departing from what game theorists usually talk about, but thinking about this in a larger context—you can raise these questions at different levels. So if we're thinking about coordination problems, we could think about them from an individual point of view. Here I am, in Britain, let's say, and I'm thinking, "How do I coordinate with others?" Well, I have to drive on the left. I'm supposed to walk on the left.
- 2.63** But there are all sorts of other conventions that now I'm expected to follow—some of them linguistic conventions, what words I use for certain things, some conventions about other things. Things are just done differently there in various kinds of contexts, and you need to understand the local patterns. Or you go to Latin America, and it's not just that people are speaking Spanish or Portuguese. There are also different ways of greeting people when you first meet them, for example. There are different ways of behaving at dinners and various other things.
- 2.64** There are different times when you're expected to sleep or be awake, something which I found very, very difficult when I spent some time in Costa Rica. Costa Ricans are, from Americans' point of view, party animals. But it's not that they don't sleep; they just sleep at different times. When Americans go there, we aren't used to sleeping at those times, and so it feels like we're getting no sleep at all. But in any case, it's something that requires you to adapt to a different pattern.
- 2.65** You can think about this from a group point of view. It's not just, how do I coordinate with other people here, but how do we coordinate with one another? Let's say we are designing— we've started a company, and we're thinking, "What should the rules and policies of our company be?" We need to think, "We're going to face all sorts of

coordination problems as we go. How do we solve those coordination problems?" We need to coordinate the solutions to coordination problems, and think, how are we going to address these things?

- 2.66** Then there's the question of institutional design. How do we structure an institution to promote coordination, to make it easier for people to coordinate? And so, think about these with respect to transportation. As an individual, I can think, how should I drive in this context? Should I drive on the right? Should I drive on the left? What are the basic expectations of the other drivers? I have to think about the formal rules and about the informal patterns of how people drive here. And that can vary, even though the rules don't change at all.
- 2.67** The rules just between cities like Pittsburgh and New York, the laws are basically the same, but the patterns of driving are quite different. In New York, you have to merge. It's actually remarkably civilized. Everybody faces it all the time. It's just you go, you go, you. It's just absolutely—it goes very smoothly. In Pittsburgh, it's like, honk, honk, honk. It's very—this is competition; our masculinity is at stake, and so on. Whereas in New York, it's like, what a schlep, and we just do this automatically. And so, the laws are identical, but the social patterns are very different.
- 2.68** Then there's a question of how we design the traffic laws. You're in the legislature, you're thinking, "How should we set up the laws here?" You're trying to think about minimizing the amount of time it takes people to get places. You're thinking about safety and minimizing the number of accidents. The same thing is true with institutional design. How do we design roadways in such a way to facilitate safe transportation? How do we design an entire transportation system for this region, for example?
- 2.69** That's a way of thinking about coordination from a different point of view. So, I can think, "How do I coordinate with you? How do we coordinate with one another?" and then, "How do we design this system so that people do coordinate effectively?" All of those are coordination problems. They're all game theory problems, but they're looked at from different points of view. If I'm in the system already, there are already rules, and I know what all the other players are doing. If I'm in a group, or especially if I'm involved in institutional design, I'm thinking, "How do I design this game, so that people actually coordinate their activities and incentives to coordinate?" That's a matter of game design more than it is, how do I play the given game?
- 2.70** Now, I've already mentioned that many of these problems are solved by social conventions. What is a convention? I've got an AI-generated image of David Hume there for you, precisely because it was Hume who first developed this idea of a convention and the importance of conventions in solving these kinds of problems of how we get along together. What is a convention, from his point of view? It's a widely shared rule or regularity. It might be a rule that we actually write down. It might be an informal thing we all do, without even thinking about it.

- 2.71** It's chosen, but often unconsciously, implicitly. We just do it. We don't even think consciously about doing it, by some group of people. If you think, what do you mean, unconsciously? I mean, it's not unconscious, unconsciously, that I learn about traffic rules and learn to drive. Well, yeah, you go through driver education and learn the rules, but a lot of it comes from just observing drivers and picking up the conventions and the way people do things, like how far in advance should you turn on your turn signal and so on.
- 2.72** And so, you think about these things partly formally, but partly informally. A lot of linguistic rules, for example, we really, for our own native language at least, learn unconsciously and implicitly. I didn't, as a baby, sit down and start studying books of English grammar. I just picked it up by being in an English-speaking community. I learned them unconsciously. Some of them tend not even to be written down in the grammar books, even once linguists go at it.
- 2.73** For example, there are rules about the order of adjectives. I can say, "Oh, the little red one," but I can't say "the red little one." That sounds really odd. Why can't I say that? They're logically equivalent, but still, one is perfectly acceptable in English, one is weird-sounding. You know what it means, and yet you're thinking, "You don't know this language very well, do you?" And yet, you won't find in a grammar book a set of rules about this, typically. So those are things we just learn unconsciously.
- 2.74** Now, these are stable solutions. The point of us doing it as a social convention is that we no longer have to arrange this individually. We don't have to create a system of communication, one for you, one for you, one for you, and so on. Instead, we have a general solution to this sort of coordination problem. We don't need all of those individual negotiations. Sometimes it can be written into law or into a formal rule, but most times it's informal.
- 2.75** So, think about Oxford, my walking on the left—that was an informal rule. That was a pattern that was not a question of a formal rule. It was just figuring out how people do things. In other countries, it's a matter of eye contact, let's say. That didn't seem very different in England, but on the other hand, in Denmark, let's say, it seemed very different. I had the sense when I was in Denmark that either people didn't like me or I was really ugly by Danish standards. Maybe that's true, I don't know. But in any case, nobody makes eye contact there in the same way that Americans do.
- 2.76** It had not occurred to me until I traveled in Europe in various places that Americans are super easy to pick out. I think part of it is just our faces and the way our faces are formed by speaking English as opposed to some other language. But some of it is, we make eye contact in certain ways and we smile. People in many other cultures don't do that. There are no formal rules about this. It's just a matter of how people act.
- 2.77** In Denmark, I'm walking around, often in chinos and a white shirt, bright colors or very light colors, making eye contact with people and smiling at them. And Danes are

dressed in dark colors and are looking down and certainly not making eye contact. And if they do, they act like, "What, you just got off a boat from the Bahamas or something?" I mean, they look like, "What kind of weirdo are you?" And it's all a matter of informal conventions.

- 2.78** David Lewis, a philosopher who recently died, was the person who has, in recent thinking, done most to advance the theory of conventions. He says here's what's true of a convention. Everybody in the group has similar preferences, and that's going to be important. We've been talking about knowing what kind of game you're playing, and knowing what the preferences are of the other players. Presupposed by this idea of solving things through a convention is that we all have pretty similar preferences.
- 2.79** For example, in traffic, we want to be able to get where we're going safely. We want to get there pretty efficiently. We want to be able to spend as little time driving as we have to, to get where we're going. We want it to be a fairly pleasant experience. We don't want to be constantly shouting out the window and cursing at people and so on. And so, we have these various preferences, and we pretty much share those. We also conform to this regularity ourselves. We tend to fall into this pattern of behavior, and we expect other people to do that, too. If they don't, that sticks out, and they're violating our convention, violating our expectations, and we prefer that everybody else conform to it.
- 2.80** Now, it might be that multiple solutions to this problem exist, but nevertheless, we have a certain way of doing it, and we prefer that other people do it that way. So, Hume, for example, thinks this is the way to explain property and why people have a right to property. He says, "I observe that it will be for my interest to leave another in possession of his goods, provided he'll act in the same manner with regard to me. He's sensible of a like interest in the regulation of his conduct."
- 2.81** So, in other words, we have similar preferences. I prefer to keep my stuff. You prefer to keep your stuff. I expect that you'll leave my stuff alone, and I expect that I will leave your stuff alone. I prefer that you leave my stuff alone. I prefer that I leave your stuff alone. In short, we have similar preferences. We have similar expectations, and so a social convention exists between us, whether or not there's actually a rule against stealing.
- 2.82** So he says, "When this common sense of interest is mutually expressed, and is known to both, it produces a suitable resolution and behavior." So it is important that it be known to both. That is to say, we understand not only that we have these preferences and these expectations, we know that you do, too, and you know that we do, too. Otherwise, we might be going for different Nash equilibria. I know you'd like to get there safely. You know I'd like to get there safely. But you also know my solution to that is to drive on the right.

- 2.83** If you don't, if you're thinking, "Gee, I don't know what the traffic laws are here," maybe we're suddenly in New Zealand and we realize, "I don't know whether they drive on the right or the left in New Zealand," neither do you, so you're guessing, well, there are two good Nash equilibrium solutions. On the other hand, we don't know which one the other person is following, so we're in trouble. So we have to know enough about the preferences and expectations and patterns of behavior of the other person.
- 2.84** If that's all true, if "the actions of each have a reference to those of the other and are performed on that supposition, that something is to be performed on the other part." In other words, if I know you're expecting me to behave this way, and that you prefer it, and actually, I prefer we have a solution to this problem, too, then there's an obligation on my part to do the same. So, actually, this expectation turns into an expectation— sorry, turns into an obligation—for me to actually abide by that convention. There's a mutual preference, a mutual expectation, and that becomes an obligation.
- 2.85** It's important that we expect others in the group to conform to this, and that does give each member of the group a good reason to do it. It actually means, look, everybody solves it here by driving on the right or by walking on the left or by saying this when you meet a new person and so on. And that actually gives me a good reason to go along with it, too. It makes it an obligation on my part. So, the sense of interest has become common to all our fellows. It gives us confidence in the future regularity of their conduct.
- 2.86** That confidence is important. That's why being in a high-trust society matters a lot. If I can't trust whether you're going to do it or not, then this convention is something that's pretty weak. I realize, I don't have then so much reason, I don't have so much interest in following it myself. But if I can count on you to do it, then that gives me strong reason to do it as well. Now, these are highly context-sensitive. Something Hume makes clear, and that I think Lewis makes even more clear, is that we've got to solve certain kinds of coordination problems, but we don't all solve them the same way. For example, there's a certain set of conventions about how you act in a restaurant, and in fact, different kinds of restaurants. Most restaurants, you go up and you wait to be seated. But some might have a sign, "Just sit down wherever you like." Or some, you might have to go up to the counter to get any food. You can sit there all day, and nobody's going to come over and take your order. And so, you have to know what the rules are for this particular kind of establishment.
- 2.87** A different set of expectations might exist in a tavern, where you think, "Hey, how do I behave here?" Well, here I probably do have to go up to the bar. Here, there are different conventions about how loud I can be, for example, or whether I can walk around and so on. I go to church, and there's a certain expectation for how people behave in church. It's a question of—well, it's rather different from a bar. I don't just get up and start wandering around and drinking beer and so on in the middle of a

church service. It may depend on the church. From one church to another, there may be a different set of conventions.

2.88 Then, I'm in a museum. A certain set of expectations are on me, as part of what it is to be in a museum. The same thing is going to be true in, let's say, a classroom or a lecture hall. It may depend, again, what kind of classroom. Is it a big lecture hall? Is it a small seminar room? What are the expectations in this context? The same thing is true in language. What do you call a carbonated beverage? Is that a soda? Is it pop? Is it Coke? There are different conventions in different parts of the United States for this, and they're actually surprisingly variegated.

2.89 Or what is the plural second person pronoun? Is it "you," actually what the grammar books say? Is it "you guys"? Is it "y'all"? Is it, well, where I grew up, it was "yinz." It might be "youse" in certain parts of New York and New Jersey. Those are all ways of indicating, by the way, that it is not a good idea to have the same linguistic form for singular and plural in the second person. People feel a need to mark second person singular, you individually, you as a group. People find ways of doing that in some way or another, but it's different in different parts of the country.

2.90 What do you call a large truck? Is it a tractor-trailer, a semi? Is it an 18-wheeler? That too varies by part of the country. There are different linguistic conventions for this. Is it a drinking fountain, a water fountain, or a bubbler? I've got to admit, if people said, "Oh, there's a bubbler over there," I would have no idea what they're talking about, until I actually encountered this. But that's something where, sometimes we are familiar enough, with enough different conventions, that I know what people mean if they say a semi or a tractor-trailer.

2.91 But in other cases, I might not know about that convention. It might be highly localized. Indeed, the Pittsburgh accent has been calmed down over time, but among the people I was growing up around, a lot of what they said would be simply unintelligible to the average English speaker. It would be something that was filled with—for example, there are rules there that you can just leave out prepositions, you can leave out "th"s, you can leave out the verb "to be." It's very efficient speech. On the other hand, it's highly unintelligible to somebody who is not part of that.

2.92 In any case, there are linguistic maps that indicate those different regions and different sets of conventions, and it actually is quite complicated. It's becoming much more uniform because of the influence of radio and television, film, and so on. But on the other hand, it's still somewhat varied and used to be much, much more varied.

2.93 This brings me finally to that idea of social capital, something we're going to be talking about quite a bit more as we go along. How easy is it or how difficult is it for us to find solutions to these coordination problems? Think about the obstacles. Let's say, going back to our driving question, we're thinking about whether we do it this way or do it that way. In this case, we just have to make a decision. You might say, look, there are

two good equilibria here. We have to, as a group, select one. And so, we say, "We're going to do it this way." Sometimes that's what happens.

- 2.94** Within a company, it might be, look, we have to have a policy about what happens when this kind of situation arises—arises all the time. Maybe it's a regular feature like budgetary frameworks. We have to decide on the budget for the year, what's going to be our annual budget process. And so, there are going to be coordination problems to be solved, but we have to set it up now in a way that we're going to facilitate those solutions. Then we have to be able to trust that people will follow our policies.
- 2.95** Sometimes—I don't know if you've encountered this in organizations, but I have—these solutions end up starting out very reasonable, very sensible. Here's our method, here's our procedure. Good. But then somebody says, "Well, what about this problem?" Oh, let's tweak it. "What about that problem?" Let's tweak it. "What about that person who once did that?" Oh, yeah. Then it becomes so complicated that nobody can follow the procedure.
- 2.96** There is a danger that the better and more fine-tuned you make that convention, the harder it is for people to follow it, and the more people just ignore it. I think, earlier, I was talking about rationalizability of certain choices, saying that simple heuristics tend to be better in this respect than complicated holistic judgments, just because they lead to more consistent outcomes. The same thing can be true here. If you've got simple procedures, simple policies, simple conventions, people can follow them. If you have immensely complex ones, then it's very difficult for people to follow them.
- 2.97** There are certain games, actually, that are like this, that start out relatively simple. I'm thinking of a game that a friend was very fond of for a while, called Killer Bunnies. It starts out quite intelligible. You know what to do. You're trying to kill each other's rabbits for no apparent reason. But anyway, okay, the rules are pretty simple. But then they introduce this complication of the game, and this complication and that one, all to try to sell you more card packs of these rabbits. But in the end, the game becomes so complicated, nobody knows how to play it.
- 2.98** That becomes a problem in organizations all the time. We have to not only have enough social capital that we actually can trust each other to actually abide by the conventions, the rules, the policies, but we also have to have policies and rules and conventions that are simple enough that people can actually do it without expending huge amounts of energy.
- 2.99** There's a book called *Three Felonies a Day*, claiming the average business person commits three felonies a day, not because they're such evil people, but because the rules governing businesses are typically so complicated that no one can even know what they are, let alone actually abide by them all the time. When rules become that complex, when conventions become that difficult to follow, you have real problems.

- 2.100** For example, here would be a solution to the driving problem. Let's agree to drive on the right Mondays, Wednesdays, Fridays, but Tuesday, Thursday, Saturday, Sunday, we'll drive on the left, except in leap years when—okay, nightmare. It's not so much that we don't trust each other. It's actually like, what day is it today? What side of the road am I supposed to be on? Is it a leap year or not? Oh, gosh, it's January 1. Now we're in a leap year. Anyway, this is something that would be a recipe for accidents.
- 2.101** In general, then we have to think not only how do we establish these networks of trust, but also, how do we design solutions to these, leading us to the same kind of Nash equilibrium, which we want? We want it to be a stable solution. We don't want anybody to have a unilateral incentive to defect. But on the other hand, we've got to make it possible for people to do it and give them encouragement to do it, without making it immensely difficult. Sometimes, often, luckily, we succeed in doing that. But sometimes we don't, and that becomes a serious social problem.
- 2.102** I've been talking about this idea of social capital, which we'll return to, but I haven't actually defined it for you. I've said something about it. Coleman, in his famous 1988 paper, actually, [1988] paper, where he talks about this, says, "There's a variety of different entities, with two elements in common. They all consist of some aspect of social structures. They facilitate certain actions within the structure." And so, he began talking about this idea very early in his career, developing it more and more as he went. It became one of the most foundational concepts of sociology and the social sciences in general.
- 2.103** There are features of social organizations, Putnam has said, "such as networks, norms, social trust that facilitate coordination and cooperation for mutual benefit." So it's that existence of trust, of norms of trust, and of norms of communication. Fukuyama has put it this way—it's the "ability of people to work together for common purposes in groups and organizations." There has to be a set of informal values or norms. In other words, those shared preferences, they've got to be shared among the members of the group in such a way that they can coordinate, they can cooperate, and it encourages all of that.
- 2.104** Those conventions that Lewis talked about and that Hume had talked about, we have to have networks that make it possible for people to form those. They are highly specific to activities. What applies in an organization, a large company, may be different from a small company. That might be different from a university. That's different from a restaurant, different from a bar, different from a church, and so on.
- 2.105** It also may be highly relationship-specific. Being a friend of this person might make that person trust you more and this person trust you less. It's highly relational, so it depends on the kind of relationship we have. Is this a friendship? Is it a romantic relationship? Are we brothers and sisters, or parent and child, or just coworkers? There are going to be different conventions about all of those. But it's immensely valuable.

- 2.106** A high-trust group just accomplishes way more than a low-trust group, because it can find its way to Nash equilibria in common, so that we all end up in the same place. The large question is—it's very important—it's how we solve these coordination problems. It's pretty easy to identify in games, when we have this kind of game, what those possible solutions are. The hard part is leading us all to the right one where we're all at least in the same one.
- 2.107** How do we do that? How do we develop social capital, so that we build up those norms? That's a topic we're going to look at a little bit later. I end with a cliffhanger. There's something that all of this depends on, these abilities that we have to trust each other. How do we develop that? How do we strengthen it? Future lectures will answer that.

Lecture 3: Cooperation Games

At a glance — Cooperation games.

- 3.1** I want to talk now about cooperation games. Now, this is not just the idea of a cooperative game. In fact, it's quite different. A cooperative game is one where players can cooperate and form coalitions. But what I mean is a game where we're simply given incentives to cooperate with other players. This is typically, actually, officially a non-cooperative game, where we're not forming coalitions. On the other hand, we have to cooperate to get good outcomes.
- 3.2** And so, I'm interested in my welfare. You're interested in your well-being. But if we cooperate, we realize we can both do better than we could do on our own. This is a case where we're not, in effect, forming a coalition and having joint preferences—you have yours, I have mine—it's just that we can actually coordinate our efforts and cooperate in a way that allows us both to win. And so, this is a kind of win-win scenario-type game.
- 3.3** It's, from one point of view, a really good kind of game, therefore, because, hey, we can mutually benefit. We have a lot to gain if we actually cooperate in this way. On the other hand, if we don't cooperate, we find ourselves in a situation where, even though we're each pursuing our own self-interest, we get in each other's way, we end up harming each other, and so we end up worse off than we each could actually be in some other scenario.
- 3.4** So, as Aristotle has said, we have an impulse for this sort of community, a community where we all get together and we can thrive together. We can thrive as individuals, so it's not as if we put aside our self-interest and become altruistic or something; it's just that we can work together, so that we're all better off. We all have an impulse toward that sort of community, so how do we achieve it?
- 3.5** Well, I think we can start here. We're going to lead to this classic case of game theory. If people know anything about game theory, they typically know about the prisoner's dilemma. So I want to start not exactly there with the situation of these hypothetical prisoners, but instead with the situation that Thomas Hobbes writes about in *Leviathan*. Hobbes was born in 1588. His mother had him prematurely when she heard about the approach of the Spanish Armada. He quipped that, "Fear and I were born twins." It does seem to affect his entire philosophy.
- 3.6** He paints a picture of the state of nature. What would it be like if we had no government, no political authority over us at all? He says, well, it would be "a war of every man against every man." Everybody would actually be self-interested. They would be trying to grab whatever they needed and whatever they wanted. People would not get along with each other at all. There'd be constant fighting. It would become a war of all against all.

- 3.7** And so he says the result of that would be terrible. Life would be "solitary, poor, nasty, brutish, and short." We'd constantly be fighting, constantly striving, each for his own welfare, not caring about the welfare of others. We'd all be poor because we couldn't cooperate. Life would be nasty, brutish, and short. Well, this is the classic form of a prisoner's dilemma, as it's become known. But in Hobbes, it's simple in a way. Here are my choices, faced with another human being, and here I am as a caveman.
- 3.8** So I've got—cavemen didn't dress like this, but I imagine myself as a caveman, and I think, "Another man approaches. I can fight or I can seek peace. So what should I do?" Here's my first choice. Since I'm purely self-interested, I think, "I hope he is a wimp and he seeks peace, because then I will beat him and take his stuff." So my first choice is that I fight, and you don't fight back. I just get to dominate you and take your stuff.
- 3.9** My second choice is that we seek peace together. Then we can actually each do our own thing and neither one is harmed. My third choice is, we both fight. I mean, if you're going to fight, then I want to fight. My worst choice is that I'm the victim, that you fight and I'm not prepared to fight, and you just clobber me. I think, now, I don't know what you're going to do, but suppose you're seeking peace. Then I'd better fight. We could get along well, if I sought peace with you as well. But on the other hand, I can take advantage of you. I'm better off if I take advantage of you. So, if you're not going to fight me, I can take advantage.
- 3.10** So, let's say, put this in the context of countries and not in the case of cavemen. You're a neighboring country, and I think, "You're weak. You're not going to fight. You don't have much of an army. I can just take over your territory." Why not? I'm willing to fight. You aren't. I'll just take your stuff and I'll conquer you. So that's the idea. I'm better off actually fighting, if you're not going to fight back. Why wouldn't I get to take your stuff?
- 3.11** Now, suppose you're willing to fight. Well, if I don't fight you, you're going to beat me up and take my stuff and conquer me. So I'm better off fighting. So suddenly I realize, "Wait a minute. If you are seeking peace, I'm better off fighting and dominating you. If you're fighting, I'd better fight back. I'm better off fighting, no matter what you're going to do." And so, fighting is my dominant strategy. I am better off fighting you, no matter what you're going to do.
- 3.12** But you apply the same logic to me. You think, "Well, wait a minute, wait a minute, wait a minute." Not only are you at least as well off, you're in this case better off if you fight, but so is the other person. Because the other person is thinking, "If you're seeking peace, I can fight and dominate you. I can conquer you and take your stuff." But suppose you're fighting. "Well, I'd better fight back or else you're going to just conquer me and take my stuff." So the other person also thinks, "My best response, no matter what you do, is fighting."
- 3.13** So each thinks, "If you're going to seek peace with me, I can dominate you. I can fight. If you're going to fight, I'd better fight back." So fighting ends up being the norm. So in

short, there's one Nash equilibrium here. Both players have fighting as their dominant strategy, and here is the situation where we're each giving a best response to what the other is doing. We're both fighting. It's a war of all against all. "Life is solitary, poor, nasty, brutish, and short."

- 3.14** That is a classic prisoner's dilemma. We both realize, "I get to take advantage of you, if you're not going to oppose me. But if you are going to oppose me, I'd better fight back or else you'll take advantage of me. So I'd better fight. I'd better actually try to struggle against you." So we both do that; we get constant war of all against all, very destructive. In this case, we each get our third choice. I can think of this purely in terms of preferences. I don't even need numbers attached. I just realize, "Okay, we're both settling for our third choice, mutual combat."
- 3.15** But wait a minute. We'd both be much better off if we sought peace. Then we'd each be getting our second choice instead of our third choice. I'm doing what seems rational for me. If you're seeking peace, I'm better off fighting and taking advantage of you. If you come in fighting, I'd better fight back. I'm better off fighting either way. Same thing applies to you, so we fight. But we'd both be better off living in peace with each other.
- 3.16** So we've got an odd situation here, where what seems rational for each of us, pursuing our dominant strategy, leads us to an outcome that is clearly suboptimal. We could both do better by seeking peace with each other. And so, there is a Nash equilibrium; there is a highly stable situation here. On the other hand, it stinks. We're trapped in a suboptimal equilibrium, in an equilibrium where, wait a minute, there's a much better outcome sitting right over there. Now, why don't we do it?
- 3.17** Well, first of all, if we're in this Nash equilibrium, notice that is a situation where we cannot unilaterally improve our situation. So we're engaged in this mutual combat, and we both say, "Life is solitary, poor, nasty, brutish, and short. This stinks. Can't we do something better?" I think, "Well, I could. Maybe I should be the one who, instead of fighting, says, 'I will seek peace.'" But if I do that, then you conquer me and take my stuff. I'm worse off. I cannot make myself better off unilaterally.
- 3.18** You can't do that either, because if you stop fighting, then I'll beat you up and take your stuff. It looks as if, wait a minute, neither one of us can act individually, but if we both act together, if we say, "Look, you seek peace, and I'll seek peace," then we actually can achieve a better outcome. You might think, well, wait a minute, then why isn't that a Nash equilibrium? Because it's highly unstable. Suppose we're living in peace, and now I think, "Can I unilaterally make myself better off?" Yeah, you're peaceful, so I can attack you. I can bully you and take your stuff.
- 3.19** And so, it's highly unstable. That's the problem. We're in a situation now, we're seeking peace with each other. So it looks like life is much better, right? Except each person has an incentive now to actually break the peace and do better for themselves. Yet if we both do that, we're back in the suboptimal equilibrium. And so, it's very difficult to

maintain a situation that is not a Nash equilibrium. There is a better outcome here, and our goal is to find it, but it's really hard to do it.

- 3.20** In this case, Hobbes says, "Look, let's lay down some basic laws of nature." What he means by that is something that is a law that we adopt for ourselves because we're realizing, "Okay, I need rationally to respond to this kind of situation. What should I do?" He says, "Seek peace and follow it." That should be my first rule. The best outcome is if we mutually seek peace. On the other hand, he says, "By all means we can, defend ourselves."
- 3.21** In other words, I should ideally be in that better Nash equilibrium, a situation—sorry, it's not a Nash equilibrium. I should get out of that bad Nash equilibrium into an outcome where we are mutually cooperating, where we are seeking peace with each other. That should be what I really strive for. But I've got to be ready to defend myself, because if the other person is willing to fight, I've got to be willing to fight too. And so, I'm in this situation of tension. I've got an equilibrium situation where we're both fighting.
- 3.22** Well, my first choice would be to get out of that and find one where we're in an outcome where we're both seeking peace. But I realize that's highly unstable. I'd better be ready to fight if you do. And so, he lays down these laws. He says, "Every man ought to endeavor peace, as far as he has hope of obtaining it." On the other hand, since this is a highly unstable situation, we can't count on being able to do that.
- 3.23** And notice, I can't change the situation unilaterally by changing my preferences. Suppose I decide, "You know, this caveman life is really not a good thing. You know what? My first choice is no longer to bully you and beat up you and take your stuff. Instead, I want to live in peace. I am going to be a pacifist caveman. I seek peace above all." And so, I am no longer going to say, "If you are seeking peace, my first choice is to beat you up and take your stuff. I renounce that form of aggression." I am going to say, "My first choice is to live in peace with you."
- 3.24** That seems very noble. I say, "Yes, mutual cooperation will be my first choice." But that doesn't change the situation at all, because it turns out then, well, fine. You're going to say, "Okay, you're seeking peace. But I can do better if I fight and take your stuff." My unilaterally changing my preferences and deciding to be a peaceful, civilized caveman does nothing to change the circumstance. We're still back in mutual combat.
- 3.25** There is a better situation now, one where I get my first choice. Yay, we live in peace. You get your second choice. Better off there than the mutual combat, the war of all against all. But on the other hand, it's still unstable, and still we're likely to end up back in that situation of mutual combat. Life is going to be solitary, poor, nasty, brutish, and short. I can't change that all by myself. And so, Hobbes, recognizing that, says, "When he can't obtain peace, he may seek and use all helps, all advantages, of war."

- 3.26** Historically, we can think, yeah, this is part of the reason that attempts to eliminate war have not been very successful. There have been groups that were pacifist groups that said, "No more war." In fact, in the 1920s, the Kellogg-Briand Pact was actually something in international law that outlawed aggression, outlawed war. Yeah, gee, 1928, let's outlaw war. That didn't work out terribly well. But why not? Why don't people say, "Let's live in peace"?
- 3.27** Because setting yourself up by being a pacifist and saying, "I won't fight," enables bullies to take advantage of you. It gives other people an incentive to attack you. So it doesn't help to eliminate war. And so, suppose I do unilaterally decide to become a pacifist, not just a civilized caveman who says, "Mutual peace is my first choice," but I say, "You know what? I seek peace as my dominant strategy. That will be my first choice. That will be my second choice." First choice is that we live in peace together. Second choice is, if you attack me, I will not fight back. I would rather be conquered by you than stain myself morally by fighting.
- 3.28** Now there is a Nash equilibrium, and it's exactly that one. You're going to attack me, and I will remain morally pure, but I will lose the war. I will be conquered by you. And so, it doesn't actually lead to peace. It just leads to my submission to you. And so, becoming a pacifist unilaterally doesn't do anything for me. I get clobbered as a result, and so it would require a mutual change.
- 3.29** Well, when Neville Chamberlain came back from Munich in 1938, saying, "I have achieved peace [for] our time," making an agreement with Hitler, it was only a few months before Hitler violated the terms of the agreement. And Chamberlain himself said, "We have been utterly taken advantage of." And Hobbes was recognizing, look, that's the normal pattern of human affairs. There is a better situation. Living in peace is better. On the other hand, you can't achieve it all by yourself. It requires everybody to cooperate.
- 3.30** And so, that's the problem of this kind of situation. We can do well together, but we have to cooperate. Each one has to put our own welfare aside and say, "As a group, we do best this way." Now, I, as an individual, given that other people are doing that, could do better this way. I could take advantage of people, but I've got to forego taking advantage of them in order to achieve that better group outcome.
- 3.31** So, in this case, my individual welfare and the group's welfare really come into conflict. I've got to set my own rational goal, you might say, aside, and say, "I'm going to settle for my second choice and not my first." You have to do it, too. If one of us does it, it doesn't accomplish anything; we have to do it together. If we do, we can actually achieve a better situation.
- 3.32** Now, there are then possibilities of cooperation, but it's a little different from these coordination games I talked about last time, because there I was able to point to Nash's existence theorem and say there are Nash equilibria, which means that these

coordination games are solvable. In this case, I've got to say, there's a Nash equilibrium, but it stinks. There's a situation where, basically, our own self-interest draws us into this equilibrium where we can no longer unilaterally do any better, but it's not good for the group.

- 3.33** We'd all be better off, not only as a group, because of group harmony or something abstract, each one of us would be better off individually in a group that was like this. But given a group like that, each one of us has an incentive to defect and, as it were, cheat the rest of the group, take advantage of the rest of the group. That's why these are so tricky. This, too, was actually one of the examples in Nash's 1951 paper, even though there, he didn't draw attention to it; he just—it's example two. It has this structure, but he doesn't really give it a name. He doesn't notice, as it were, that this is the circumstance.
- 3.34** Let's talk about this in the form that has become more familiar. It's called the prisoner's dilemma, because here is the classic hypothetical case. You and I have been involved in a chain of robberies. We go around, let's say, Scottsdale, Arizona, stealing from art galleries, and there are lots to choose from. So we go to this one and we make sure we don't go back to the same one, so they don't recognize us and so on. But we very cleverly pretend to be very high-income people, seriously interested in buying art. And then while you distract them, I run off with some object of art or something like this.
- 3.35** We've been doing this for a while, pretty successfully. Then we get caught by the police. Now, they've caught us for this one, but they've had the sense, look, there is this gang ripping off these Scottsdale art galleries. They've known about this for a while. Finally, they think, "Aha, we've got them." But on the other hand, they have pretty good evidence against us, but only on this one robbery, and it was a small thing. It was like, "Wow, what I'd really like to steal here are those huge things of insects." But we can't very well just walk out with a giant insect, so I'm going to steal this little thing, and we get caught.
- 3.36** It's a pretty small offense. It's not worth a huge amount of money. The police know, "You guys have been stealing a lot of stuff. This is minor." So what do they do? They take us into separate rooms. And students who have run into trouble with the police before they became philosophy majors have told me, "Yes, this is exactly what happens."
- 3.37** The police put the people in separate rooms, and talk to each one, and say, "Look, we've got very solid evidence against you on this robbery. We can send you away to jail for a year, given what we have. On the other hand, we know you guys have been involved in a string of robberies here in Scottsdale, and this is pretty minor. So, look, here's the deal. If you testify against your buddy and say, yes, he's been involved in these robberies, then we're going to let you off for cooperating with the police. You'll get a suspended sentence. You won't have to go to jail for a day. On the other hand, we're going to throw the book at him. But you get off completely free."

- 3.38** So you're thinking, "Hm, okay, what should I do?" I can either confess and give witness against my buddy, or I can stay silent. Now, if I remain silent, what happens? Well, they've got us on this robbery, so we each go to jail for a year. That's not great. But if I testify against him, then I can walk out of here scot-free. So I'm thinking, "Hm, I don't know what he's going to do," because undoubtedly, they're down the hall, offering him exactly the same deal.
- 3.39** So, let's say he remains silent. He doesn't testify against me. Then if I don't testify against him, I go to jail for a year. If I do testify against him, I get off scot-free. I'm better off testifying against him. Now, suppose he actually testifies against me. Well, in that case, I'm in trouble if I don't also testify against him, because they made it clear they're going to go lighter on me if I actually give evidence against my buddy. And so, I realize if he's testifying against me, I'm going to go to jail for 10 years for this whole string of robberies, unless I get some credit for actually cooperating with the police, and maybe I go to jail for only five years.
- 3.40** So I start thinking, well, then if he's testifying against me, it's either 10 years in jail, if I remain silent, or five years in jail, if I testify against him. I'm better off testifying against him. So in short, if he stays silent, I'm better off testifying against him. I walk out of here scot-free. And if he's testifying against me, I'd better testify against him because then, instead of a 10-year sentence, I've got a five-year sentence. So I'm better off either way if I confess and implicate him.
- 3.41** Of course, he's offered the same deal and thinks about it the same way. "I don't know what that guy's going to do, but I'm better off testifying against him, no matter what he's going to do." So I testify against him, he testifies against me, we both go to jail for five years. And we think, "Wait a minute. If we had both remained silent, we'd have just gone to jail for one year." So this is a terrible Nash equilibrium. On the other hand, I couldn't have gotten out of it individually, because if I hadn't testified against him, and he was testifying against me, then I'd be going to jail for 10 years.
- 3.42** And so, that's the nature of the prisoner's dilemma. It's like, wait a minute, I've got an incentive to cooperate with the police and actually testify against my partner, whatever he's doing against me. And yet, if we both do it, we're both much worse off than if we had remained silent. So it's the same structure as Hobbes' problem. It's like, in this case, testifying against the other person is like fighting in Hobbes' scenario, and seeking peace is like remaining silent in this case.
- 3.43** I realize, "Oh my, I am in a situation which is now kind of a war of all against all—in this case, a war of testimonies and actually implicating somebody in a crime, a whole string of crimes. I'd be much better off cooperating with my buddy and not cooperating with the police." And so, again, I'm in a Nash equilibrium that's suboptimal. We would both be much better off in a different scenario. So, we're both in a situation where confessing and testifying against the other is our dominant strategy. It's our best

response, no matter what the other player is doing. And yet, the Nash equilibrium that results is bad, bad for both of us.

- 3.44** So what do we do? There is a general rule you might think about, given our conception of rationality, which is, wait a minute, do what makes you better off. And so, if I've got a dominant strategy, where I'm better off no matter what the other player is doing, I should play that dominant strategy if I have one. And in fact, I should eliminate the dominated strategies. For example, if remaining silent is worse for me no matter what, well, then forget about that. I shouldn't do that.
- 3.45** And yet, if each of us follows that general rule of rationality, which seems plausible, I am better off no matter what the other player is doing, so I'll do it. But if we each do that, we end up in a terrible scenario. And so, that's the trick behind the prisoner's dilemma. We each do something that's completely rational from our own individual point of view. But as a group, taken together, it seems highly irrational. That's a terrible outcome for the group. And so, we end up sitting in jail for 10 years for this whole string of robberies. Headlines blare, "These guys, finally, we've captured the Scottsdale thieves." We realize, "If only we had remained silent."
- 3.46** The difficulty here is that the police do this with both of us. Interestingly, if you watch TV dramas about crime, in the American dramas, they do it this way. Classic prisoner's dilemma, they separate the prisoners. In British ones, intriguingly, they don't. They often actually bring people together and try to get them to contradict each other, which strikes me as an interesting difference between British culture and American culture.
- 3.47** I have no idea whether the British police actually do that, but at least on Midsomer Murders, they do that. Maybe it's better drama. But in any case, "Wait, wait. No, no. But you said that. You're betraying me." Actually, then you get to see the prisoners' shock on their faces as it happens right in front of them. So it is better drama, but on the other hand, it's maybe less effective at inducing these testimonies against one another.
- 3.48** The difficulty is precisely that each one is offered the same deal. And of course, in this case, they can't communicate. Now, in the British dramas, they communicate, but what they communicate is this sense of betrayal from the other. If you can't communicate, though, it gets harder. Now, if you could communicate, you might think, well, then isn't the problem solved? I mean, I can be looking at you, and we're both saying, "Shh." And so, communication would seem to solve the problem.
- 3.49** Except we might both be there together, saying, "We're not going to say anything. Okay, don't testify, don't testify, don't testify." Then you leave the room, and I say, "Wait a minute," because I'm still better off if I betray you. And so, while I might not be willing to betray you to your face, I can still betray you in other ways. So, again, we've

got a situation where we're both much better off this way. Why don't we just do it? Well, because each one has an incentive to break that and betray the other.

- 3.50** So, we are again in a suboptimal equilibrium. We can get out of it by agreeing and by communicating or saying in advance, "Hey, we're part of this gang. Don't betray me. When I get out of jail, I'll find you and I'll kill you or something." That's a way of changing payoffs. But on the other hand, we still have an incentive. Unless you are sure that I'm going to be able to pull off that threat, I have an incentive to actually betray you, and you'll have an incentive to betray me.
- 3.51** Now, intriguingly, people have actually presented these scenarios to students in social science courses and also to prisoners, to find out what they do. What's interesting is that, among prisoners, 56% opted to cooperate. In other words, the majority of prisoners faced with this hypothetical scenario will say, "I'm not going to testify. I'm not going to testify against my buddy." But only 37% of students said, "I won't testify against my buddy." In short, only 13% of the students paired up randomly actually got the best outcome, where they remained silent. Prisoners got that about 30% of the time.
- 3.52** Interestingly, prisoners have higher morals than students at colleges and universities. That's kind of a silly way of putting it. But in a way, it's like, prisoners have thought through these scenarios and realized, "Wait a minute, I know this game. I've played this game in real life and I know what I'm supposed to be doing." Whereas students hypothetically think, "But I'm still better off." And so, students, a vast majority of students, are going to play their dominant strategy. But prisoners have realized, "Yeah, that's what got me in here in the first place. I'm not going to do that this time."
- 3.53** Now, there's a very important concept I think we need to really understand what's going on here. The person who came up with this concept is Vilfredo Pareto, an Italian economist of around the turn of the century, working around 1900. Jeremy Bentham, in his theory of utility, assumed interpersonal comparability. He assumed that we can evaluate my utility compared to your utility. Sometimes there is a common measure. Money is a good example. We can measure all sorts of things with money and say, "How much of a share of the company does each partner have?" and so on. Or, "Hey, what salaries are we going to pay our various employees?"
- 3.54** Sometimes it's easy, but a lot of times it's really not. Think about us in the ice cream shop, and we're thinking about our various preferences. I prefer this one; you prefer that one. Well, how much is that worth to you, and how much is this worth to me? And so on. It may be hard to make those comparisons. How much enjoyment do you really get out of that mint chocolate chip? Give me a number, 0 to 10. Or actually, in a hospital, nurses will ask patients this. "Rate your pain on a scale of 0 to 10—0, you're feeling fine, 10, excruciating." This person says, "I'm about a seven." This one says, "I'm at maybe a three."

- 3.55** Can we compare those? Do they have the same scale in mind? It's, after all, all subjective. You can say things to people like, "Oh, three is like hm, seven is hmm," or something. But still, what that feels like to a person, that may be different. So this is very approximate. So you might think, yeah, I don't trust interpersonal comparability.
- 3.56** Indeed, most economists over the last century really haven't trusted it. But Pareto developed this other idea. He said, "This can substitute. It can still let us make comparisons." Now, it's going to be weaker than utility, but nevertheless, it will substitute and allow us to come to a lot of conclusions. So here's the idea.
- 3.57** A state is called Pareto optimal, Pareto stable, Pareto efficient— all of those are equivalent terms—if no change can make anybody better off without making somebody else worse off. Is there a way of improving the situation for some of us that doesn't hurt anybody? That's the idea. If you can, this is better in Pareto's terms. It is Pareto superior. But usually, there are no solutions, only trade-offs, and so improving these people makes some of these worse off. That tells you you're already at a situation that's Pareto efficient, Pareto stable, Pareto optimal.
- 3.58** So the idea is, can we improve this for some without hurting anybody? Sometimes we can make it better for everybody, and nobody suffers at all. So that's clearly Pareto better, better in Pareto terms. It's better for some people, at least, maybe for everybody, worse for nobody. There are typically going to be a lot of these situations that are Pareto optimal or Pareto efficient. We call the set of those the Pareto frontier. The idea is, this is a situation where there are no longer any easy solutions.
- 3.59** Sowell is absolutely right on the Pareto frontier. There are no solutions anymore. There are only trade-offs. Any improvement you make for some makes some worse off. Instead of your second preference, you get your first preference. But now, instead of my second preference, I get my fourth. That indicates, look, if any change is going to involve that some people suffer, then we've got an indication that we're in a situation that is on that Pareto frontier. We're Pareto optimal.
- 3.60** So, if we take some simple examples, we start realizing, I get the feeling of this. For example, here are a number of possible distributions of welfare in this small three-person group. Let's say each person has one unit of welfare. There's another scenario where each person has zero, one where somebody has two, the other two have zero. Then one where person one has one, the other two have two. Then two people have five, one has zero. Which of those are Pareto optimal or Pareto efficient?
- 3.61** Well, you know zero, zero, zero isn't. Every other one improves things for some people without hurting anybody, so clearly rule that out. That's bad in Pareto terms. What about one, one, one? That improves the situation for everybody there. So it's clearly Pareto better. Is there any way of getting better than it? Well, yeah, one of those had one person getting one and the other two getting two. So two people benefited, and nobody got hurt. So that's Pareto better than that one.

- 3.62** What about a situation where five, five, and zero? Well, that's clearly better than the two, zero, zero. Somebody's improved from two to five. One went from zero to five. Nobody got hurt. So that's also a Pareto superior one. So if we think this all through, we realize there are two scenarios there that are actually Pareto efficient. One, two, two, and then five, five, zero. Five, five, zero is better than some of the others, like one where we all get zeros, because two people are better off, nobody suffered. One, two, two, everybody's better off, nobody suffered.
- 3.63** But on the other hand, those two we can't compare, because now, yes, we can improve this guy from one to five, this one from two to five, but this one falls from two to zero. So those two end up both being on the Pareto frontier. Neither one is Pareto better than the other. All right. So the idea is, this concept of Pareto superiority or Pareto efficiency, Pareto stability, Pareto optimality, it's very weak. It's the sort of thing that says, look, it's pretty rare, actually, that you can improve things for some people without anybody suffering.
- 3.64** I mean, maybe sometimes Sowell is wrong, maybe there are some solutions, but actually, society is pretty efficient at finding Pareto optimal solutions, and so they're pretty rare. Really, most of the time, we are in a situation where there are no solutions like that; there are only trade-offs. In that situation, it tells us that this is not exactly saying this is best. It's just saying, yeah, there is no simple way of improving this without some cost. Any improvements now have a cost.
- 3.65** So imagine that two people are at some little pizza shop. They order some pizza and some wings, and they're thinking, "Let's think about our distributions here." It may be that there are some things that are Pareto efficient, that once we make certain improvements, we can't get to any more. For example, let's say I eat most of the pizza and most of the wings. Well, one improvement before we started was that we share them evenly. But too late now, I ate most of them. And so, there were Pareto optimal solutions there where you ate your share, and sorry, the pizza is gone. That's no longer available.
- 3.66** In short, if I make some Pareto improvements, they're going to make others impossible later. The order in which we do things matters. Like we start out, neither one of us has had any pizza or any wings. There are all sorts of Pareto optimal solutions here. At some point, we've eaten everything. Well, now there's no further improvement to be made. Anything that I would have eaten is now something you wouldn't have eaten, so we're in a situation where any improvement from my point of view would have been a cost to you.
- 3.67** In short, the order of things makes a difference. You might think, "Oh, look, if there's a Pareto better thing, hey, we can improve some people without making anybody worse off. Let's do it," be careful because there might be multiple ways of doing that. If you do some first, the others become unavailable. Anyway, that's just a general caution.

- 3.68** So if we're thinking about these situations of prisoner's dilemmas, we realize that's a way of putting it. We want to be in a situation which is Pareto efficient or Pareto optimal, where basically, there's no way to make both of us better off. And yet, we end up in the Nash equilibrium here where we're both going to jail for five years, and we realize, "Oh, we could both have been much better off going to jail for just one year if only we had shut our mouths."
- 3.69** The same thing happens with Hobbes' state of nature. We realize we're in a war of all against all. We'd have been better off living in peace. And yet, we're in this situation now, where we're clearly in a circumstance where we're suffering, we're fighting, life is solitary, poor, nasty, brutish, and short. We could have been living in peace, but instead we did this. It was rational for each one of us to do it, but together we're now in a situation where we could both be better off.
- 3.70** There are lots of applications of this that have been explored. So one of the main ones that motivated the development of this theory in the first place was the arms race. We're now at the beginnings of the Cold War, and we realize, "We've got a superiority in weapons. We have the atomic bomb, the Soviets don't." But then it turns out, the Soviets acquire the atomic bomb. Now we realize we're in competition. We've got to build more nuclear weapons, more missiles, more bombs, and they're doing the same.
- 3.71** We start thinking, same situation. We could either limit our production of weapons here or we can build more. They can limit them or they can build more. Suppose they limit their production. If we build more, we have nuclear superiority. We're better off. Suppose they're building; we'd better build to catch up. We don't want them to have nuclear superiority. So we both think, "What is our dominant strategy? Build more missiles."
- 3.72** And so, the result is an arms race where we're both building as many missiles and creating as much destructive power as we can. We'd both be better off not spending the money doing that, not building all these nuclear missiles. But the fact is, each one of us has an incentive to do this. Now, the arms control treaties of the 1970s were all designed to actually get us from the Nash equilibrium here, which is clearly suboptimal, to the Pareto superior situation, where we're spending only a limited amount on defense and not doing that.
- 3.73** On the other hand, how well did people abide by those nuclear arms agreements? Not very well. There was massive cheating. Why? Because both sides have an incentive to cheat. The Soviets weren't about to abide by this. They think, "You're limiting your arms? Well, I don't know if you're actually abiding the terms of the treaty or not. But if you are, we gain superiority over you by building more and cheating. If you aren't and are actually building yours, we'd better build, too."
- 3.74** And so, even though there was a treaty that said we're going to limit these, it wasn't very effective. The same thing has happened with peace agreements all throughout

history. "Oh, yes, we'll settle. We won't do this, you won't do that." There's an incentive on both sides to cheat, which is why these things are so difficult.

- 3.75** Here's another simple homemade example, the roommate's dilemma. Two roommates are there. They could either work to clean the apartment they share or they could relax. Well, each one thinks, "My first choice is that you clean up while I relax. Second is that we both clean. Third is that we both relax and the place remains a mess. My fourth is that I have to do all the cleanup and you just get to goof off. And so, gee, if you're going to clean it, I'm better off relaxing and not helping you. And if you're not going to clean it, I don't want to be the only one stuck cleaning it up. So, in short, my dominant strategy is to not clean up."
- 3.76** And the roommate thinks about it the same way, so the kitchen ends up looking like a complete mess. Nobody does the dishes. And as somebody who has occasionally been in student apartments, I can tell you, yeah, this is very common. This happens all the time, because each one says, "Yeah, I hate living with all this mess. But on the other hand, I'm not going to be stuck being the only one to clean it up. If my roommate were cleaning it, then I wouldn't have a problem. That would be my first choice."
- 3.77** In political campaigns, it can be like this. I can take the high road and stress a positive message, or I can go on the attack and stress negative campaign messages against my opponent. I think, suppose my opponent is taking the high road, talking about general issues and ideals, then I can gain an advantage by attack ads, hitting him hard. Suppose he's already launching attack ads against me. Well, if I don't fight back, I'm going to lose, so I'd better fight back. So the dominant strategy for each candidate is to go negative and attack the other candidate. So they both do that, leading to nastier and nastier political campaigns. We've all lived through plenty of that.
- 3.78** Here is an example that Plato thinks is a deep problem with democracy. He says, look, at a certain stage, people in a democracy are free to live the way they want, and so they are very happy. He says, in a way, it's the most beautiful of all circumstances. But on the other hand, it tends to devolve and, eventually, collapses into tyranny. Why? Part of the reason is this: people want more and more. They want more and more from their government.
- 3.79** And so, they think, faced with a situation where the government—in this case, it's the people. I'm imagining here a direct democracy, where people just directly vote on things. They think, "We could forego this and balance our budget, or we could spend more and do this." They think, "Well, I want that." So suppose other people are in favor of balancing things; I can gain an advantage by adding to the budget, by saying, "We'll spend on this and this," my favorite things.
- 3.80** Let's say I'm a big baseball fan, so I say the government should subsidize baseball stadiums, and you're a fan of the arts and you say, the government should subsidize museums and concerts and art exhibits and blah, blah, blah. Then somebody else says,

"I am concerned about the welfare of the poor. We have to create welfare programs to take care of the poor. We should spend money on that."

3.81 What happens is, we all think, if other people are spending on their pet projects, then I want my project funded. If they're not spending, hey, there's money for mine. In short, we all say, "Yes, spend money on what I want." The result is that lots of money is spent. The result is government debt soars, and in the end, the country goes bankrupt. And so, he says, look, basically, a democracy lets people just exhaust the treasury, and eventually it collapses, partly for that reason.

3.82 Now, there's an analogous problem, he thinks, with power, because actually, if we're spending money to do all these things, we've got to have more and more government workers who are going to do all these things for us, more and more administrators, more and more managers, more and more bureaucrats. He calls them all drones, imagining this is a big beehive. He says, we've got all these drones. They become this additional powerful class in a democracy. In the end, they're the ones making all the decisions.

3.83 The decisions they make have less and less to do with what the people actually want. So there's a huge conflict between the bureaucrats and the public. So populist leaders arise, the bureaucrats do their best to destroy them, and in the end, you get tyranny out of this, either a tyranny of the drones or a tyranny from the populist leader who has to adopt extra-legal methods against the drones in order to stay alive. In any case, it's a very bleak picture.

3.84 All forms of government end up collapsing into tyranny from Plato's point of view, unless philosophers become kings. Having sat in many meetings with philosophers, let me say, you don't want philosophers to be kings. Maybe kings should become philosophers, but it's definitely not true that philosophers should become kings. Anyway, there's a real dilemma here, and it's basically just the prisoner's dilemma, adapted to that kind of setting.

3.85 There are others within a republic such as ours. James Madison was very worried about factions and having a government that can actually withstand factions. It's the same thing. Various factions within the community are going to seek to dominate others. He says, look, the factions might cooperate or they might seek to dominate. In each case, it is the same scenario. If others are seeking to cooperate, my faction gains an advantage by trying to take advantage of their cooperation and dominating them. If, on the other hand, they're trying to dominate us, we'd better fight back. It's a Hobbesian scenario among the factions.

3.86 Madison's point is, this is a very serious problem within any society, really, but also within a republic. You have to design mechanisms to try to restrain the factions and keep them all in check. Hence, the whole system of separation of powers, checks and balances, and so on, all a way of trying to say, "We need to make it hard for any faction

to dominate others. We have to make the payoffs for cooperating higher than the payoffs for defecting."

- 3.87** That is one of the classic ways of trying to solve a prisoner's dilemma. If you're on the outside, let's say you're a member of the criminal gang, realizing, "Our guys are getting into trouble with the police a lot. We'd better have a method for dealing with," you've got to change the payoffs then, by saying, "Hey, look, if you actually remain silent, we're going to give you a reward. Or if you rat us out, you're going to suffer a severe penalty from the gang, maybe not from the police, but we'll get you." You have to change the payoffs.
- 3.88** That's what, in a sense, Madison was trying to do with the constitutional framework, saying, "We've got to change the payoffs, so that factions have an incentive to cooperate, so we get to the Pareto optimal solution, and not to the war of all against all at the level of factions."
- 3.89** In trade, it's much the same situation, because we might have free trade or we might enact protectionist legislation. Well, in a sense, we're all better off if we're all trading freely than if we're all protecting our own industries. On the other hand, if others have free trade, we can gain an advantage by having protection. If they've got protectionist measures in place, then we're going to be taken advantage of unless we do it, too. The result of that is, everybody's dominant strategy is to protect their own industries.
- 3.90** So there's a natural pressure for protection, for tariffs, and so on. We'd be better off in a situation where trade was freer, but on the other hand, we're also constantly given incentives to do the opposite. And so, it becomes a very difficult problem. Now, I should say, this is generally true, and most economists think that's a pretty simple picture. There are probably all sorts of complications.
- 3.91** For example, free trade in some things makes you highly dependent on your trading partner, so it's going to depend critically on trust. You don't want to—for example, suppose Britain had said, "You know, we can't produce weapons in England as well as Germans can, so let's just offshore our weapons production to Germany." That would have been a really terrible strategy in the 1930s. You don't always want to do this. There are times when the payoffs end up being very, very different.
- 3.92** I've mentioned environmental concerns. That's another classic case where people talk about the prisoner's dilemma. You've got a situation where you can clean up your own act, let's say, as a factory, or you can pollute. If others are cleaning up, you gain an advantage by polluting, because you don't have to bear the costs of the environmental controls. On the other hand, if everybody else is polluting, the air and water are becoming nasty, so you're having trouble attracting people to work at the plant, because now the conditions are pretty bad.
- 3.93** On the other hand, they're gaining an advantage by polluting. You've got a strong financial incentive to pollute, and so your dominant strategy is to pollute and to try to

evade EPA regulations, for example. And so, what do you do? Well, everybody pollutes. You'd be better off if we had environmental controls, but again, that's a Pareto optimal solution; that's not a Nash equilibrium. Even if we have regulations, even if we have expectations, people have an incentive to cheat.

3.94 And that brings us to this general problem of the tragedy of the commons. I've already mentioned that as an example of a mutual hostage situation. It's also an example of a prisoner's dilemma, because each of us can pursue our self-interest. I'm better off, no matter what the other farmers are doing, if I graze my sheep on the common pasture. But the result of that is a destroyed pasture. That's a suboptimal outcome. It would be better for us if we didn't do that.

3.95 But here's why it's partly a generalized prisoner's dilemma. If each of us pursues our dominant strategy, we make ourselves collectively worse off. But it's also, in a sense, a mutual hostage game, because I can look and say, "What difference are my sheep making?" It's imperceptible. I mean, it doesn't matter that I take them out Saturday mornings and they graze on this grass. I look at the pasture, it's still fine. It's no worse off. Now, it is a teeny bit worse off. But on the other hand, it seems imperceptible. It seems like, look, what I'm doing doesn't affect me. Actually, I don't think it even affects you guys, really.

3.96 But the truth is, it does. So, the sheep farmers have some private land. They share this pasture. Well, each has an interest in grazing a sheep on the common pasture, and so they do that as much as possible. If the others aren't, their sheep gain an advantage; if the others are, they'd better not fall behind. So either way, they're better off grazing their sheep there a lot on the common pasture. But that does deteriorate the pasture, even though each individual act seems minor. Each grazes their sheep on the public land, and with that much grazing, it suffers from erosion. It produces less grass for the sheep to eat. It becomes less useful, maybe eventually completely useless.

3.97 Each one was making a decision in his own self-interest. It didn't seem to produce any larger-scale harm, but really, it did. So this has exactly the form of the prisoner's dilemma. I won't go through the details. It's kind of obvious. Each one benefits by grazing more, but they'd be better off collectively if they grazed less.

3.98 Now, this brings us to an interesting sort of case, because you might say, if it's true that we have a tendency to destroy the commons in this way, because each one has an incentive to basically use as much of the commons as possible, then how do we solve it?

3.99 One way is to try to regulate. One way is to try to take turns. Say, you get to graze on Saturdays, you get to graze on Fridays, and so on. One would be to simply sell off the common pasture. Let's have property rights. Or maybe we actually collectively do this. We collectively buy that common pasture, and then we charge people for grazing their

sheep on it. In any case, we find ways. Some of those ways are governmental ways, some are market-driven ways, but we do it in such a way as to change the payoffs.

3.100 I now might be violating the law if I graze my sheep there. Well, that changes my payoffs. Now it becomes risky to do that. Or I realize, now I have to pay a fee to do that. Okay, that changes it. Or I have to buy the land; that again changes payoffs. All of them are ways of getting us out of this tragedy and out of a prisoner's dilemma. But it does it, basically, by changing the payoffs. It says, look, that thing that was a little too easy for you to do? We're going to make that more costly for you. And as a result of that, we're going to try to make that your first choice and not that other Nash equilibrium. So we actually make the good outcome the Nash equilibrium. We do it by changing people's payoffs.

3.101 Now, you might think one of these solutions is better than another. You might be a pro-government type or a pro-market type. Elinor Ostrom has argued, look, there is no universal solution to this. What solves the problem here may not be what solves it there. What works for sheep farmers in Peru may or may not work for, let's say, industrial pollution in the upper Midwest. That may not work for water use in the desert Southwest. That may not work for some other kind of problem like this.

3.102 In each case, people will work out the best form. Sometimes it's government action, sometimes it's privatization, sometimes it's sharing and cooperatives. But in any case, there's some need to cooperate in a way that changes people's incentives, so they're no longer incentivized to cheat.

3.103 There's another way of looking at this. It looks maybe like a mutual hostage situation, but it's really not. It's like, "Hey, my sheep aren't doing any harm," or I walk across the grass rather than on the sidewalk to take a shortcut. I say, "Look, my footsteps didn't make a difference."

3.104 Well, they made a little difference. Okay, it's imperceptible, maybe, but nevertheless, these very small effects add up. Lots of people do it, lots of the time, that adds up. So what difference does it make if I didn't recycle, or if I walked across the path, or if my factory polluted a little bit? Well, it made a little difference, and many little differences make a big difference.

3.105 So Derek Parfit, for example, has a chapter of Reasons and Persons he entitles "[Five] Mistakes in Moral Mathematics." And this is one of them, thinking, "Oh, it's an imperceptible thing, so I can count it as zero." No, you can't. A very large number multiplied by a very small fraction is not zero, and it's important to realize that the fact that, oh, what difference does this make? It makes a little difference, and many little differences add up to a big one. So, there are various ways in which the farmers can change this. They do come down to changing the incentives.

3.106 Here's a silly example. This applied to my own children when they were young. There was a Dr. Seuss book that said, "Day, play. We play all day. Night, fight. We fight all

night." Or there could be fight club where you say, "Here's the full problem. You guys didn't enjoy the war of all against all. You should have enjoyed it. Hey, it's great to fight." And really, for some brothers and sisters, I'm convinced, yeah, they take joy in fighting. And so, that's one way of saying, "Hey, we'll reward you for actually being in this quadrant where you are cooperating, rather than in this one, the Nash equilibrium initially, where you're defecting."

- 3.107** In other words, here's a way of affecting the payoffs. Give rewards for the peace solution, for the cooperative solution. Here's another one, though, to penalize people for being in that Nash equilibrium, saying, instead of that being my third choice, that's going to become my second choice, because the payoffs have changed. Either the benefits of cooperation have increased or the penalties for non-cooperation have increased. In both of those ways, we can actually try to change things and turn a prisoner's dilemma into a prisoner's delight, basically, by saying the Nash equilibrium is actually the one that's best for us all.
- 3.108** That's the goal, really, of government regulation, or of privatization, or of various sharing schemes, or all that kind of thing. It's really a way of trying to say, "Let's reward the cooperators and punish the defectors." We'll talk more about that later in situations where we have repeated prisoner's dilemmas, because then it may be vital that we find ways of getting people out of those cycles, of nasty seeking of the Nash equilibrium, where everybody's defecting and harming one another. We say, "Let's find ways of trying to get people to cooperate."
- 3.109** Now, there's another kind of scenario that's much more optimistic than the prisoner's dilemma. It's called a stag hunt. This actually originates from Jean-Jacques Rousseau, who thought this was actually the main way in which people develop social conventions and social solutions to social problems. It's also something that Brian Skyrms and others have tried to advocate as now a good model of how we solve collective action problems.
- 3.110** Here's another sort of situation. Here, there are two Nash equilibria, but one's much better than the other. It, too, appears in Nash's original 1951 list of examples, but again, unlabeled in any way that let people see the significance. Here is Rousseau. "If it was a matter of hunting a deer, everyone well realized he must remain faithful to his post; but if a hare happened to pass within reach of one of them, we cannot doubt he would have gone off in pursuit of it without scruple."
- 3.111** So the idea is, we're a hunter-gatherer society, and we're trying to decide whether to work together to try to hunt a stag, or whether we should at least go our own ways and hunt rabbits. So we decide, look, there's a lot more meat in a stag than in a rabbit, but it takes a lot of us to surround it. I mean, they're fast. No one hunter with a bow and arrow is going to be able to do all that well, so really, we need to work together for that. We're better off collectively if we work together to get the stag.

- 3.112** But look, if some people, as Rousseau says, give up on the hunt, if they're at that post so the stag can't go that way without getting into trouble, and they go off hunting rabbits, then we're never going to get this stag. If some people give up the hunt, we're all going to end up getting nothing, except maybe they'll get a few rabbits. So we've got to work together. So here's the idea. I think, "What's my first choice?" That we work together to get the stag. There's a lot of meat in the stag. We could feed the whole village.
- 3.113** On the other hand, suppose you're going to go off hunting rabbits, then I'm going to be left going hungry, because I'm going to be sitting there trying to get this stag by myself. You're not helping, I've got no chance. So that's my worst-case scenario. I commit to hunting the stag, you go off and leave me with nothing.
- 3.114** Now, suppose you are hunting the stag with me. My first choice is definitely to hunt it with you. That's better than going off hunting rabbits myself because there's a lot more meat in a stag, and my share of the stag is still going to be way better than just getting a rabbit.
- 3.115** But now, do I trust you? Because, after all, suppose you're going to go off rabbit-hunting. I'm better off rabbit-hunting also, so at least I get a rabbit. Because if I'm out there hunting the stag, and you've gone off chasing a bunny rabbit, then I'm going to get zero. So my first choice is, we hunt the stag together. Second choice, I hunt a rabbit, you're off hunting the stag, but at least I've got a good chance of a rabbit. Third choice is, we both hunt for the rabbit, but now we're competing for the rabbits. And then fourth choice, you're rabbit-hunting, and I'm left with nothing.
- 3.116** Well, in this case, there are two Nash equilibria. If you're hunting the stag, my best response is to hunt the stag, too. But if you're hunting rabbits, my best response is to hunt rabbits. So two Nash equilibria, we're both doing our best response to what the other does. On the other hand, one is way better than the other. We get much more out of hunting the stag than we do out of hunting rabbits individually. We'd like to get ourselves to the better Nash equilibrium, the one that's Pareto superior to the other. We're both better off hunting the stag.
- 3.117** So why wouldn't we do it? Now, in a healthy group, we would. We'd say, "Let's hunt the stag together," and we'd trust each other to do it. But that's the real obstacle here. Can I trust you to withstand that temptation that Rousseau talks about, where you're going to see a bunny rabbit and you're going to say, "Oh, well, there at least is a certain dinner," and you go off chasing the bunny? Then I go home hungry. So I've got to trust you. I've got to trust that you will remain at your post and you won't run off against that rabbit. If I can't trust you, I'm better off chasing rabbits instead.
- 3.118** So, this is the dilemma. How can I trust you enough to trust you to cooperate with me in getting the better Nash equilibrium? If we're there, I can't unilaterally make myself better. It's a stable solution. But it takes trust to get there. And so, that's hard, because

sometimes violating a trust would be in your self-interest. You might be out there thinking, "The rabbit's a certainty. The stag, even with all of us working at it, is kind of risky. We might or might not get one. I want the certain thing, of going after the rabbit." And so, if violating the trust is to your self-interest, you might do it.

- 3.119** Or on the playground, bullies might say, "Hey, I can benefit. I can steal your lunch money. I can violate the trust and be better off." So why don't we do it? Hume faces this problem and says, look, here's the thing. We sympathize with potential victims. We're motivated to restrain ourselves. We think, "I don't want to be victimized by a bully, and I don't want to go off and end up hunting a stag to no avail and come home hungry." So we understand the value of cooperation. He says, "That means we agree to actually be true to our word, to keep our promises, to become trustworthy."
- 3.120** Why? Because the whole system depends on it. We recognize that we want people to actually keep their promises to us. Now, he says, "Here's the problem, though. On a large scale, it's really hard." If I make promises to you, and you make promises to me, then they're pretty well defined, and I know what happens if I fail. But suppose we've got thousands of people, millions of people, involved in this. Can I trust that all million, or even enough of the million, are going to cooperate, that I can trust that we'll get the best outcome? So he says, "The bigger the group, the harder this is to achieve."
- 3.121** And he talks about a farmer's game. My crops will be ready next week. Yours will be ready two weeks later. Each would be better off if we helped each other harvest our crops. We can finish mine by the time we get to yours. And you'll help me if you expect help in return. But here's the thing. I'm selling my farm when my crop comes in and I'm moving to Florida. So, suppose you help me. Now, I could help you, but I've got no incentive in terms of future interaction with you. What will I do?
- 3.122** This has the form of a prisoner's dilemma. One player is thinking, "Well, should I help you? Should I agree? I don't know. Because after all, I'm better off if you help me get my crops, yes, but I'm also better off not helping you." And so, it's exactly the form of a stag hunt, really. I end up thinking, "Yeah, there's a better outcome here, but there's also a worse one." We can also view it as a prisoner's dilemma. Call it the farmer's dilemma, where we realize, "Wait, I've got an incentive not to help you, because I'm moving to Florida, but you've got an incentive then not to help me, because I'm not going to help you." And so, really, it turns out, we're stuck in this suboptimal equilibrium where we don't help.
- 3.123** Now, all of this, you might say, is due to the fact that we're trying to decide in advance. But the truth is, because some crops come in before others, one player has to go first. And so, we're going to talk about this later in the context of sequential games, where one farmer has to move first and then the other has an incentive. This really depends on that sequence and what we assume about that sequence.

- 3.124** But in general, my moral is this. Both with respect to stag hunts and with respect to the prisoner's dilemma, the question is really, "Can I trust the other person?" The police are questioning you off there, and I'm thinking, "Can I trust you to remain silent?" Okay? Really, we're better off together if we both remain silent. Yes, I could gain some advantage by testifying against you, if you're not testifying against me. But I recognize, ultimately, that's destructive. Really, I should remain silent if I can trust you that you're doing it, too.
- 3.125** And the same thing with the stag. I'll hunt the stag, if I can trust that you'll help me. And so, a lot of things depend again on trust and on social capital. But we've already talked about what that is and the importance of it. We'll talk more about how to build it later.
- 3.126** With game theory, would you say that it's inclined towards prescriptive articulation of the world or reality in games and how things work, or descriptive, if that makes any sense?
- 3.127** Oh, that makes complete sense, and it's a crucial question, because, if we think, what do people really do here? If this is predictive, if the idea is, we're going to think about what rationality is and then predict what rational agents will do, it works sometimes, but it doesn't work a lot of the time. For example, a majority of the prisoners would remain silent, and at least 37% of the students would remain silent, so a little over a third. So, one out of three students is trustworthy. Five out of nine prisoners is trustworthy.
- 3.128** But if you think about things that way, you realize it's not actually great at predicting what people will do. And in general, people cooperate much more than this would predict. Because the fact is, we do trust each other a lot in society, at least in a healthy society, we do. And consequently, most of the time, faced with a prisoner's dilemma situation or a stag hunt situation, people do go for the Pareto-optimal solution. They don't get stuck in the Nash equilibrium. They don't play their dominant strategy, and so, if it's a predictive theory, it's not very good.
- 3.129** Now, in some settings, for example, arms control, where this was originally developed, it's actually pretty good. I mean, yeah, countries cheat, partly because verification is very difficult. Because it's hard for me to know whether you're really secretly producing nuclear weapons in that underground lab or not. Whereas I know, for example, the friend says, "Yeah, I'll meet you at the bar at seven o'clock," I show up, and the friend's not there and isn't coming, well, I find out. I know whether he's there or not. Whereas I don't know whether Iran is actually making nuclear weapons or not, etc.
- 3.130** It turns out that in settings where it's very hard to verify, the predictive aspect is pretty good. But in cases where you can verify it, it's actually not very good. That brings us to the prescriptive side. Are we saying that this is what you ought to do? That doesn't really work either, because if you think about the prisoner's dilemma, I'm not saying,

yes, you should follow your dominant strategy. Testify against that other guy, pollute that stream, etc. It seems terrible to do that.

- 3.131** On the other hand, do I want to say, always you should seek that— I mean, that makes me a little nervous, too, for various reasons, partly because it may be that there are certain things we don't—we just let people play their dominant strategies because it doesn't matter enough. In other words, yes, there would be a better solution in a certain sense, except it's not worth it. What do I have in mind?
- 3.132** We could let the kids run around in the playground and amuse themselves, or we could try to organize a game, and maybe we'd be better off organizing the game. We teachers get together and think, "Let's have all the fourth graders play kickball or something." Maybe it would be better if we could do that. But really, they're having lots of fun as it is. And so, we say, "Yeah, this is the Nash equilibrium solution. It's Pareto suboptimal. But you know what? It's a lot of work to get to that one. It's not worth it."
- 3.133** So, sometimes I think we just say, "Here's where intensities matter." We say, yes, there's a Pareto improvement, but it's small enough that, forget it, we don't worry about it. And so, I don't think it's exactly prescriptive. It's certainly not prescriptive to say, yes, follow your dominant strategy. Just lean into those Nash equilibria. But neither do we want to say, so we always have to get people out of it. It's tricky. Sometimes we think people's freedom of choice is just worth it.
- 3.134** So, I think, given that, you might say if it stinks as a descriptive theory, and if it's not very good as a prescriptive theory, what is it then? I don't think game theorists have a great answer to that, in a sense. Here's what it is, really: an investigation, as I see it, into the nature of rationality. It turns out that rationality is complicated. What it is to be rational as an individual? What it is to be rational as a group? It's just a hard question.
- 3.135** And at the very end, I'm going to talk about group decision-making, voting, but also generally, how you make decisions. It's going to turn out, there is no perfect method of doing that, no perfect method of coordinating. You can't have everything you want. But what I take from it all is that the idea of rationality is really complicated and is a kind of ideal that cannot possibly be achieved. You're simply never going to be a fully rational individual. You're never going to have a completely rational group.
- 3.136** It's not because, "Oh, we're imperfect. I mean, we Calvinists believe in the total depravity of man. That's why." No, it's because of the nature of actually putting things together and trying to act as groups. There is something about that that immediately means some of these problems can't be solved. My own view is that, in the end, it doesn't really work as a descriptive theory or as a prescriptive theory.
- 3.137** What it does is tell you, there are traps. As soon as we get together, as soon as the results of my actions depend on what you do, there are all sorts of traps here, and there's no simple way out. Often, I think it's a matter of context. What's the best

solution we can come up with here for this kind of problem? But yeah, I think it more is a—it's a problem-finding tool than it is a solution tool.

- 3.138** In secularism, the idea is that there are certain topics that we don't need to settle in order to come to agreement on things. But it seems like not knowing the truth of the matter about some of those, like, does God exist, would affect how you understand the game you're in. So is that a limitation on secularism?
- 3.139** Oh, that's a very interesting question. Yeah, you might think, look, why don't I just attack the other person? There's this pacifist. Maybe let's go back to my playground scenario where, all right, the teachers are just letting us run amok on the playground just before lunch. I look at that kid and I think, "That kid's a nerd. I could go beat him up." Now, actually, in real life, I was the nerd. But anyway, so I think, "Hey, I could take his lunch money." You might think, "Hey, it's my dominant strategy. I win, I win."
- 3.140** Now, why would I not actually benefit from that? One way is that, aha, the teacher might catch me, and I might get in trouble, sent to the principal's office. But another way is, God might see me, and I might be punished for my sins. And suddenly I realize, "Oh, wait a minute." So, if you think about this in the context of, let's say, an atheistic worldview, you might think, hey, all the things that fill in the boxes are known right here. They're just these simple, finite things. Okay, I gain a \$1.10 by stealing his lunch money or something.
- 3.141** But if you start thinking of it from that point of view, you realize, "Oh, wait a minute." There is this other huge thing on the scale. I am thinking here about the short-term immediate consequences; I'm not thinking about the longer-term spiritual consequences. For example, you might do that in a religious context, where you are saying, "But God is looking down on you and seeing your sins and seeing your immoral actions and so on," rewarding you for the good thing, rewarding you for the cooperation and punishing you for the defections.
- 3.142** Or you might do what Plato does and, say, take a long-term perspective, even forgetting, I mean, Plato was a theist, but leaving that aside even, Plato is thinking, think about what it's doing to your own soul, to your own character. In the end, it does not benefit you. He imagines the tyrant as not the happiest of people, the one who can do whatever he wants, get whatever he wants. He says, "Is that the happiest person?" No, it's the most miserable person, because this is a person whose lawless desires are running amok in his soul and constantly in tension with other parts.
- 3.143** This is somebody who—it's as if they have an intense itch. Yes, they can scratch it as much as they want, but the fact is, it never stops itching. He says, "That's actually the most miserable person." So I think you can think of that in terms of divine rewards and punishment. You might also think of it more in terms of character as Plato does, or as, for that matter, Hinduism does in thinking about a doctrine of karma. It's not this magical, "hey, what goes around, comes around; the universe will punish you."

- 3.144** It's more like that act of stealing, for example, or testifying against your buddy falsely, let's say, or doing this or that other thing, that is corrupting your own soul. That is making it more likely you're going to do similar things in the future. Turning yourself into an untrustworthy person is not good for you. That's a long-term consideration that's independent, in a way, of God's punishing you for it. It's more like the world will punish you for that, even independently of that, because you have to take this long-term perspective and think about what it does to your character, what it does to your soul, not just what it does to your bank account.
- 3.145** And so, yeah, I think it's a combination of a secular view, but then also a very short-term view, that thinks, "Hey, I win here." Well, yeah, you won there, but what did that do to you? That perspective, I think, is often missing from these considerations and missing from our culture in general. It used to be there. People used to think, "What kind of person do I become if I act this way?" To the extent that we've lost that, it does force us into short-term thinking, which makes it all the more attractive to just follow your dominant strategy in terms of the short-term payoffs. That inclines you to get trapped in more and more of these suboptimal Nash equilibria.
- 3.146** I think it's a very important question, and one that means the shorter-term and more limited your perspective on things, the more you're going to get trapped in these ways that are really collectively self-defeating.

Lecture 4: Competitive Games

At a glance — Competitive games and what people compete for.

- 4.1 Today, I'm going to be talking about the good things that people compete for in competitive games. There are times when games are there about coordinating. We talked about a driving game that's like that. There are other ones that are more about cooperation, like prisoner's dilemmas and stag hunts. But there are many games that are really competitive. If you think about most of the games that immediately come to mind—football or tennis or chess or checkers or a board game—people are competing against each other, so the class of competitive games is really important.
- 4.2 What it tells us, when we start thinking about the nature of different kinds of games, is that it's really vital to know who you're playing against, understand them, but also know what kind of game you're playing. Are you competing? Are you cooperating? Are you coordinating? What are you doing? We ordinarily have an intuitive sense of that, so we don't really have to work very hard at this in much of ordinary life. But there are contexts where it's not so clear whether we're in competition or whether we're trying to cooperate.
- 4.3 It can easily happen, as we mentioned last time, that one person thinks they're playing one kind of game, and another person thinks they're playing another, and that causes problems. So, to form a strategy, you need to know what kind of game you're in. Are we trying to coordinate? Are we competing? Are we cooperating? What exactly is our relationship here, and what's the nature of the game? It's really easy, even if we're competing, to misunderstand the nature of the competition. I have to know what your aim is.
- 4.4 In games like chess or checkers or tennis, it's not very difficult to figure out what your aim is. But in much of life, it is actually hard. I have to know what you want and what you value. I have to know what your preferences are, to really understand the nature of the game. That can be hard. How much information are we going to have about the other players' decisions, about the other players' preferences? Sometimes we have a lot, sometimes we don't have all that much.
- 4.5 We have to know how much our own preferences coincide with those of the other players. Are my interests and their interests in alignment? Are they directly opposed? I have to understand that. I've got to know to what extent our interests overlap. If they overlap enough, we can cooperate and maybe jointly achieve a positive outcome. If, on the other hand, they're pretty opposed, we're forced to maybe compromise, or it's just a competition. One of us has to beat the other.
- 4.6 Sometimes, as in business negotiations, a compromise makes sense. On the other hand, if we're playing football, a compromise doesn't make much sense at all. "Hey, we're about to throw a long pass, but tell you what, we'll only throw a short one if you

guarantee that you'll let us catch it." That doesn't happen. And so, you have to understand the nature of the game and whether the other person is willing to compromise, whether they're willing to cooperate, or whether this is a direct head-to-head conflict.

- 4.7 If I stop and think, "What are the possibilities here?", I realize that, at least if we have two players and we don't allow ties, so we assume that I've got options, let's say, four outcomes that are possible, and I think, "Here's my first choice, second choice, third choice, fourth choice," then I look at you and think, "What exactly are your preferences with respect to these outcomes?" Then I can think, "It could be that they exactly match mine, one, two, three, four, just as mine are." They could be directly opposed, like, the thing I like best is the thing you like least.
- 4.8 In many competitive games, that's exactly what's happening. Our interests are directly opposite of each other's. But there are many other possibilities in between. And so, if you analyze this, you realize, wait a minute, as long as we don't allow ties, to simplify things, there are 24 possible combinations here. Maybe you agree with me exactly on preferences. Maybe we disagree completely; we're absolute opposites. But it could be all sorts of things in between. Maybe what I like best is really your second choice. My second choice is your first choice, and so on. And so, there are 24 possibilities.
- 4.9 Here's a case. Let's say Alice ranks things exactly coincidentally with mine. I say first choice, second choice, third, fourth, and she does, too. Then we can cooperate perfectly. We can coordinate. We say, let's both get our first choice. Bob maybe has exactly the opposite choice. What I like best, he likes least. Then we're in a situation of perfect competition, perfect conflict. But there can be all sorts of things in between.
- 4.10 Let's say Carol has the same first preference as me, likes that one best. But on the other hand, my second choice for her is her last choice. And then my third choice, her second. My worst choice is her third. There's a partial agreement here. We both agree that one is better than four. We agree that one is better than two, which is better than three. But she ranks the fourth option that I put at the bottom much higher than I do. There's partial overlap here, but partial competition. We have to know the relevance of that. How important is that?
- 4.11 Then Dave, let's say, takes what I consider the best option, and he only ranks it third. My second choice is his worst-case scenario. He actually likes the thing I like least. There's a little bit of agreement there. We both agree that three is better than four. On the other hand, we also disagree about everything else. We have to understand, to what extent are we on the same team? Are we on opposing teams? Is there a partial overlap? If so, what is it?
- 4.12 There's a little more math that can help us here, because if you think about the possible forms of games, at first it looks complicated. It looks like there are many of them, but you start realizing, wait a minute. It doesn't really matter which column we put in which

order. It doesn't really matter which row we put in which order. It doesn't matter whether you're player A or player B, we should be able to switch them. There are all sorts of axes of symmetry here. When you take that into account, you realize, as long as we disallow ties and so on, we get 72 non-isomorphic games. In other words, if we've got two players and we're just thinking about preferences, so first, second, third, fourth, and not actual utilities or cardinal numbers, then there are only 72 possibilities.

4.13 Seventy-two from one point of view is a lot, but from another point of view, we can say, all right, it's not that hard maybe to actually figure out what game we're playing, as long as those constraints are satisfied, because it's a relatively small, finite number of possibilities, and they fall into certain groups. Six of those 72 have the form of a prisoner's dilemma, and six have the form of a stag hunt, and so on. So, they fall into groups in a way that's actually pretty natural.

4.14 If this is true, we can say, how many games are there, where we are in full competition? My first choice is your last choice, and your last choice is my first choice, and so on. Then we think about this and we realize, there are actually only three. There are these three basic forms. If I rank things one, two, three, four, and you rank them four, three, two, one, there are only three ways that can really happen. So, you can analyze those games and start thinking, if we're really directly opposed, here are the possibilities.

4.15 In two of them, there actually is a Nash equilibrium, so we can kind of predict in advance what's going to happen, given that sort of conflict. In one, however, there isn't. So let me try to illustrate a little bit. Here we've got one of these pure competition games. Let's imagine that our players are orange and red. Orange doesn't like red. Orange is a hard worker, quite a good worker, thinks that red is lazy. Red's a slacker. Orange wants this job done and is eager to work on it, but prefers not to work with red, like, "Why do I have to share the credit with this guy? He's not going to contribute anything."

4.16 It turns out, he's right about red. Red is lazy. He prefers not to do it. You can see here there is a kind of competition. Who's going to actually do the work? You might say, look at it from orange's point of view. The first choice is that orange wouldn't really have to do it at all. He might say, "Look, I'd love it if somebody else did this. But on the other hand, I'm not a fan of red. I don't think that red will do a good job. But still, my first choice is, I don't have to do it. Let somebody else do it."

4.17 Second choice, though, is that I work on it and that red works on it, too. I'm going to share the credit, but at least he'll get a little help. Or third choice, I do it all myself. I'd rather do it all myself in a sense than let it not happen. The worst-case scenario is that it doesn't get done at all. Then red might think, "Look, my first choice is that nobody does it. I don't care about this. I think it's a dumb thing to do in the first place."

- 4.18 If you've ever been in a highly bureaucratic organization, you can sympathize with red here. There are a lot of things that aren't worth doing. I've had projects where I actually changed the icon on the little file folder in my computer to a fist, slamming a keyboard, because I thought, "This is stupid. It's a lot of work and there's no reason to do it at all." Red is like that. He's like, "I don't think anybody should be doing this."
- 4.19 My second choice, though, is that at least I don't have to do it. Let that guy do all the work. Third choice is, all right, we both work on it. The worst-case scenario, from my point of view, is that I have to do it all myself. I don't think it should even be done. What's going to happen here? If we think about those options, we realize, from orange's point of view, the best-case scenario is really that somebody else does this thing, because orange kind of agrees that it should get done. We have to get it done. On the other hand, I don't really want to deal with it. It's not very interesting.
- 4.20 So, orange says it'd be great if somebody else did it. And so, if the other guy is going to do it, I don't think he'll do a good job, but who cares? I'll let him do it. However, suppose the other person is slacking off. Then orange says, "It needs to be done, so I will work on it. That's my best response." Our slacker says, "Look, if that guy is going to work on it, good, I don't want to have to." So we can predict what's going to happen, which is, orange does all the work. Because the best response, if orange is working, from red, is to say, "You do it. I'm not going to get involved. I'm not going to do it."
- 4.21 So, before we start thinking about the actual competition, we realize there's already an equilibrium here. We don't have to really puzzle out how the competition is going to go. Orange is the one who most wants it done. Orange is going to do all the work. Here's a little illustration of that. We see that in all sorts of contexts, not only in bureaucratic organizations, but in others, where one says, "Yeah, I've got to do this," and others just kind of stand around and lend, at best, moral support.
- 4.22 Here's a case which is in a way more interesting, because in this case, there isn't a pure-strategy Nash equilibrium. We can't really predict how it's going to come about in the first place. Here's our scenario—and we're going to come back to this later because it's interesting, I think, from a number of points of view. We have a conformist and a nonconformist. Our conformist has opposite color preferences from our nonconformist. Here's the idea.
- 4.23 We have somebody who loves blue and is also a conformist. She wants to paint her own house blue, and she wants other houses in the neighborhood to be blue, too. She loves blue and she wants everything to be the same color. She hates this idea that her house is blue, somebody else's is white, somebody else's is red, and so on. She wants them all to look the same. But our nonconformist says, "Look, I'm not that fond of blue, but also I want all the houses to be different colors. I don't want them to all look alike. I don't want to look like I'm just in one more box, in all these identical boxes. I want individuality."

- 4.24 And so, if you think about what happens here, you realize that, from the point of view of our blue lover, conformist, the best-case scenario was that both houses are blue. Second choice is that, now you have to decide, is the love of blue stronger or is the conformity stronger? But suppose it's the conformity. If I can't have blue, at least I want them all the same color. Second choice is, they're the same color. It's not blue, but still they're the same.
- 4.25 Third choice, that her house is not blue, but the other one is. At least she looks out her window and sees blue. My house isn't blue, but I don't usually see the outside of my house. I see the outside of that house and I want to look out and see blue. The worst-case scenario is that her house is blue, but the others aren't. So, she doesn't get her conformity and she looks out the window and doesn't see blue. So, those are her preferences.
- 4.26 Our nonconformist, on the other hand, says, look, my first choice is, we just do our own thing. You want a blue house? Paint your house blue. I want it some other color. I'll paint it that color. Second choice would be, mine is blue, yours is not blue. I'm not that fond of that, but at least they're different. I want them to be different. Third choice is that we're both not blue, because I don't really like blue that much. My worst-case scenario is, they're both blue because then, this uniformity, I don't like that, and I don't like blue.
- 4.27 In this case, there is no Nash equilibrium. The best response for each player is just the opposite of what another wants. For example, our conformity lover, if the other person's house is not blue, she thinks, then I want mine not to be blue. I want mine to match that one, whatever color it is. Suppose the other person paints her house blue. Then she says, then my best response is to paint mine blue. That's what I wanted in the first place. But if our conformist paints hers blue, the other's best response is to go for some other color. Suppose she paints it not blue, then our nonconformist, the best response is to paint hers blue.
- 4.28 So there is no pure-strategy Nash equilibrium. There's no square in that little game form, where both are giving their best response. As we'll see, you might think of this easily in terms of a sequence rather than a simultaneous thing. It'd be a little weird if they both are there at Home Depot, picking up paints at the same time, without paying attention to what the other is doing. Typically, what happens is, one paints first, then the other paints. As we'll see, then it's a highly unstable situation.
- 4.29 I paint my house blue. She paints her house white. I think, "I want conformity. I'll paint mine white." Then she said, "I don't want to match you. I'll paint mine blue," and so on and so forth. We're in a situation where a lot of paint gets sold until finally one of them gives up. But in any case, that's our second pure strategy, a complete opposite situation. There is no Nash equilibrium now. It's going to be highly unstable.

- 4.30 The third one actually looks like a minor variant of a prisoner's dilemma. We're again in complete opposition. But in this case, we actually get a Nash equilibrium, and I think it's a really interesting one. In economics, there's something called Gresham's law. The idea is, bad money drives out good. Suppose somebody comes in and cheapens the currency. Let's say the coins were made with pure silver before, or almost pure silver, and now we mix it. We say it's part silver, part nickel, or part something else.
- 4.31 Then the value of the money has really gone down. What this law says is, pretty soon all the money in circulation is going to be the cheap, the bad money, because people are going to hoard the good money and not be willing to part with it. This problem goes all the way back to Aristophanes, who writes about this in *The Frogs*, and he talks about many aspects of this. He says, look, this isn't just a problem about currency. It applies to that, but it applies to every dimension of life.
- 4.32 This is why bad music drives out good, bad literature drives out good, bad employees drive out good employees, and so on and so forth all the way down the line. He ends up saying, "This is the explanation for all social decline." You might think old people always look at things and say, "Oh, things are going downhill. It was so much better when I was a kid." But Aristophanes says, in effect, what if it's all true? What if it has been all downhill? What if there is this general pattern of social decline? Why? Because people have these opposed interests, and bad money drives out good. Bad stuff drives out good stuff.
- 4.33 From one point of view, this sounds like "make Athens great again." Indeed, he has a bit of that atmosphere in *The Frogs*. But here's the underlying game theory. He's saying, suppose we're in a situation where there's a buyer and a seller, and each one has a choice. I'm going to use good money for this transaction or I'm going to use bad money. The idea is, the buyer hands over a certain amount of money to the seller, gets the good, but then gets something in change.
- 4.34 And so, we have to think about that monetary exchange. I give, let's say, \$20, or I guess, to bring it back to Aristophanes' time, 20 drachmas or something like that, and I get a certain number back as change. So I'm thinking, as a buyer, what do I want to do? Here would be my first choice. I give the seller bad money, and he gives me back as change good money. I've got something more valuable. Maybe it's less money, but nevertheless, those coins are worth more. I think that's my best-case scenario.
- 4.35 But from the seller's point of view, that's the worst. Wait, I'm giving change in the good money and I'm getting bad money from the buyer? I don't want that. That's my last choice. If I'm the buyer, I think, what's my second choice? I am a net loser of money in this, because I am giving you a certain amount of money. I'm getting some back in change, and I'm getting the good that I'm purchasing. But overall, I'm giving you more money than you're giving me. So I am a net loser here.

- 4.36 Do I want to lose good money, or do I want to lose bad money? I'd rather lose bad money. So even if you're going to give me bad money in exchange, I'll give you bad money. What's my third choice? That you give me good money back, and I give you good money in the first place. Then, since I'm a net transferrer of money to you, since I'm buying this object, I'm giving up good money for it, but at least I'm getting the object.
- 4.37 The assumption is that the object's worth enough that I'm willing to do this either way, so I'm a net gainer, given that I'm buying this thing. But on the other hand, in terms of the money, I'm actually, in this case, giving up more good money than I'm getting back in change. My worst-case scenario was that I give you good money, and you're giving me the cheap, not-very-good money in exchange.
- 4.38 So, the seller's perspective is completely the opposite. His first choice is really, look, I get good money from you. I give you back bad money, and so on. Now we can think, what's going to happen here? How does it work out? Think about the best response of each person. The seller, if the seller is actually operating with good money, and I'm the buyer, and I think, "I'm going to get good money in return; this guy only deals in good money," I can do better if I give him bad money. Then I give him a certain amount of bad money. He gives me back some good money. Yay! So, I benefit in that way. My best response, if he's using good money, is to give him bad money.
- 4.39 But now, suppose he's giving me back bad money. I certainly don't want to give him good money and only get back in change bad money. So my best response is to give him bad money. But now, look at it from the seller's point of view. The seller says, "If you're coming in and giving me good money, these coins are pure, oh, fantastic. I can give you bad money in exchange," and I end up benefiting.
- 4.40 On the other hand, suppose you're giving me bad money. I certainly don't want to give you good money back. It's best for me to give you bad money. In short, both of them have a dominant strategy. Use the bad money. The result of that is that all the transactions end up being conducted in the bad money. The good money stays at home in the drawer or under the mattress or in people's pockets, or wherever in ancient Greece you store coins that you have around the house. That's where the good money is going to be. People will hoard it. All the exchange is going to be in bad money.
- 4.41 Aristophanes says, look, this isn't just a financial thing. Yes, it does apply in finance, but it applies in all sorts of dimensions of life. Let's say you're working really hard on your music. You're creating these symphonies and so on, they're complex pieces, and somebody else has three-chord rock, which I've been hearing a lot of in my hotel because it was built in 1956, and they take you back to 1956 in all the public areas. This morning, I was listening to "Charlie Brown" and all sorts of other songs like that.
- 4.42 You might think that's all fun. It's not nearly the accomplishment of that symphony, but very quickly the bad music—and here I shouldn't say bad music; I like that music—but

nevertheless, the simpler music, the easier-to-produce music, drives out the music that's harder to produce. Same thing with literature, same thing with television, same thing with really every cultural phenomenon. The idea is, really, look, bad stuff drives out good stuff. Why? That's the Nash equilibrium. If I am working really hard in doing this and then you're achieving it much more easily, then I might as well do the same.

- 4.43 Notice, in other contexts, this isn't necessarily a bad thing. In other words, what happens with money or other cultural products is quite different from what happens, let's say, with technology, where you start using a better technology, and all of a sudden, I've got an incentive to use that good technology, too. You might think, when it comes to technical advances, then it operates just the bad way. The good technology drives out the bad technology. It's much less efficient. It doesn't do as well.
- 4.44 And so, what in technology and the advancement of industry, you might say, is a good thing, the Nash equilibrium there is a nice thing, here, in cultural exchanges, it tends to be not a nice thing. After all, in this case, it's the stuff that takes less work, driving out the stuff that takes more work. The stuff that has less value, driving out what has more value. And so, in those kinds of contexts, it turns out to be this decline.
- 4.45 All right, so that's the basic picture. When I've given this, by the way, as an assignment to my students, saying, "Figure out what the Nash equilibrium is and tell me what it means," they don't tend to get that the buyer and the seller are in different positions. They say, oh, it's just a prisoner's dilemma. But it's not, because the amounts of money going in the two directions are not the same. I give you the 20 drachmas, maybe I get three back in change. I'm giving you 20; I'm only getting three. So, there's an asymmetry there that a lot of people don't grasp, but it's really important to understanding the case.
- 4.46 Let's think about a very simple, pure competition game, much simpler than economic exchange or some of these other things. This is just rock, paper, scissors. I can play paper, I can play scissors, I can play rock. The idea is, scissors cut paper, paper covers rock, rock smashes scissors. And so, I've got this interesting kind of cycle about how this goes. That creates an instability. Right away, you realize, oh, what do I do? There's going to be no pure-strategy Nash equilibrium.
- 4.47 I mean, what that means is, suppose I just say, like Bart Simpson, I'll play rock. Nothing beats rock. That's a terrible strategy in this game. If I go in constantly saying, "I'm playing rock," you can immediately say, "Paper, paper, paper." You can beat me every time. And so, if I start thinking about this, I realize, let's say you play rock. My best response is to play paper. Suppose you play paper, my best response is to play scissors. Suppose you play scissors, then my best response is rock.
- 4.48 But I don't know what you're going to play in advance. What do I do? There's no pure-strategy Nash equilibrium here. There's no case where we're both playing our best option. Mine go that way, but yours, of course, are quite different. If I'm playing

scissors, you should play rock. If I'm playing paper, you should play scissors. If I'm playing rock, then you should play paper and so on. So, it's going to turn out, we have to mix it up. In the next lecture, I'll be talking about mixed strategies.

4.49 But here it's really intuitive what the mixed strategy is. It's not complicated at all. It's not like, with what probability should I play rock? The answer is one-third. We're going to take that next time and really talk about the significance of that and what it tells us about finding mixed strategies. But here, what we can do is say, "I'm not going to play the same thing all the time. I should randomize. I should do each one about a third of the time."

4.50 What I find interesting about this is that there are actual rock, paper, scissors competitions. You might think, "This is purely random, right? This is dumb." But actually, it's not dumb. Some people are much better at this than others. And so, it's an interesting question to say, "Why is that true? How can you be better at this game than somebody else?" I mean, you're just doing it randomly, right?

4.51 There are two points. One is, think about what's happening. I decide to play scissors. That takes a certain amount of time. It doesn't take much time, and most of us aren't going to notice and be able to change and adapt, given the milliseconds that are involved in this, but maybe some people can. I mean, there are people who can hit 98-mile-per-hour fastballs. I don't understand how it's done. I've gone to batting cages, and if it's 90 miles an hour, I can't even see it. It's not like I can't hit it. I can't see it. It's just [whoo] so I can't respond that quickly.

4.52 But suppose you're the sort of person who can, maybe you get a little information as you see that hand coming down. That's one way you could be good. But the other factor is that it's actually really hard to randomize your actions in this way. Even if it's something as simple as our two-finger morra game, I just do that or that, and I want to randomize it 50% each, I think, how do I do that half the time? It turns out, people are not very good at randomizing their actions. They tend to fall into patterns.

4.53 And so, another way you could be good at this game is recognizing the other person's patterns, studying what they do and realizing, "Oh, he always plays two rocks and then goes to paper or something." If you can pick up on those patterns, you can do better. There's no pure-strategy Nash equilibrium. There is a mixed strategy equilibrium, easy to know what it is, hard to do it in practice.

4.54 You can complicate this all you want. The Big Bang Theory popularized rock, paper, scissors, lizard, Spock. That's more complicated. And you could do that, so, rock, paper, scissors, I don't actually know how you do lizard. I don't remember. But that's a way of actually setting up a game that has a similar cycle. You can make it even more complicated. See, there's the way that game goes in terms of what beats what.

4.55 You could also have rock, paper, scissors, lizard, Spock, Spider-Man, Batman, wizard, Glock. People have spelled that out. It's really complicated, but you could do that.

There are the hand signals that indicate exactly what this is. You play, I don't know, Spock, and I play this. Anyway, it gets complicated, but it's the same basic problem. Those games are familiar, but they're pretty simple. There are games that are not so simple.

- 4.56** So, here I want to take us back to the 1950s, James Dean and Natalie Wood, in *Rebel Without a Cause*, and a game that gets played in that movie, a game of chicken. This has become now one of the standard games that's studied in game theory. Here's what's happening. We are in our cars, our mid-1950s hot rods that we've souped up, and we are going to have a choice of either going straight or swerving. Two variants of the game, actually. One is, we're headed straight for each other. You can either go straight. If you both go straight, you know what happens. Bam. Horrible crash. You both die. That's both players' worst-case scenario.
- 4.57** You could both swerve. You avoid the accident, but you show that you're a chicken. Really, what you'd like to do is, you're the brave one who goes straight and impresses the girl, and the other person swerves to show he's the chicken. You'd rather be the chicken than die, to be clear, but you really don't want to be in a situation where you lose to the other guy if you can avoid it. Your first-case scenario is, you go straight, the other guy swerves. Your second choice is that you both swerve, so at least you're both chicken. You don't lose any comparative or relative social status.
- 4.58** Third choice is, you end up being the chicken and swerving. The worst-case scenario for you is, you go straight. The other guy does too. You both crash. There's also a variant where you're headed toward a cliff, but that's really bad because you think, if you keep going straight, you die. In that case, you're like, there has to be a place where it's a question of who swerves first.
- 4.59** But in any case, this is a highly competitive game. But notice, the players' interests are not completely opposed. Neither one wants to die. Neither one wants the crash. And so, this is a bit different from the games we've thought about. It's highly competitive. They're almost opposed, but not quite. So if we start thinking about what happens in this game, we realize, what is my best response? If you're going to swerve, then my best response is to go straight, so I am being the courageous one.
- 4.60** But suppose you're going to go straight, then my best response is to swerve and avoid the accident. It's just the same for you. If I'm going to go straight, your best response is to swerve. But if I'm going to swerve, your best response is to go straight. So there are two pure-strategy Nash equilibria here. One person goes straight; the other swerves. But who's it going to be? The competitive element is, who's going to swerve? Or in the other variants, who's going to swerve first? That becomes the question.
- 4.61** In this case, we don't want the scenario where we crash, but each one would like to be the one who goes straight, and therefore ends up showing courage; the other person ends up being the chicken. In that case, notice what we have here. We have two Nash

equilibria, but unlike in a stag hunt, where only one of them was Pareto optimal, the other was clearly inferior. I mean, two of us getting rabbits, not nearly as good as bringing down the stag.

- 4.62 But in this case, they're both Pareto optimal. That is to say, I'm in a situation where I'm the one going straight. You're swerving. You can't do anything unilaterally to improve this. If we switch places, one of us is benefiting, but the other is suffering. There's no pure gain without a compensating loss here. The players are in really different positions. So one of them is going to win, one of them is going to lose, but which one? And the danger is, we both say, "It's not going to be me. I'm going to be the one who wins." But if both make that choice, bam, and they both lose big.
- 4.63 So that ends up being not just of interest in thinking about a movie from, [1955] but it's also of interest in thinking about all sorts of other scenarios. So there's a variant of this called the snowdrift problem, and really, it applies to any group project. It's the same pattern. We could both goof off, but then the project doesn't get done. That's bad. We could both work on it. That's each of our second choice. But really, each of us has the first choice; let the other guy do it. What's going to happen here?
- 4.64 If you're going to work on it, my best choice is to slack off. On the other hand, if you're going to slack off, I'd better do the work. And so, it's the same for you. In short, there are two pure-strategy Nash equilibria. In one, I'm doing all the work. In the other, you're doing all the work. But in any case, one person is just doing all the work. But who's it going to be? That makes it complicated. You might say, here's a situation where you could compromise. Let's both do our part. That's each of our second choices.
- 4.65 That would be a reasonable compromise here, kind of like saying to the other person in the chicken game, "Listen, let's both swerve." Then neither one of us loses face, and neither one of us dies. That would seem good. But now, why doesn't that work as a solution? We just compromise. We'll both swerve. Because the moment you know you're going to swerve, you can go straight and you can benefit, so that's highly unstable. It's not a Nash equilibrium.
- 4.66 Similarly, in this case, the other person says, "Oh, I'll do this part." But then they've got every incentive not to do it, because if you're going to do a part, then, hey, why don't you just do it all? I'm not going to do my part. And so, it becomes a highly unstable situation. One person can unilaterally improve their situation, and yet if both try to do it, boom, the project doesn't get done, or in the case of chicken, we get the crash.
- 4.67 And so, this is a scenario where we've got seemingly an ideal compromise, except it's highly unstable. Each person has an incentive to break that compromise, to break their word. If they both do, disaster results. So, it becomes really difficult to deal with. We've got these two Nash equilibria, and they're both Pareto efficient. If this person does

unilaterally improve their position, this one is worse off. Neither player has a dominant strategy, so I can't say, "No matter what you're going to do, I'm going to go straight." No, then we'll crash. Or, "No matter what you're going to do, I'm going to slack off." Maybe then we'd both fail at this project.

- 4.68** I think this is actually highly general. I, in school, always hated group projects. My university periodically tells us, "Have your students do joint projects," and I never do. In fact, in one meeting about this, I was asked, "So, Dan, how do you build group projects into your ethics courses?" And I said, "I don't." And they said, "What? You don't?" I said, "No. Why do you think I became an academic? I don't like working with other people." That's a little too strong, really. I was exaggerating.
- 4.69** But the fact is, with group projects like this, there is a huge tendency for things to happen exactly the way they happen in *The Hangover*, the movie, where one guy does 99% of the work. Somebody has no idea what's going on and doesn't help at all. Somebody says he's going to help, but he doesn't. He's the one who says, "Hey, I'll go along. We'll compromise," and then actually unilaterally defects. Then there's somebody who disappears at the beginning and doesn't show up until the very end.
- 4.70** Indeed, in the few cases I've actually seen group projects happen, that's exactly what goes on. I've talked to my students, and they say, "Yeah, here's why I don't like group projects. I end up doing all the work." Or at least the good ones admit to that. The bad ones stay silent about this because, presumably, they're the ones who let the other guys do all the work. But there's an incentive here. There's a simple explanation. Why does that happen?
- 4.71** Because the compromised position where everybody pitches in and gets involved is highly unstable. Everybody has an incentive to let somebody else do the work. In the end, it's the person who cares most about it getting done who ends up doing virtually everything. There are a lot of other games that are near-perfect competitions. Not exactly. There's some overlap between the players' interests, but there's a lot of opposition as well.
- 4.72** Here's one. I'm going to call this the older brother game, and I'm making that up. That's not general in the literature, but I think it's an interesting game, where there is a pure-strategy Nash equilibrium, and it has a lot of similarity with this snowdrift problem. Here's the situation. You've got two brothers. The younger brother wants to do this thing, and preferably with his older brother. It's like, "Let me tag along. I want to do it, too." The older brother, let's call him Wally. Here, I'm really showing my age, going back to *Leave It to Beaver*.
- 4.73** But the older brother, yes, he prefers to cooperate, but he'd really rather do it himself. So the younger brother says, "My first choice is to do this with my big brother. My second choice is, at least I get to do it, even if big brother doesn't come along. Third

choice is that big brother does it alone, and I'm not allowed to go. And the worst-case scenario is, none of us actually get to do this fun thing. None of us go."

- 4.74** The big brother, on the other hand, looks at this and says, "Look, I'd really rather do this thing myself. I don't want little brother tagging along. My second choice is, you can come along. Third choice would be, neither one of us go because I don't want my little brother going if I can't participate, too." And so, in this case, they do have opposing interests to some extent.
- 4.75** The younger brother's first choice, "Hey, let's do it together," is the second choice of the older brother. Indeed, that turns out to be the Nash equilibrium here. Because why? If I'm the younger brother, I think, "Hey, my first choice is going along. If he's going to go, I want to go, too. That's my best response. If he's not going to go, I still want to go. And so, I'm going to have a dominant strategy. I want to go, whether big brother's there or not."
- 4.76** What about big brother? He says, "If little brother is coming, I guess I'd still rather go. If little brother isn't coming, then great. That's what I really want, to go by myself." He also has a dominant strategy, which is to, in this case, go along, and in the end, goes along with younger brother. They both cooperate with each other. They both go on the activity. And so, younger brother says, "Me, too. I want to go. I want to go," and he gets his way.
- 4.77** For those of us who have a younger brother, you realize, this is not an unfamiliar pattern, but it works in all sorts of other scenarios, too. This person wants to come along. They want to be involved, too. The other person thinks, "I'd really rather do it myself, but okay." So, it might be older brother, younger brother. It might be mom in the kitchen having the 3-year-old say, "I'll help you, mommy," and having to deal with that problem. But in any case, what tends to happen is, both people get involved, whether they'd really like to or not.
- 4.78** Here's another near-pure competition situation. You've got two roommates, and one of them is a cleanliness freak. The other is a slob, like *The Odd Couple*, if you remember that movie or that TV show. So one of them, let's call him Felix, wants to actually clean up and wants the other to participate in cleaning up, too. Like, "Hey, this place needs to be cleaned up. Come on, help me out." That's my first choice, that we both get involved in cleaning the apartment.
- 4.79** Second choice is that you do your part. You clean up. Then I won't really have to because you don't do anything, you slob. If you only did your part, then there wouldn't be this mess in the first place. Third choice is I'll do it myself. I'll clean up the apartment. The worst-case scenario from our clean freaks' point of view is that neither one of us does this, and the apartment's a mess.
- 4.80** But now, put yourself in the shoes of Oscar. He's our slob. He says, "What's my first choice? That you clean up. You want to clean? You do it. I'm not going to do it." Second

choice now is, "All right. I admit it's a pigsty, so I'll help you." The third choice is, "Let's just let it be a pigsty. I really don't care very much." My worst-case scenario was that you expect me to clean it all up and you don't do anything, because after all, I don't really value having a clean place all that much.

4.81 You can predict in this case what's going to happen, because if I'm the clean freak, if I'm Felix, I think, "You're going to help me? Good. That's what I want." That's my best response: to also pitch in and clean up. Suppose you're not going to help me. I still want to do it. I'm better off cleaning myself. So I have a dominant strategy: cleaning the place.

4.82 But what about Oscar? Oscar is saying, "If you're going to clean it, I don't want to. You do it. You care about this. I don't really." So, my best response is to defect, to be a slacker, and say, "I'm not going to help." Suppose you're not going to do it. I'm certainly not going to get stuck doing it. I don't even care if there's a mess. My best response is not to do it.

4.83 It turns out, they both have dominant strategies. Our clean freak says, "Cleaning is my dominant strategy. I want to clean, whether you're helping me or not." And our slob, Oscar, says, "I'd rather not clean, whether you're cleaning or not." What's going to happen? Felix is going to clean up and Oscar isn't. And you could extend this to all sorts of situations. This could be a workplace scenario, too. This person really cares about getting this job done. This person doesn't really care. Guess what? The dominant strategy for the person who cares is to do it. Dominant strategy for the person who doesn't care is not to do it. Consequently, one person ends up doing it all.

4.84 Here's another scenario that's a little more complicated. Here, I'm taking this over inspired by the TV show, *The Office*. One person wants cooperation. He wants agreement, and he certainly doesn't want to be the one who messes things up. His first choice is that we both get involved, we both cooperate. But the second choice is, I want us to agree. If we're not going to do it, let's agree not to do it. Neither one of us will do it. Third choice is, I'll do it and you don't. And fourth case is, you do it, and I don't. Because look, I don't want to be the one who's accused of messing things up. I want to be the one who's viewed as responsible.

4.85 What about the other person? Let's call him Michael. He wants to take advantage of the other person's cooperation. But if there's a defection, he wants it to be the first person's fault. So he says, look, here's my first choice, that you do the work and I don't, but at least it gets done. My second choice is that we both get involved and do it. My third choice is that I do the work and you're the defector, because I want you to get the blame if things go wrong. I want to look like I'm the responsible character and you're the slacker. My worst-case scenario is that neither one of us do it, because then we're both slackers; we're both going to look bad.

- 4.86 What happens here? If I am Dwight, if I'm the one who wants the cooperation and agreement but doesn't want to be blamed for things, then if you're going to work on it, I want to work on it, too. I don't want to be the one who seems to be the slacker. But if you're going to goof off, I want us to agree then we're not going to do it, so I won't do it. But what about you? You're going to say, look, if you're getting involved and you're doing it, I'm willing to let you take it over. I'm not going to get involved. I'm not going to do it.
- 4.87 But on the other hand, if you're not going to do it, then my best response is to do it, because I don't want us both looking bad. So at least I can blame any defects on you for not doing your part. In this case, there is no pure-strategy Nash equilibrium. Notice what happens here. There is no case where my best response and your best response line up. In every case, there's a sort of divide. It's like that game of pure competition, where there's no pure-strategy Nash equilibrium.
- 4.88 That means it's a highly unstable situation. Who's going to do what? We're not going to be able to predict that. It's going to be kind of like a game of chicken in a way, just because it's not going to be clear who makes what move here. But on the other hand, if we look at this and say it's even worse, it's not like we have a pair of equilibria, and now it's a question of which one.
- 4.89 There is no equilibrium at all here. And so, each one is going to have to play some sort of mixed strategy. Maybe I do it, maybe I don't. As we've seen, that tends to be really bad in situations where you expect coordination and cooperation, because then I can't count on you, you can't count on me. That's why it's such a great thing for a sitcom, because you realize a lot of conflict, a lot of interesting scenarios, can arise out of this.
- 4.90 So that's a situation where we end up predicting, there's going to be trouble. And indeed, that seems to be, in many workplaces, exactly what happens. There are other kinds of scenarios that are competitive, but in ways that give us surprising results. So far, I think it's really been useful to think, what is the Nash equilibrium here? Things are going to tend to go in that direction. Even when we can compromise and seek some better solution, you might think, it's highly unstable.
- 4.91 Here's one where there is that instability, and yet the result, the Nash equilibrium that comes about, seems completely counterintuitive. So here's one called the traveler's game. Two travelers go to a foreign country. They bring back the same souvenir in their suitcases. The suitcases are lost. The lost baggage clerk says, "All right, we'll compensate you for the value of the loss." But what is this object worth? Let's say it's somewhere between \$180 and \$300. But it's not clear what it's exactly worth.
- 4.92 And so, here's the idea. You can each declare a value. But I know you're going to be inclined here to exaggerate the value. You're going to be inclined to say, "Oh, 300, 300." I want you to tell the truth. We separate you. We take you into separate rooms and say, "Give me the value that you think it has, some number between 180 and 300,

and we'll compensate both of you according to the lower value declared." If you give different prices, you're going to be fined \$5, if you're the higher price. Whoever guesses the lowest price will get \$5 added. So there's a penalty for being the higher number.

4.93 What is the Nash equilibrium here? What amount do you propose? This turns out to be a sort of weird question, because you might think, "I'm going to say 300." On the other hand, if I say 300 and that person says 299, then I end up getting only 294. They get 304 because I get penalized. Then maybe I should say 298. But then you think, "Oh, wait a minute. If I do that, then suppose they guess 297. They're the lower number. So they jump up five, I jump down five. I'm worse off than they are," and so on.

4.94 And so, in this case, there is a unique Nash equilibrium, a pure strategy one, anyway, which is that we each propose the lowest amount, 180. But that's ridiculous. Why would I do that? After all, suppose we both put 300. Then nobody's higher than anybody else, so we'd each get 300. Suppose I put 300 and you put 299, then I suffer the penalty, but I still get \$294. Why would I propose 180? It doesn't make any sense. And so, if you have people actually respond in this scenario, almost nobody says \$180.

4.95 A game theorist can say, "Oh, the Nash equilibrium is \$180," as I did right away. But on the other hand, then you think, "But that's really stupid. Why would I do that?" People, reasonably enough, don't do that. Even though, if I pick any number between 180 and 300, you can pick one lower and do better, so it looks like the Nash equilibrium is where we just propose the minimum, but that's clearly suboptimal. So, this has the character of, in one sense, a competition. But from another point of view, it's like a prisoner's dilemma. The Nash equilibrium is actually the worst-case scenario for both of us. Actually, it would make more sense to put almost anything higher. That's bizarre.

4.96 People have done empirical studies about this. What do you do? What do you declare? It varies country by country, which I think is really quite interesting. For example, in the Netherlands, most people say, "Oh, 300." Even more say that in Israel, and in the UK, even more than that. In Canada, it's about the same as the Netherlands. In the US, only about half the people say 300. A significant number actually say 180, and even more say 180 in Canada. Then in Thailand, actually, a minority say 300, and a significant number, 30%, say 180, and actually give you the Nash equilibrium.

4.97 So it's surprising. The average declaration varies country to country. It's highest, actually, in the UK, \$281. It's lowest in Thailand, \$250. But I find it interesting that, in different cultures, people respond differently to this problem. In some cases, almost everybody declares the maximum value or very close to it. In other cases, a significant number of people actually do choose the minimum, the Nash equilibrium.

4.98 Here's a related game. It's called the ultimatum game. It's interesting because here, too, there's been a lot of empirical research. What do people actually do? So, I say to you, "I'm going to give you \$100. You and somebody you don't know, somebody chosen at random, can share this \$100." It's up to you to divide it between the two of

you. You can take a certain amount from the \$100 and offer them a certain amount, but they get a chance to either accept your proposal or reject it. If that person rejects the proposal, neither of you gets anything. So what sum do you offer the other player?

- 4.99** You've got \$100, but now there's somebody else who is going to have the chance to accept or reject, and you get to offer them a certain amount, hoping they will accept it. So what do you do? It's not obvious, but here, too, there's been a lot of study, and the most common answer is, let's split it 50-50, which I find interesting. That's actually most common in the US. It's a bit less common, it's just under 50% in some countries, just barely over in some other countries. In the US, it's almost 60% who will just say, "Let's split it evenly."
- 4.100** What is the Nash equilibrium here? It's actually to say, look, you're better off if I offer you even \$1. Why don't I just keep \$99 and offer you \$1? You're still better off with \$1. But here's what happens if you do that. A significant number of people will say, "Forget it. I'm not going to accept this." Why? Because it seems unfair. In part, it seems unfair. In part, it's like, "I don't want to encourage this kind of behavior where you are given a chance to split this with me, and you're only willing to give me \$1. Yes, I'm \$1 better off. But actually, I feel belittled by that insulting proposal," and so they will turn it down.
- 4.101** What the theory predicts is, look, but you're better off. Your best response is to accept whatever you're offered. If I know that, then my best response is to offer you as little as possible. And yet, that's not what people do. There is a certain minority of people who will do that. For example, in the Netherlands, to offer you either nothing at all or \$1, that's about 15% of the people who respond. That's interesting. In the United States, it's less than 10%. It's even fewer than that in Thailand. In some countries, a significant number of people will actually offer the person a dollar, say. But in most countries, by far, the most common response is 50%.
- 4.102** Interestingly, there are a few people who say, "Just take it all," which is rather bizarre, actually, from a game-theoretic point of view; then you don't benefit at all—except, I suppose, you benefit by feeling good about helping somebody else. In any case, what's weird is, what would you do if you were on the other side? We've looked at it from a point of view of what would you offer, but now, you're on the other side, what would you accept?
- 4.103** Suppose the person offers you \$10, and they're keeping \$90. Do you say, "At least I have \$10, better off"? If I say no, neither one of us gets anything. So either I can have \$10 or not have \$10. I'm better off with the \$10. That would seem to be my best response. On the other hand, a significant number of people will say, "No, that's really unfair."
- 4.104** Here's a more extreme game. This is called the dictator game. In this case, again, you and a person you don't know can share \$100, but you're the total dictator. This time, they take whatever you give them. They have no choice at all. They cannot turn it

down. And so, you're just the dictator. You split it however you want. What do you choose?

- 4.105** In this case, you don't have to give them anything. They have no power to reject or accept this proposal. What would you do? It turns out, again, that a significant number of people will offer them at least something, typically less than if they have the power to accept or reject. But still, a lot of people, it turns out, will say, "Let's split it 50/50," which suggests that people are not purely utility maximizers. From game theory's point of view, people will say, "But there's this notion of fairness that I have to think about. It would be just unfair to keep the whole \$100 for myself."
- 4.106** By the way, why might you do that? For a person you don't know, it's like, they're not even going to know who you are, so why would it matter? But there are scenarios where it's a bit like this in real life. Let's say the boss is given \$100,000 to distribute to his employees in this division of the company as bonuses. The boss says, "The best employee in this division is me, so I'm going to keep the whole \$100,000." What happens? And this, by the way, there's a real-life scenario I have in mind here. What happens when the employees find out? They're going to be furious, right?
- 4.107** Whenever there's a situation where you're realizing this, you're thinking, but there are real people here. Yes, they have no power. But do I want to be that unfair with them? Do I want to anger them and alienate them that much? I think it's easy to think, "No, I don't want to do that." Even in this scenario where somebody has no power in this situation—you don't even know who they are, they don't know who you are—nevertheless, we have this pretty strong sense that it would be unfair. It would be a cruel thing to just keep it all myself. Relatively few people actually do keep it all to themselves.
- 4.108** There is one last kind of situation. This has a lot of ramifications, actually, of a competitive or near-competitive situation. Harold Hotelling devised what he called Hotelling's law. It's a principle of minimum differentiation. The simplest case is where we have two shops or two hotels. That's kind of a nice way of thinking about this and remembering his name. Imagine that there are these two hotels or two shops or two whatever it is, that are in different parts of town.
- 4.109** The thesis is, they are going to tend to end up next to each other. Why does it so often happen that, in the town, there are these two gas stations and they're right opposite each other on the same corner? Or there are two banks and they're right there together. There are two ice cream shops and they're very close together and so on. Hotelling said, look, there is a rational reason. That's the Nash equilibrium.
- 4.110** So suppose you've got shops on opposite ends of town, and they're relatively similar shops. They sell whatever it is, or they are hotels that are very similar. Who's going to go to this one? The person for whom that's most convenient, that's closest to. I am interested in maximizing the number of people who think I'm the closer one. If I'm over

here and the other person is over there, my incentive is to say, "These people are closer to me. Those are closer to him. If I move in this direction, more are closer to me, fewer are closer to him."

- 4.111** So I can do better if I move toward the center, move toward the position of the other person. But that person says, "Gee, I'm getting pushed off into this little corner of town. No, I should be moving toward the center, too." The idea is that each shop does better if they move toward the center, so that they get more of the people getting to them first. So both apply this logic, both end up right next to each other in the middle.
- 4.112** In one sense, that's entirely rational. Each one says, "I can be more convenient for more people by heading toward the center." But wait a minute. If you start thinking, "But how long does it take the average person to get to one of these shops?", actually, this is not the social optimum. It'd be much better to have them over here, so that on average people would have to drive less distance to get to the shops. However, from the shop's point of view, it makes more sense to be right next to each other.
- 4.113** So, that's true, at least if these shops are very similar. But what if they're not? You might say, "This end of town is a bit different. This end of town has people who prefer blah, blah, blah. I'm going to specialize in that. If you want my kind of thing, you have to come over here." If, on the other hand, you prefer this kind of thing, you have to go over here. So in many cities, for example, there is a motor mile where a lot of car dealerships gather. But often in that motor mile, you find the ones you'd expect to find. You find General Motors, you find Ford, you find Toyota maybe, and Kia, and a variety of others.
- 4.114** But where's the Porsche dealer? Almost never right there. The Porsche dealer is way over there. Why? Because look, it's not like you're thinking, "Am I going to buy a Kia or a Porsche?" They're different markets, and so they don't have to be close. In fact, to some extent, they want to differentiate. "We're not among that group. We're over here." Hotelling said, look, the result of this lack of differentiation really is not only a question of physical location. There's going to be a question of just making things similar to appeal to more people.
- 4.115** An actual quote from him is, here's an example. "Methodist and Presbyterian churches are too much alike; cider too homogeneous." Things get too similar to other things. That, by the way, is the church I go to, that's pictured there. If I just showed you that, what kind of church is it? I don't know. It looks just like a generic Protestant church. As it happens, it's Presbyterian, but you can't really tell that. There's almost no differentiation.
- 4.116** I want to close by saying there's a really interesting application of this in what's called the spatial model of voting. The same type of thing can happen in political elections. So, voters are going to decide for everyone, and each voter bears only a part of the cost of whatever they vote for, but also only a part of the benefit. It's not the same as a

marketplace exactly. Also, there's a vast amount that voters don't know. If I am buying a television set, I can research and find out pretty much the characteristics of the average set of this kind. But on the other hand, if I'm voting for somebody, especially a new person who hasn't been in office before, I'm kind of guessing about what they're going to do.

- 4.117** I have shortcuts. I can say, "What's this person's ideology? What party do they belong to?" and so on, but still, that's only an approximation. The equivalent of Hotelling's law in politics is called the median voter theorem. It says if voters and candidates are distributed on a one-dimensional spectrum, like those shops, and if voters have single-peaked preferences, which is analogous to, "I'm at this location, so what's most convenient for me? What's the shop I get to first?", it's like, ideologically, I'm here, so what candidate do I get to first, then any voting method compatible with majority rule is going to elect the candidate preferred by the median voter.
- 4.118** People are going to have a tendency, just like the shops heading toward the middle of town, candidates are going to head toward the middle of the political spectrum and try to appeal in that way to the most voters. It's easy to see why it's really the same principle as with the shops. It works in a lot of cases. It works if they're on one dimension. It works even if there are multi-dimensions of the right kind, and the candidates are distributed in the right way, and so are the voters.
- 4.119** But you might think, here's a way we could generalize it. Think of being in this space where we care about a variety of things. Here I am in this multidimensional space. Think of a distance function that says, how far away is that candidate from my position? We could measure things in those terms, too, and talk about the median voter, even if things are not one-dimensional.
- 4.120** So, combine this with majority or plurality voting, where whoever gets the most votes wins, it naturally produces a two-party system. There's a lot that this can explain. This is why candidates start looking alike, why people could say the parties seem too similar. This is why we end up with two parties in the first place. It's all explained by the Nash equilibrium; that's the way things tend to go.
- 4.121** If this works, and in many historical periods, it is like this, what happens? Moderate policies tend to win. Moderate politicians win. There's a general dissatisfaction because most people aren't right there in the center. Like everybody, "Aw, I've got to drive downtown to get this stuff." The analog here is, "I really would rather have somebody who's more liberal or more conservative than this guy, but okay." There's very little effect on the extremes. It doesn't really matter.
- 4.122** But on the other hand, you might think a lot of this is because it's just a question of the number of votes. It's not the intensity of preferences. Moreover, there isn't always a median voter. First of all, this doesn't always work. There isn't always a median voter. It turns out that, basically, voters have to be pretty symmetrically distributed. If they're

not, for example, if some voters are way over here and others are way over there, then it's not like there's anybody in the middle to even vote for. And so, it won't work in some scenarios. There just isn't anybody in the middle.

- 4.123** Also, if the extremes stay home, it wouldn't really make any effect. But if we can manipulate turnout, then we can change this. Then there won't be a median voter. If the media is strongly biased in one direction or another, there won't be a median voter because that voter will be swayed in a certain direction. As if everybody in the town produced weird maps in such a way that everybody thought that shop was really far away, even though it wasn't, or there was a weird malfunction in GPS things so that that shop didn't even appear. Indeed, it does look like the media is like that in the United States—nothing compared to college faculties that are much worse in terms of their bias.
- 4.124** But if you look at this historically, you can look at the pattern of US elections, for example, since the Second World War. And notice, in the 1940s and 1950s, there's a cloud, and one end is blue and the other end is red, but it's one cloud. The median voter theorem was actually working really well, and it kept working well until the late 1980s and especially the '90s, when all of a sudden there tends to be a separation. And now it doesn't look like one bubble with two ends. It looks like two different communities altogether. It's almost as if the town is no longer one town, it's two towns. It used to be one thing, and now it's Dallas and Fort Worth, and they're really different, and there isn't much in between.
- 4.125** That political polarization is something that can't be explained, in a sense, by the median voter theorem. Something is different, but what? Maybe voters simply aren't distributed in the way they used to be, so there's no median voter. Maybe there is such bias that it skews things in a certain direction. Maybe it turns out that we're just evolving into two different communities, and that could happen.
- 4.126** It turns out, most votes in Congress can be explained on one dimension. Most of the time, this principle of minimum differentiation and the median voter theorem, the political equivalent, really do work on a lot of issues. But if there are many dimensions, maybe not. Think about also the price differentiation, the product differentiation. Maybe it's like that.
- 4.127** How many people would think, "I don't know whether to go to Maria Maggiore and have a Catholic Mass, or go and listen to some great Christian rock at Grace Covenant." Two different audiences. Similarly, there are many people who say, "Should I buy this little mini car, the least expensive car on the planet, or should I buy the most expensive car on the planet?" Two different groups of people. And so, maybe the country is simply coming apart. Maybe temporarily, as these historical graphs indicate.
- 4.128** Sometimes it looks like one community, and sometimes it looks like two different communities. In fact, the historical norm is actually more the two different

communities. It was kind of an artifact of the World Wars and the Cold War, that we all came together as one, and this principle of minimum differentiation really worked in politics.

- 4.129** Anyway, I leave that as a question. When there are times of political polarization, this median voter theorem doesn't seem to work. All of a sudden, things aren't tending toward what you'd think is a Nash equilibrium. Why not? What else is happening? It's not like a situation like, for example, the ultimatum game, where you say, "The Nash equilibrium is obviously stupid. Don't do that."
- 4.130** In this case, it's not like that at all. Something else is going on to disrupt what would otherwise be the expected pattern. What is it? I have no idea, but I leave you with the question, and it seems to me that thinking about things in these terms is actually really helpful for helping us understand the nature of the puzzle.

Lecture 5: Mixed Strategies

At a glance — Mixed strategies.

- 5.1** I'm going to be talking to you now about mixed strategies. We've referred to them a number of different times, and now I'm going to try to teach you how to actually compute them. I know I said at the beginning, I was going to try to do this in a relatively unmathematical way. After this, you may be saying, "I was told there would be no math." There's going to be a little math. On the other hand, it's entirely Algebra 1 type math. It's very, very straightforward. I'm hoping it will not be too intimidating, also hoping it will not be too boring, because it could be that there are dangers in both directions.
- 5.2** In any case, I want to go back to something we said near the beginning. If we're modeling people's strategies for playing various games, we can say, what is a strategy? It's a rule for determining how I'm going to play the game, what my moves are going to be. A pure strategy will specify actions, things like cooperate or defect or move toward the center of town, or offer less money, or whatever it happens to be, use bad money, and so on. The strategy will tell you what to do.
- 5.3** A mixed strategy, however, is one that is not going to tell you, no matter what, do this. It's going to say, do this with a certain probability. Do that with a certain probability. It's going to give you probabilities over actions, not just simple actions. Think about something like rock, paper, scissors. You're not going to say, "Oh, always play rock." Instead, you're going to say, "Play rock one-third of the time." That's what a mixed strategy is going to look like. With what probability will I do this? With what probability will I do that?
- 5.4** The pure strategy, to repeat, is something that's going to provide a complete definition of how you play the game. In this situation, do this; in this situation, do that. You're going to have the same response throughout in that situation. But a strategy set will be just the set of available pure strategies, and when we think about a mixed strategy, we're realizing, then I'm going to take the ones there that at least I might play in this scenario and assign them probabilities. It might be paper one-third, rock one-third, scissors one-third. That way, I'll say, those are my possible actions. I assign them each a probability.
- 5.5** In some larger context, it might be zero. For example, first play of the football game. One thing we could do is a Hail Mary, but in the very first play, we're not going to do a Hail Mary, so that just gets probability zero; we kind of leave that out. But later, that might start having a significant probability. And yet, I don't want it to become so high that the defense can just say, "Oh, I know what's coming, a Hail Mary," although there might be scenarios like that. We've got three seconds left and we're on our own 30. We're going to throw a Hail Mary. There's nothing much else to do.

- 5.6 Anyway, a mixed strategy assigns a probability to each of the pure strategies, and that allows us to move in different ways, even in the same situation. That's good. The idea here is to pick a pure strategy, but randomly, so that the other player doesn't know what I'm doing. That, in general, is the point of these. I want to keep the other player off balance.
- 5.7 That's why I start talking about this now, after we've talked about competitive games, because in a non-competitive game where we're trying to cooperate or trying to coordinate, it's important that I know what you're doing. But if we're competing, it's pretty important that I not know what you're doing. If I know what you're doing, I can make my best response to it and gain an advantage over you. So, we're going to have a lot of these mixed strategies that are very useful in competitive contexts.
- 5.8 That is especially true anytime you think the essence of this is deception, maybe it's warfare, maybe it's football, maybe it's something else, but in any case, I want you not to know what I'm up to. In tennis, I don't want you to know where I'm going to hit that serve. I don't want you to know whether I'm going to hit it toward the back of the court or just barely over the net. Similarly, in football, I don't want you to know what play I've just called. If I'm on defense, then I don't want you to know whether I'm blitzing or not.
- 5.9 I want to be able to keep you guessing. The same thing is true in pitching, for example, in baseball; the pitcher does not want the batter to know what's coming. In all of these respects, there's an advantage to keeping the opposition unclear about what it is you're doing.
- 5.10 If you go through and think about board games that are commonplace, some of them, it's fine to play pure strategies, but most of them involve some element of deception, or at least keeping certain things hidden from the other player.
- 5.11 Do you want them to know what your destination is that you're trying to get your trains to go to? Do you want them to know actually how you're going to try to go about attacking their flag or doing whatever it is you're supposed to do in that game? In some simple children's games, it's not really a function of that. If you're playing Candy Land, there's not a lot of strategy, and you don't really have to worry much about probabilities. On the other hand, in most games for adults, or at least for older children even, there are going to be things where you have to mix it up, so that the opposition doesn't know what you're doing.
- 5.12 When we're trying to coordinate, as I've said, or when we're trying to cooperate, this is not very useful. There will be mixed-strategy solutions in many cases, but they're almost always bad. However, when we're competing, they're often good. Think about what makes this desirable. In a pure-strategy Nash equilibrium, each player's choice is designed to be a best response to what the other player is doing.
- 5.13 But in a mixed-strategy Nash equilibrium, I don't actually know what the other player is doing necessarily, and so at least one player, and often both, are randomizing. They're

putting a positive probability on more than one strategy. There isn't just one thing that's a best response. What's the best play in rock, paper, scissors? There is no best play. I've got to think, what is the best mix of possible plays in this scenario? It's not so simple to say, "Ah, my best response." In this case, what do I do to figure out the best way of arranging the probabilities?

- 5.14** I want my various pure strategies in the mix to give exactly the same expected payoff against the opponent. That at first glance might seem strange, but I don't want you to know what I'm doing. I want you to think—and again, we're in a game of perfect information, so you know what my payoffs are; you know what my various strategies are, possibilities are—and you're thinking, what's he going to do? Is he going to hit it forehand or backhand? Is the defense going to blitz or not? If I'm defense, I want it to seem as if you have no idea. You're not sure which I'm going to do.
- 5.15** Suppose, in this situation, it seems clear. "Oh, you should blitz." Then you can easily try to defend against that, run the best play against it, maybe a screen pass or something like that. You know you're likely to do this. I can respond. But if I can keep you completely uncertain about what I'm doing, then you don't have an advantage like that. You can't say, "Oh, well, then screen pass, or he's not going to blitz. So, instead, let's throw it longer down the field." I don't want you to have any guess about what I'm doing. How can I keep you completely uncertain?
- 5.16** I want it to look like I am completely indifferent among my options and arrange the probabilities in such a way that the payoffs are the same, so you have no idea. Suppose one of these options actually paid off better for me, then I could just do that, and I would shift all probability to it. For example, suppose I am a football team and I know my passing game is much stronger than my running game. Intuitively, you might say, "Well, then, pass." But if the defense knows, "Oh, you're just going to run a pass play every time or almost every time," they know just what to do against me.
- 5.17** So, it's better for me to mix it up and to say, "I want it to be unclear to you what I'm going to do here, because I want the payoffs to look equal from your point of view. I don't want you to know whether I'm passing or running on this down." Behind this is something known as the "indifference principle." The equilibrium probabilities are the ones that make the opponent indifferent. So both players are willing to mix. I don't want you to know enough about what I'm going to do that you can play in a way that advantages you and disadvantages me.
- 5.18** I want to make you indifferent. I want you to say, "Look, I have no idea whether he's going to run or pass, so I don't know which defense to use." So I'm going to start by finding pure-strategy Nash equilibria, if there are any, and then I'm going to look for mixed ones. This will be especially useful where there are no pure-strategy equilibria defined. Then I'll have to find these mixed ones. The thought is, if my mixed strategy is a best response, then all the pure strategies in support have to be best responses and payoff equal. If one had a lower one, I'd just drop it out and go with the others.

- 5.19** The thought is, actually, as I'm describing it now, abstract, and it may be like, "I'm not quite following." But if you think of an example, it becomes really clear what I'm talking about. Because what I want is to say, "Look, I want you to be completely unclear about what I'm doing." So, think about that simple game, two-finger morra. I can put out one or I can put out two. Same for you. One of us calls even or odd. Then if the total ends up being even, even wins. If it's odd, the odd person, the person who's called odd, wins.
- 5.20** In that game, what do I want to do? What should my mix be? My two options are play one, play two. Suppose I think two is better than one, so I just always play two. You can always beat me, right? Suppose you call odd, then you know I will always play one. You will always win. So, if you know what I'm about to do, I lose. Consequently, I think, "Well, what am I going to do?" I've got two options. The payoffs are the same. It doesn't matter. It's not like I get double points for winning with two. I just say, it's 50-50. In other words, I want to make you indifferent.
- 5.21** I want you to think, "I have no idea what he's about to do." You're going to say, "So should I put out one finger or two finger to win? I'm indifferent. I have equal probabilities for both. I don't know." That's my goal, to make you unclear what you ought to do. The moment I start tending to do one thing or another, suppose you observe my play for a while and you start thinking, "He puts out one finger twice as often as he puts out two fingers," you can then beat me.
- 5.22** You can, at least over the longer term, say, "Okay, great, I've called odd. He tends to put out one finger, so I know I'm going to tend to put out two fingers, and I'm going to win a significant amount of the time." Consequently, I don't want my tendencies to be known by you. If I do have a tendency to go in one direction, I want to counteract it in such a way that you're unclear about what I'm doing.
- 5.23** The same thing is true in rock, paper, scissors. I want to do each of those one-third of the time. Why? Because the payoffs are equal, and I want to keep you completely uncertain about what I'm going to do. You're thinking, "Should I play rock, paper, or scissors?" I want you to be indifferent, in the sense that they're all equal for you. If you get a sense he tends to play scissors more often, then you have a way of gaining an advantage by saying, "Then I'm going to play rock more often." I don't want you to have any awareness of me having that tendency. I want to keep all of them indifferent from your point of view.
- 5.24** So, that's the key idea of the indifference principle. My goal in playing rock, paper, scissors is to do each one a third of the time, so you have no advantage in choosing one over the other. Given a scenario, it could be any of the three with an equal probability. So, your best response is just to play one of those with equal probability, too. You can't play one more often and gain any advantage over me. In fact, if you start playing one more often, then I am going to gain an advantage over you.

- 5.25 In more complicated games, it's the same thing. Rock, paper, scissors, lizard, Spock. If you think, "Spock's smart. I'm always playing Spock," then it's easy for me to defeat you. You can't allow those tendencies to become often. Nobody's going to play Spock all the time. But if, on the other hand, I start noticing, "Oh, you like Spock, you're going to play that more often," then I can gain an advantage over you. Lizard poisoned Spock, so I can start playing lizard more often.
- 5.26 If we go back to Nash's 1951 paper, we find a competitive game there, which is one that has mixed-strategy equilibria. It doesn't have a pure-strategy equilibrium. It's interesting to take that as an example and say he was already there, giving us the technique for solving for mixed strategies and understanding the nature of such games. I'm going to take his, which is just an abstract thing with numbers, and make it a little more poetic.
- 5.27 I am Hannibal, thinking about attacking Rome. I've got my elephants. I'm ready to go. What are my options? I'm thinking, I can go by sea and just sail directly from Carthage toward Rome, or I can go over the mountains. I can take my elephants over the Alps. That seems like a crazy thing, a very difficult thing to do. On the other hand, the Romans are not going to be expecting it.
- 5.28 If I'm a Roman general, I think, what can I do? I can keep my army south of Rome, to try to defend Rome against a sea attack coming up from the south, but I could also move them north, in case he lands in northern Italy or in case he tries to go over the mountains. What do I do? I don't know whether Hannibal is going to actually go by sea, the direct route, or take this indirect route over the Alps. If I'm Hannibal, I don't know whether the Roman army is going to be waiting there in the south or whether it's going to move north, anticipating that I might actually try to go over the mountains.
- 5.29 I think that means I have to mix up what I'm doing. I can't just say, "Here is the best response to whatever they do." In a certain sense, I can figure it out. If they're in the north, then I should just go directly by sea. That's my best response. If they're in the south, then I'd be better off going over the mountains. But on the other hand, I don't know which they're going to do. It's the same for the Romans. There is no pure-strategy Nash equilibrium here. But I can think, "Well, what should I do? With what probabilities?"
- 5.30 Of course, it's only going to happen once. I'm not going to say, "Well, I'm just going to keep trying and trying and trying." To some extent, it can look a little silly if it's a one-time thing, as opposed to a continuing thing like a tennis match or a football game. But in this case, I think, what is the likelihood of them having troops in the north? What's the likelihood of them having troops in the south? The Romans can think the same way about me. They can think, "What's the likelihood he'll come by sea? What's the likelihood that he'll come over the mountains?"

- 5.31 We might think of it this way. Suppose—and here I'm just assigning arbitrary values. They happen to match the values in Nash's example, but they're pretty plausible in this case, in a general way—suppose I, as Hannibal, attack by sea. That's easier. I'm not going to lose any elephants going over the mountain. Pretty straight shot from Carthage, North Africa, up to Rome. I can do that.
- 5.32 If he's got his armies in the north, Rome is ripe for the taking. Let me say 5 is the value I assign to that one, and for the Romans, minus 3. They're in trouble. I'm going to be able to move into Rome. Their army can still move from the north, back down south, and counterattack, but I've got a huge advantage over them.
- 5.33 On the other hand, suppose their army is already in the south. Then they've got the advantage because I'm going to be trying to land near Naples, say, and they've already got positions on the shore—that's going to be really difficult. Let's say that's minus 4 for me, 4 for them. They're going to gain a significant advantage over me. I think, as Hannibal, maybe I could take my army over the mountains, come over the Alps. That'll be a big surprise.
- 5.34 On the other hand, if they're waiting for me in the north, that's really bad for me. Let's call that a minus 5, because now I've probably lost some troops, some elephants, going over the Alps, and now we come down off the Alps. This has been a difficult trip, and now we have to face the Roman army directly. That's really bad. Very good for the Romans. Call it plus 5 for them.
- 5.35 Suppose that army is down there in the south, waiting for us to come by sea, but we've gone over the mountains instead. Then, that's good for us. We gain an advantage, but it's less than the advantage we'd gain by sea if they're not there, because after all, we've got to go over the Alps. That's really difficult. We're going to be tired. We're going to lose some soldiers. We're going to lose some elephants. Let's call that a plus 3 and a minus 4 for the Romans.
- 5.36 Now I start thinking, what are the probabilities that I could expect here? I think, what is the expected utility of my attacking by sea? I don't know whether their army is going to be in the north or the south. Let's say they're in the north, with probability q , and then in the south with probability $1-q$. There's some likelihood they'll be up in the north expecting me to go over the mountains. I think that's probably low, but I don't know. Let's call it q .
- 5.37 Then what is my payoff if I attack by sea? It's $q \cdot 5$. That is to say, 5 is the value of me going by sea and catching their army up in the north. I get a big advantage, so it's $5 \cdot q$. On the other hand, if their army is in the south, probability of that is $1-q$, then it's -4 for me. So, my expected utility of going by sea is $5q + (-4)(1-q)$. What does that mean? For the moment, it means nothing.
- 5.38 But I then think, what is the expected utility of my going over the mountains, taking my troops over the Alps? Then I think, with probability q , they're going to be waiting for

me up in the north. That's really bad for me, $(-5) \cdot q$. But then, with probability $1-q$, they're going to be down south, and I've gained an advantage, so $3 \cdot (1-q)$. I add those up.

- 5.39** Now, here's the key indifference move. I set those two together and equate them. I say, I want it to be looking as if, from their point of view, it's indifferent what they do. So I want the expected utility of me going by sea to be equal to the expected utility of me going over the mountains, so I set those quantities equal. One of them simplifies to $9q-4$, the other to $3-8q$. I set those two things equal and solve for q . In this case, that just gives me $17q = 7$, or in other words, $q = \frac{7}{17}$.
- 5.40** I think, what's the probability then that they're going to have the army in the north? Basically, it would be rational for them, knowing what my payoffs are, to do that 7 out of 17 times. The army will be in the south, then 10 out of 17 times. That is what would be rational for them. The Romans can think about this as well, and they can think, we want the expected utility of our positioning of the armies in the north to be equal to the expected utility of putting them in the south. What is the utility for us of putting them in the north?
- 5.41** Now I think, I don't know what Hannibal is going to do, but suppose I put my armies in the north. Then, let's say he goes by sea, with probability p . Then my payoff, if that happens—I'm in the north, he comes by sea—that's -3 for me. Here it's $-3 \cdot p$. Then, what if he's going over the mountains? I'm in the north, ready and waiting for him, so that's going to be great for me. My payoff is 5, so it's $5 \cdot (1-p)$. What is my payoff here? $-3p + 5 \cdot (1-p)$.
- 5.42** Then suppose I put my troops in the south. What's the payoff there? If he comes by sea, then I'm ready for him. I'm ready to fight him. I'm right there in position. I've got the advantage. It's 4. So it's $4p$. But then what if he comes over the mountains? Then it's $-4 \cdot (1-p)$, the probability he's coming over the mountains. Again, I take those two quantities and set them equal. I think, "I want him to be indifferent," so I want it to look as if the expected utility of these are equal. I do that.
- 5.43** One simplifies to $5-8p$, the other simplifies to $8p-4$. I solve for p . $16p = 9$, so $p = \frac{9}{16}$. I think, "Aha." So I have to assume that if he's rational, he's going to attack by sea 9 out of 16 times, over the mountains, then 7 out of 16 times. So he's more likely to come by sea. On the other hand, Hannibal has done the calculation and thinks, the Romans are more likely to position their armies in the south.
- 5.44** You might think, if both think of it that way, they think, "What's the opponent probably going to do?" Probably what this is going to end up being, more likely than any other scenario, is that I come by sea, and he's got his armies there in the south. If I'm Hannibal, I think, "That's bad. I don't want that to happen because then I'm walking right into the trap." Yes, an easy journey from North Africa to Rome, from my point of

view, but he's got his armies in place ready to defeat me. I won't even be able to complete the landing. That's not good for me at all.

- 5.45 There's a probability, some chance he's going to have his armies in the north, but it's less of a chance. It's more likely they're going to be in the south. What if I go by mountain? From his point of view, he has no reason to expect me to go one way or the other. But if I'm thinking about it, I might think, if I have to place a bet here, it looks like more frequently he's going to have his army in the south. So, I'm going to gain maybe some advantage by going by mountain.
- 5.46 Hannibal does this. It does not work out well for him, as you know, historically. Nevertheless, you can think, that is what I would expect the opponent to do, so this is what is going to happen for me. If we say, we've been calculating all this and setting them equal and figuring out the probabilities, but what does that tell me about the expected utilities? Hannibal, what is the expected utility then of going by sea, or of going by the mountains, of attacking Rome altogether?
- 5.47 We put those things together. We said $q = \frac{7}{17}$. Plug that into the formulas and we get the expected utility, whichever way he does it, is $-\frac{5}{17}$. In other words, this does not work out well for Hannibal. We can already calculate that. We realize the expected utility of attacking Rome here is negative. That's a bad decision. Don't do it. There is another alternative for you, which is don't attack. Unless you have reason to think that's very bad, like worse than $-\frac{5}{17}$, then you shouldn't launch the attack in the first place.
- 5.48 Suppose you're Rome. You say, what is the expected utility then of having my armies in the north or in the south? Calculate, and the answer is one-half. The important part here is, it's positive, at least. It's positive for you. Rome has the advantage here. The expected utility of this war is higher for the Romans than it is for the Carthaginians. History bears that out. It turned out to be a bad idea to attack. The Romans had the advantage from the start, if those numbers are at all accurate.
- 5.49 It's hard to put numbers to this. What's the likelihood of winning? How badly will my army be defeated? I may estimate this. I can't know it for certain. But at least if I've got some kind of projection about what might happen in these various scenarios, I can do this calculation and think, is it worth attacking at all? Or as the Romans, I can think, should I put my armies in the north or in the south? What's my likelihood of winning? If you're a Roman general here, you're going to look at these numbers and say, "I feel pretty good about this." If you're Hannibal, you should say, "I feel pretty nervous about this. This is not a good idea."
- 5.50 Last time, I talked about this scenario in the office, where we have two people who are partially in competition, partly trying to cooperate to get things done. All sorts of mayhem ensues from that. Let's go back to that situation. We noticed there was no pure-strategy Nash equilibrium. Dwight wants cooperation and agreement, and he

doesn't want to be the one who messes things up. He wants to look at least responsible, responsible in the virtuous way. He doesn't want to be responsible for failure.

- 5.51** Neither does Michael. Michael wants to take advantage of his employee. But if something goes wrong, he wants to be able to blame it on Dwight. He doesn't want to take the blame himself. We got this sort of scenario, where we were just thinking, what's a first choice? What's a second choice? Just from that ordinal information about preferences, we can't compute anything about a mixed strategy, because we have to know values. In order to get these probabilities, we need to not just know, "Oh, this is my favorite option," but how much more valuable is it to you than the next option?
- 5.52** We need cardinal numbers. So, let's just assign some cardinal numbers that would go along with those preferences. I've chosen these more or less at random, but suppose that the value of this, mutual cooperation, Dwight gains 10, Michael gains 5. If Dwight cooperates and Michael defects, then Dwight gets 3, Michael gets 10. If Dwight defects, but Michael does the work, he cooperates, it's 0 for Dwight, but 4 for Michael. Then if they both defect, it's 5 for Dwight and 0 for Michael. That would correspond to the preferences that we had before. It's just an estimate of what the intensities might be.
- 5.53** Now let's do the same type of calculation. Again, a little bit of boring elementary algebra, but the idea is, Dwight says, "I'm going to cooperate with probability p ," and Michael cooperates with probability q , and now we begin to calculate. We look at it first from Dwight's point of view, and we think, "What is the expected utility of my cooperating, by really trying to do the work here?" If he cooperates with probability q , it's 10. So $10 \cdot q$, plus then I get 3 if he defects, so $3 \cdot (1-q)$.
- 5.54** What if I defect? Then it's $0 \cdot q$, because if I'm not doing any work, I get no credit at all. On the other hand, if neither one of us does something, then I can at least claim some and say it's not my fault, so 5, $5 \cdot (1-q)$. Now I set them equal. I think—and here's the part that's a little bit perverse—I don't want the boss to have any idea what I'm going to do. That's not a great strategy for getting ahead in your career. But in this case, if you view it as a competition, then you could say, I don't want him to know what I'm going to do. I want to set those two expected utilities equal.
- 5.55** That means $10q + 3 \cdot (1-q)$ is going to be set equal to $5 \cdot (1-q)$. Simplifies to $7q + 3 = 5 - 5q$. So, $12q = 2$, so $q = \frac{1}{6}$. In other words, how often am I going to do the work? Only 1 time out of 6. I'm not going to let him know for sure, but odds are, I'm going to defect. I'm going to slack off, because I figure, I don't have a lot to gain, but I don't have that much to lose either.
- 5.56** The other way, glancing at it, it looks like—wait, wait, wait—but still, q . I'm thinking, what are the odds here that Michael is really going to get involved, really going to do much work? Really, not very high. I tend to assume he's not going to help me. What is

the expected utility I get out of all this? I plug $\frac{1}{6}$ in, I get 4 and $\frac{1}{6}$. That's what I expect for my utility.

- 5.57** Let's look at it from Michael's point of view, the boss's point of view. He's thinking, suppose I cooperate. My payoff is $5 \cdot p$ —the probability that Dwight also cooperates—+ $4 \cdot (1-p)$, the odds that he doesn't. Then what is the expected utility of my defecting, my goofing off and not doing it? It's $10 \cdot p + 0$. I set those two equal. I don't want Dwight to know how I'm going to respond. When I do that, I get that $5p + 4 \cdot (1-p) = 10p$, so $p+4 = 10p$. In other words, $9p = 4$, so $p = \frac{4}{9}$. I plug that in. What is my expected utility? 4 and $\frac{4}{9}$, just a little bit higher than Dwight's, notice.
- 5.58** I figured the odds of Dwight doing the work are almost $\frac{1}{2}$, not quite. On the other hand, the odds of Michael doing the work are only one out of six. You think, here's a mixed-strategy solution, and it suggests that the payoffs for the two are 4 and $\frac{1}{6}$, and 4 and $\frac{4}{9}$. If I think about that, I realize, how good is that as an outcome? Actually, they could both do better by jointly cooperating. It's not a Nash equilibrium, but on the other hand, it is better for both of them.
- 5.59** You could say, wait, this is clearly Pareto inferior to another option. Guys both do this. Both get involved. Instead, one of you is almost certain not to actually do any work, and the other is actually more likely not to than to do it. That's a pretty bad scenario. You'd be better off cooperating and doing the work. Yes, they would.
- 5.60** On the other hand, why don't they then just do it? After all, that would make more sense. They'd both be better off. The problem is, each one has an incentive not to do it, because it's too easy then for Michael to say, "But look, I can double my payoff by not doing it." He's got an incentive to defect.
- 5.61** Once you figure in the probabilities, neither one has an incentive to defect away from that, but on the other hand, it's also inferior. So it's stable, but stably inferior. You can do better, but it's highly unstable, highly vulnerable, to somebody saying, "But I'm going to defect. I'm going to not do it anymore." Cooperate-cooperate is clearly Pareto superior, but it's highly unstable. That's why both have an incentive to let the other not know what they're really up to.
- 5.62** One game we talked about that's partly a coordination game, but partly also a competitive game, is the battle of the sexes. Let's go back to that. It was a case where we had two Nash equilibria. Basically, he and she want to do something together, but they have different preferences about what to do. There are two successful, stable Nash equilibria. They both go to the opera or they both go to the football match. On the other hand, in each case, neither one can unilaterally improve. On the other hand, it feels like one wins. "Hey, we're doing my thing this time." The other comes in second. "Okay, I'll go along with you, because I care more about being with you than actually seeing the opera or seeing the football match."

- 5.63** We had that combination of preferences. The first choice of each is to do their favorite thing with the other person, but that's the second choice of the other person. It's far inferior for them to do their own separate thing, but even worse than that, that they do what the other would like. The person who loves opera goes to the football match, or the person who loves football goes to the opera. We think, if those are the preferences, then we need some cardinal numbers to actually compute probabilities and determine what a mixed strategy would be.
- 5.64** Let's assume that—to just simplify things a bit—if they each do their own thing, the payoff is just 0. Actually, the way I've set it up so that each doing their own thing is better than each doing the other's favorite thing isn't reflected by this. But on the other hand, it's going to make the math way easier. Let's just, for the moment, forget that and assume, that they are unhappy, they don't get any pleasure out of doing the things apart.
- 5.65** Let's say one of them gets 2 points, and the other gets 1 point of pleasure or value or something, out of going to the opera jointly. Then if they both go to the football match, it's just reversed. One of them gets 1 unit of value, the other gets 2 because that's their favorite thing. Then we can again assign probabilities and say there is going to be a mixed-strategy solution. We know there's always an odd number of Nash equilibria. Here are two. There's got to be a third at least.
- 5.66** We look and we say, suppose blue here prefers going to the opera. Blue will go to the opera, let's say, with probability p ; the football match, $1-p$. Then pink will go to the opera with probability q ; football match, $1-q$. We think, what should these probabilities be to give us a mixed-strategy solution? So, we do a little math. We think, what is the expected utility for blue of going to the opera?
- 5.67** The probability is q that the other will be there, and so $2q$, but then they get no benefit at all if the other isn't there, let's assume, it's just $2q$. Then what's the expected utility of going to the football match? It's $0 \cdot q$, because if they go without the other one, they won't enjoy it at all. Then it's $1 \cdot (1-q)$. In this case, it's pretty easy. We just identify those two. $2q$, we set that equal to $1-q$.
- 5.68** What will that tell us? That means $3q = 1$, or in other words, $q = \frac{1}{3}$. So, player two has to play opera with probability $\frac{1}{3}$ to make player one indifferent. So, pink in this case should go to the opera one-third of the time; the football match, two-thirds of the time. What about from the other player's point of view? Then, pink's payoff is going to be, let's see. Going to the opera, it's going to be $1 \cdot$ —the probability the other is also there—so p , and then 0 otherwise, so it's just p .
- 5.69** What is the payoff for going to the football match? It's going to be $2 \cdot (1-p)$. So, we set those two together and say, p is equal to $2 \cdot (1-p)$. It will turn out that $p = \frac{2}{3}$. So, player one has to play opera with the probability $\frac{2}{3}$ to make player two indifferent.

- 5.70** You can already see, this is weird, right? This is a very strange solution. "So, shall we go to the opera tonight, honey, or shall we go to the football match?" "Well, I'm not going to tell you where I'm going to be. You have to guess." That's a very odd relationship strategy. Anyway, if we combine them, we say, player two has got to play opera with probability $\frac{1}{3}$, and player one has to play opera with probability $\frac{2}{3}$, we think then, there's a Nash equilibrium solution for sure.
- 5.71** What's the payoff then? What are the expected utilities? Plug those into the formulas and we say, for the one player, it's then $\frac{2}{3}$. For the other player, it's $\frac{2}{3}$. Notice, that's better than the zeros. Nevertheless, it's worse than either going to the opera together or going to the football match together. It's an insane sort of solution. It is a Nash equilibrium. If we're doing that, that is stable in a way, but it's clearly kind of dumb. It's worse than either of the pure-strategy Nash equilibria.
- 5.72** If we think it through more carefully, we say, suppose they do those things with those probabilities, then two times out of nine, they'll both be at the opera. They'll enjoy it. Two times out of nine, they're both going to be at the football match. They'll enjoy it. But five-ninths of the time, they're going to be in different places. More than half the time, they'll be in different places, and they're not going to enjoy that at all. This is clearly a bad strategy.
- 5.73** It's like saying, "I'll meet you tonight, sweetie, for dinner. Guess where?" "Where?" "I won't tell you." This is not a good plan. Indeed, if we do this, we realize that's pretty bad. Payoff for each one is just $\frac{2}{3}$, worse than they could get either way. Suppose we say, but what's a really sensible solution? If a couple has different preferences like this, let's say we take turns. "All right, it's your turn. What do you want to do tonight?" "It's your turn. What do you want to do?" Then the expected payoff for each is. [1.5] Clearly, that's a better solution than doing this kind of mixed strategy.
- 5.74** Often, that will turn out to be the case. If there are pure-strategy solutions, they're usually better than mixed-strategy solutions. If we look at the payoffs as they're listed in the battle of the sexes version in Nash's paper, it looks a little bit different. He tries to take into account the fact that, it matters whether I'm at what I really like without you, or whether I'm at the thing I don't like without you.
- 5.75** If we think about it from that point of view, we could say, instead of just assigning zeros in those cases where we're in different places, let's say the one who is at their favorite thing only has -1 . Oh, I'm sorry to be doing this alone, but at least I kind of enjoy it. Whereas the other has -4 . We're in different places, and I'm not even enjoying it, so it's something like that.
- 5.76** If we have numbers that look like that and do the calculation, then it ends up being considerably worse. Actually, I said the expected utility in the case where we just assigned zeros, that was $\frac{2}{3}$ for each—pretty bad, worse than either of the pure-strategy equilibria. But in this case, it actually ends up negative. If it's like, "Oh, I really don't like

being apart from you," to varying degrees, then actually the expected utility just becomes negative, and it's easy to compute that.

- 5.77** If that's true, then they both end up worse off, not just worse off than they would be if they did one of the sensible things and took turns about where they go, but instead worse off than just doing nothing. They're going to end up someplace where they actually end up with negative utility. They're going to do that significantly. In fact, if we calculate the probabilities here, we realize almost half the time, they'll be together at, let's say, instead of talking about football and opera, we talk about seeing a dramatic movie as opposed to seeing a comedy.
- 5.78** Then maybe, almost half the time, they're at a comedy together. Maybe one time out of 10, roughly, they're going to be at a drama together. The rest of the time, almost half the time, they're going to be in different places and be unhappy. There's a mixed-strategy solution again, but as here I put it, quite bluntly, it's stupid. They're both going to end up unhappy as a result. Take turns instead. That is the general point of this. There's always some Nash equilibrium, often a mixed-strategy one. But if you've got any pure-strategy options, they're generally better than a mixed-strategy equilibrium.
- 5.79** So far, I've been talking about cases that I think bear similarity to real life. On the other hand, you might think, in this case, the mixed strategy isn't very interesting. It gives us a result—do this, with this probability—but in scenarios where it really seems stupid to actually keep the other person ignorant of what you're doing. After all, a battle of the sexes is partly a competition, but it's mostly a coordination problem, and you're sacrificing the coordination to try to gain some advantage in the competition, and you don't even do that very well.
- 5.80** Let's go back to our snowdrift problem. That is to say, the group project problem. We looked at that and we said there's a very strong likelihood that one person is going to end up doing all the work. After all, who is it going to be? Is it you doing all the work or is it me doing all the work? There is still this competition aspect in addition to some kind of cooperation or coordination aspect. It's not so bad if we're in a situation where we say, "Okay, what are you doing? Do you have time to shovel snow?" I say, "Yeah, sure." If we reach an agreement like that.
- 5.81** But otherwise, it might be, as in many group projects, each person saying, "I've really got a lot to do. I don't want to do more than I have to do on this." We get this problem where whoever cares most about the product is going to end up doing most of the work. We've got two Nash equilibria. But that suggests there's going to be another mixed-strategy Nash equilibrium. So, we can say, let's actually do the calculation. Let's put in some numbers to indicate values here, and then think about what this mixed-strategy equilibrium might look like.
- 5.82** I put in some numbers. I think if I am the one who gets to slack, I get 3 points. You doing the work, get 1 point, let's say. Or if I'm doing the work, I get 1, and you being the

slacker, get 3. If we work together, we each get 2. If we both slack, then we just get 0. The snow hasn't been cleared, or the group project hasn't been done at all. So, I start doing my calculation. I say, for me, the expected value of working is $2 \times$ the probability of you also working, so that's q , plus 1 times the probability that you'll slack off, so that's $1-q$, simplifying to $q+1$.

5.83 What is the expected utility of my slacking? It's 3 times whatever the probability is that you will actually do the job. That's $3q+0$ then. So, it's just $3q$. So I set those equal—again, the principle of indifference. I say it'd be rational to actually set these up so that you don't know what the other player is doing. So, $q+1 = 3q$. That means $2q = 1$, so $q = \frac{1}{2}$. In other words, it should be 50/50. Am I going to do the work or not? If these are the utilities, then I should do the work with probability $\frac{1}{2}$. What is the expected utility then for me? It's 1.5.

5.84 Notice, I could do better if I slack off and you do the work, or if we work on it together. But there is a danger that I slack off, and you do too, and we both get 0. At least I get 1.5 in general out of this. It's going to look the same from the other point of view. We do this half the time. Notice that's going to mean that one quarter of the time, the job doesn't get done at all. There's still a real danger of that outcome. But at least there's a three out of four chance that somebody's going to do the work. Maybe we do it together, maybe one or the other of us does it.

5.85 What if one person cares more than the other, so the payoffs aren't the same? Like, hey, I'm the one who actually is supposed to be organizing this. I'm the project leader here. I've got more depending on this than you do, as just one of the people who's supposed to be involved. In that case, it changes the probabilities a bit. Depending on how we set the numbers, it might vary them considerably or it might not.

5.86 What's weird about this is that the way this calculation goes, if A cares more than B, A still ends up working half the time and slacking off half the time. But the other person now, in order for them to appear, to make it seem indifferent, is going to have to do the work more often. They end up, given plausible scenarios where we put in additional numbers and give people an extra unit, or give A an extra unit for the work being done, it turns out that q ends up being 0.75.

5.87 For the other person to actually remain indifferent and to set it up this way, it will turn out that they actually have to be more likely to do the work. If the boss says, "Look, we've got to get this done," the boss may not change his behavior, but actually, the other people do have a stronger incentive to actually do the work, and the expected utility of working actually goes up significantly. I think that's kind of an interesting result. It may be that, if I care more, it doesn't really change my behavior so much as it changes the other person's behavior, if they're to keep things indifferent.

5.88 Let's talk a bit about stag hunts. What about that? This was another scenario where we had two Nash equilibria, one clearly better. It was better for us both to hunt the stag

than both to go off hunting hares. If we can trust each other, we can just say, "Yeah, let's hunt the stag." But suppose we can't entirely trust each other. Suppose I look and I say, "Oh, that guy, he's so distractible." It's like Rousseau says, "Yeah, can I really trust you, that if a rabbit goes by, you're not going to take off and forget about the stag and just hunt that rabbit for yourself?" No, I can't really trust you. People are too weak. They're going to go off and do that in the short term, for a short-term benefit.

- 5.89** What do I do? I can say, yes, my first choice is that we both hunt the stag. It's really my third choice and yours, that we both end up hunting rabbits. But it's really bad if you go off and hunt the rabbit while I'm hunting stag. I go home hungry, and you actually at least get a rabbit. I'm thinking about that. Again, I have to assign cardinal values. Just knowing preferences won't be enough to allow me to compute probabilities.
- 5.90** Let's say we each get 5 value if we hunt the stag together. We get 1 each if we hunt rabbits together. Then if it turns out you're hunting rabbits while I'm left there with the stag, I get nothing. On the other hand, you get 2 units of value because, after all, now you get to catch both rabbits. You're rabbit-hunting the whole time, and so you've got an advantage. You're not having to compete with me for hunting rabbits.
- 5.91** We think about those probabilities, and we say, fine, I'm going to hunt the stag with probability p , you're going to hunt the stag with probability q . How does it look? We can do the calculation this way. What's the expected utility of my hunting the stag? It's going to be $5 \cdot q$, the likelihood that you'll hunt it with me, then plus 0, otherwise, so it's just $5q$.
- 5.92** What's the expected value of my hunting rabbit? If you're hunting the stag and I'm hunting the rabbit, then I get 2, so it's $2q$. Then if we're both hunting rabbits, it's 1, so it's $1 \cdot (1-q)$. Simplify all that, and we get $q+1$. So $5q$, we set equal to $q+1$. Again, that principle of indifference. I want it to look like these expected values are the same. The result of that will be $4q = 1$, so q is $\frac{1}{4}$.
- 5.93** It's going to look the same from your point of view. The expected utility then is going to be just 1.25. It'll be slightly higher than just us both agreeing to hunt rabbits. On the other hand, what are we doing here? Each of us is hunting the stag one time out of four. So, one time out of four, we actually get the stag. But otherwise, it's three-fourths of the time you're hunting rabbits, three-fourths of the time I'm hunting rabbits. Nine-sixteenths of the time, we're both hunting rabbits, and we each get the one.
- 5.94** But the rest of the time, we're in bad shape. It turns out, we're out there, and actually, one of us is hunting rabbit, the other hunting stag. One person gets twice the rabbit, but the other person is left holding the bag with nothing. If we think about it this way, we realize, oh, wow. Actually, I said a moment ago, the probability of us together hunting the stag, it's really $\frac{1}{4} \cdot \frac{1}{4}$, so it's $\frac{1}{16}$. So, only 10 times out of 16 are we actually doing anything that's going to pay off for both of us. Consequently, we get a mixed Nash equilibrium here. We each get 1.25, but that's not very great. We'd again clearly

be better off hunting the stag all the time than having this mix. It does make us a little better off than jointly just hunting rabbits, but not much.

- 5.95** Now let's return to another competitive scenario. This is about the house, and this is going to be a sad story. Remember, we have the conformist who loves blue. She wants her house to be blue, but she really likes all the houses to be the same. She cares about that more than anything. The other is a nonconformist who says, "Well, I don't like blue, but look, what really matters to me is that they be different colors. I don't want my house to look like yours."
- 5.96** So, what happens? If we have just the preferences again, we can't do anything. But if we have utilities associated, then we can. We can assign those, let's say. For our conformist and our blue-lover, then 4 points, if her house is blue and so on. We just assign these values and we say, "Great." We do this kind of thing and say, "All right." In each case, the worst-case scenario for the person just gets 0 points, the best one gets 5 points, and then 2 and 1 for the second and third choices, respectively, and we actually start doing the calculation.
- 5.97** So, my house is going to be not blue, let's say, with probability p , blue, $1-p$, and the same for the other, not blue, q , and then blue, $1-q$, and we do a bit of calculation. I think the expected utility of my having my house some color other than blue is, what? Well, $2q + 1 \cdot (1-q)$, or in other words, $q+1$.
- 5.98** The expected utility of my painting my house blue is 0 if yours isn't, because I really want us to conform. But $5 \cdot (1-q)$, so it's just $5 - 5q$. I set those equal now, because I don't want you to know what I'm going to do, because I know you have opposite preferences, and you're going to do something just to spite me. So, I don't want you to know.
- 5.99** I set those equal, and that turns out to be $6q = 4$, so $q = 0.66$. In other words, even though I love blue, I'm going to think, "Hm, probably you're not—you know I like blue. You're probably not going to paint your house blue. You're going to paint it some other color. So I'm going to paint mine some other color two-thirds of the time, and only one-third of the time will I paint it blue." What's my expected utility then for doing either one of those? One and two-thirds. That's not quite the 2, but on the other hand, it's certainly not 5, it's not my favorite scenario, but it is a solution.
- 5.100** The other person, however, looks at this and says, "Well, what is the payoff for me if I go with something other than blue?" It's $1 \cdot p + 5 \cdot (1-p)$. That's $5 - 4p$. What is the expected payoff of me having blue? It's $2p + 0$, so $2p$. I set those equal: $2p = 5 - 4p$, so $6p = 5$, so the probability of going with not blue, p , is $\frac{5}{6}$. I realize, oh, that also assigns a probability, or an expected utility, of one and two-thirds.
- 5.101** Now notice what this actual equilibrium is. The Nash equilibrium is $q = \frac{2}{3}$, and $p = \frac{5}{6}$. In other words, the likelihood is that both are probably going to paint their house as something other than blue. For the one person, it's likely to be blue, probably two-

thirds, not blue. The probability of it being blue is actually only going to be $\frac{1}{3}$. For the other person, it's only going to be $\frac{1}{6}$.

- 5.102** So, actually, this poor woman who says, "I love blue. I want them both to be blue," what are the odds of her getting her way? One-third times one-sixth, only one-eighteenth. So she's going to be very sad. The odds of her actually getting her thing that she really wants, only 1 out of 18, not very high at all. There is a mixed-strategy equilibrium here, and it makes a kind of sense. Each is better off than they could otherwise be. On the other hand, neither one really gets what they truly want.
- 5.103** Think about a tennis match. Here's another scenario where we can talk about mixed strategies, and we can think, I might hit to your forehand, I might hit to your backhand, and so on. I think about probabilities here. Let's assume your forehand is stronger. I set these equal, and here I won't go through all of the algebra for you, but it turns out, I can calculate the likelihood that I should hit toward your forehand or hit toward your backhand.
- 5.104** I can compute this in such a way that I end up saying the likelihood of the forehand is, say, [0.3]. The expected utility in this case, 4.8. That's better than some scenarios, not as good as some others. The other person, the server, can think, "What am I going to do?" They can calculate this and similarly come up with the probability they should serve here as opposed to serve there. In the end, they have an expected utility.
- 5.105** What's interesting about this case is that, depending on the way the numbers go, we can actually calculate who has the advantage. As I've set up the numbers here, it turns out one person, the receiver, has an advantage or an expected utility of 4.8. The other, the server, 5.2. We can actually say, in this case, the server has a slight advantage. 52% of the time, they're going to be, as it were, having the advantage over the person who's trying to receive the serve. That tells us something interesting, I think.
- 5.106** Finally, one last case, and this is a little bit silly to think about as a game, but maybe not if you think about the O. Henry story, because—you're probably familiar with The Gift of the Magi. She has beautiful hair, and he thinks, "What should I get her for Christmas? It would be wonderful to have this nice expensive comb for her beautiful hair." She thinks, "He has this watch that he inherited that he values immensely. I want him to have a chain for this watch."
- 5.107** What does she do? She cuts her hair and sells it to be able to afford the chain for the watch. What does he do? He sells the watch so he can buy her this beautiful comb for her hair, and they both sacrifice the thing that they care about most for the other, but in a way that's completely useless. We know how the story turns out. What if they had adopted game theory and thought about a Nash equilibrium? Here's what happens.
- 5.108** She has the choice. Do I keep my hair or do I sell my hair? Let's assume that what happens in the story, it gives each 0 value. But if she keeps her hair and he sells the watch, then great. She ends up with her hair and the comb, so she gained some

advantage, and he gained some advantage of making her happy. Then suppose he keeps the watch, she sells the hair, and now he has the watch and the chain, so he benefits in this case. You might say, "Good." But she feels even better because she loves him so much. She actually benefits by knowing she made him really happy.

- 5.109** I've got the numbers here in such a way that actually it's that payoff for making your partner happy that adds even more value than it adds to the recipient of the gift. The giver actually gains more here, as I've set up the numbers. But you could do it either way, however you feel about that. In any case, if we do this, we can say, what's the expected value of her keeping her hair? It is $q+0$. Selling the hair: it's $0 + 2*(1-q)$, so it's $2-2q$. Set them equal, and we get $3q = 2$, so $q = \frac{2}{3}$.
- 5.110** She, in thinking about him, should think, what's the likelihood he's going to sell the watch? $\frac{2}{3}$. The expected value then is going to end up being $\frac{2}{3}$. He thinks about it and says, I can sell the watch or keep it. If I sell it, my utility is $2*p$ —the probability she keeps her hair—+ 0, so it's $2p$. What if I keep the watch? It's $0 + (1-p)$. Set them equal. $3p = 1$, so $p = \frac{1}{3}$. He's going to say, what is the likelihood she keeps her hair? Only $\frac{1}{3}$.
- 5.111** There is a mixed-strategy Nash equilibrium here. Each one gets the value of $\frac{2}{3}$. But notice, it is worse than the situation where she keeps her hair and he sells the watch, and also worse than the one where he keeps the watch and she sells her hair. In this case, it's Pareto inferior to those options, though, on the other hand, it would be better than what actually happens in the story. Nevertheless, if you think about the probabilities here, you realize, actually, it's pretty likely then that we're in a scenario where she cuts her hair and he sells the watch, and so they both end up with value 0, except knowing they sacrificed something very valuable to them for the sake of the other, but utterly pointlessly.
- 5.112** What do we end up saying about this? Here's another case where there is a mixed-strategy Nash equilibrium. Unlike the tennis match, unlike some other scenarios, it's like some of those, like the battle of the sexes in general, where we end up saying there's a mixed-strategy solution, but actually it's not very good. If you've got a pure-strategy solution, go with that. If there are different ones, then figure out, is one better than the other? If so, go with it. If not, as in the battle of the sexes or here, then what should you do? Take turns.
- 5.113** Or in the case of the story, it ruins the surprise. It ruins the story. But talk to each other. Maybe you're going to make some sacrifice to help the other person. Maybe you should let them know, or at least—anyway, I don't know. There's not a great solution to that problem, the way O. Henry sets it up. But at any rate, sometimes these mixed-strategy solutions are really, really valuable. Often, in sports and in other forms of competition, they are. But anytime they're combined with coordination or cooperation, they generally end up, even when they exist, to be a really bad idea.

- 5.114** When you don't want to have the other person know what you're doing, it reminds me of *The Princess Bride*, with Vizzini, where you can know that they think you don't know, which means you know that they think that, and you can keep going back and forth.
- 5.115** Yes, yes, yes.
- 5.116** Or even applying it to this one, where when you do communicate, but you still want to surprise the other person, so now I know what they think I'm going to do, but I'm going to do the other thing, because I want to surprise them, but they all know I'll do that.
- 5.117** Right.
- 5.118** Does that dimension get calculated out at all?
- 5.119** Great question. Because, yes, the more you think about these problems, the more you start thinking, wait a minute, we get led into a cycle, don't we? Let me go—in fact, that story is a good one. I think, wait a minute. He's probably going to sell his watch. But then again, if he knows that I realize he's going to sell the watch, then that may change what he does.
- 5.120** Actually, a simpler case—maybe this just indicates I like war better than I like love stories—let me go back to Hannibal attacking Rome, where we think Hannibal analyzes this and says, "The Romans are probably going to have their army in the south." That's the obvious thing to do. They're going to think it'd be a bit crazy for him to come over the Alps. They're not going to be expecting this. It'll be a surprise attack. There's some chance they'll think about this. But still, they're probably going to be in the south. It's an advantage for me to go over the Alps.
- 5.121** But the Romans think the most obvious thing for him to do is attack here, directly toward Rome. So we should put our army in the south. But then I think, hold on. He's going to realize we're going to have our army in the south, so he'll attack from the north. So let's put our armies in the north. But Hannibal thinks, "I bet they understand this." They think given all of this, it'd be rational for him to try a surprise attack. They're probably moving their armies to the north. I'll go by sea after all and gain a big advantage. They say, "I bet he realizes that we're going to figure this out."
- 5.122** You can keep going back and forth in this cycle. It's a bit like the scenario with the blue houses, where if you think about this as a cycle of thinking, not a cycle of actual painting, it's like, "Okay, I'm going to make my house blue, but then she's going to paint her house white. But I care about conformity, so I'll paint mine white, but she doesn't like them being the same, so then she's going to paint it blue. Well, that gives me a chance to paint mine blue, but then she's going to paint it," and you just keep going back and forth in these.
- 5.123** So, there is an interesting puzzle here that, if you begin thinking, it would be rational for my opponent to do this, so it looks like it would be rational for me to do this other thing, but if they realized that, assuming that they know I'm a rational agent, they're

going to do this. You can get led into these weird cycles. In practice, what people do after a certain point, just psychologically, is they say, "This is all getting too complicated." They stop this at some point.

- 5.124** But if we think of it from a theoretical point of view, we could say, here's one reason why a mixed-strategy solution looks attractive, because I want to keep them uncertain, hence the significance of the indifference principle. The moment you start thinking, "Oh, it's to his advantage to do that, to launch a surprise attack, so we'll move to the north," you can start getting into these cycles.
- 5.125** But if I say, "Look, all I have are probabilities," there is a slightly greater chance that he'll attack from the south than the north, but both are probable. There's a somewhat greater chance they'll have their armies in the south than in the north, but it's far from certain. Then, it's not like you can really get the cycle going. You can say, "More than half the time, they're going to do this." But there's a risk. Almost half the time, they're not doing that.
- 5.126** It's a bit like the tennis example or various other kinds of examples. Let's say the football example— are they going to run or are they going to pass? If they've got a great passing game, and I know that, then I may say, "So they're going to pass." I have a pass defense. They think, "They're going to have a pass defense, so let's run." I say, "They're going to realize I'm going to have the pass defense, and they're going to run, so I'm going to run." You can get led into this.
- 5.127** The only way out, really, is to mix it up, so there are probabilities, so that they look at it and say, "I don't know whether he's running or passing." I know his passing game is better, but that probably means he's going to run a little more often to try to get me to use a running defense, in such a way that then his passing game is even more effective. I think the mixed-strategy solutions have at least that advantage. They try to cut that off from the beginning, so the cycle can't really get started.
- 5.128** You don't want the other player to know and even to have a good guess about what you're going to do. In fact, in mixed-strategy cases where it looks like, oh, with five-sixths probability or something, you're going to do this, then you can get led into the cycle because it's too high a probability. Ideally, it works best when it's pretty much close to 50/50.

Lecture 6: Repeated Games

At a glance — Repeated games.

- 6.1 So far, we've been talking about games as if they're basically being played once, and more or less against different people each time. But really, most of life is not like that. Suppose you work at a place, you're going to be interacting with the same people all the time. Some of those games might be competitive, some are cooperative, some are coordination problems. You're going to face similar kinds of situations again and again with the same people.
- 6.2 The same might be true in sports. You're likely to be playing the same teams off and on throughout the season and across seasons. You're going to be dealing with the same people, let's say, in tennis matches, that you would face otherwise. Maybe it's just a friend, maybe it's somebody who's—if you're a really competitive tennis player, and on the circuit, you're going to be playing a lot of the same people similarly. You're going to play repeated games against the same opponents again and again.
- 6.3 That really fundamentally changes the nature of how people behave in game theory. In fact, so far we've been talking about what people do and looking at some empirical results. Broadly speaking, they suggest people are more cooperative than the theory would really predict. You'd say, here's the Nash equilibrium, here's your dominant strategy. And quite frequently, people don't play their dominant strategy.
- 6.4 In a sense, the theory is saying, but that's what a rational person would do; however, nice people don't. And indeed, one of the things we're going to be thinking about here is, what does it mean to be nice? Why are people nice in the first place? There are good reasons.
- 6.5 If you look at ancient philosophy and look at the virtues that people talk about, they talk about wisdom and courage and self-control and a variety of other virtues, but they never mention being nice. And yet, one of the first questions we ask about somebody is, well, are they nice? And it becomes a sort of sarcastic joke. If somebody does something really terrible, people respond, "He or she seems nice," and it's kind of a joke. But we think that's really important.
- 6.6 From one point of view, you could look and say, oh, the ancients had this view of the virtues that has so much more meat on it than our contemporary one. But actually, I don't think that's the right way to look at it. And indeed, game theory tells us why being nice is maybe the most important virtue. So, in a sense, what I'm going to be talking about in this lecture is, why are people nice, and why does it matter so much whether people are nice?
- 6.7 The answer is going to be that it leads to stability and to productiveness in these stable, repeated relationships that we have with other people. We're going to be facing the

same people again and again in similar games. If you're playing one time against someone, it's really different from playing against the same people again and again. Let's talk about repeated games, also known as iterated games, and think about issues about evolutionary stability and different strategies for playing again and again.

- 6.8** To some extent, the larger question here is really, how do we deal with situations where there is mutual power? We actually have to work together. I have some power to affect the outcome, but so do you. What I do is not something I can evaluate independently. I have to think about, well, what you're going to do and how those things interact.
- 6.9** So I'm going to be thinking not just, how do I prosper individually, but how do we have the group prosper? And if it's a one-off interaction, I might think, what group? I'm never going to see you again. But if we're going to be involved in repeated interactions, then I'm thinking, oh, well, you and I are, in a sense, a group. I have to think about things rather differently.
- 6.10** Let's go back to the prisoner's dilemma. In the prisoner's dilemma, you might say, speaking very broadly, I can cooperate or I can defect. If we put it in terms of the actual prisoners, I can actually remain silent or I can testify against you. If I think about Hobbes' scenario of the state of nature, I can seek peace with you or I can fight.
- 6.11** We can think about this, as we've noted before, thinking, well, if you're going to seek peace, I can gain an advantage over you by fighting and winning, taking advantage of you; if, on the other hand, you're willing to fight, I'd better fight back, so it looks like fighting is my dominant strategy. I should fight, no matter what you're going to do, and the same is true for you. Hence, Hobbes' idea that the state of nature becomes a war of all against all.
- 6.12** But it's clear that this mutual fighting is not a good thing for either of us. Life ends up solitary, poor, nasty, brutish, and short. We can do better if we seek peace. But how do we get there? If I'm interacting with you once, it's actually pretty hard to get there. But what John Locke notes, and actually Hobbes himself had noted this, he said, here's the problem: we can't really count on repeated interaction. It looks as if we're in this caveman society. Am I just seeing you this once? Then, actually, fighting is a good strategy for me. But if I'm in a repeated community with you, then that actually becomes negative.
- 6.13** John Locke, I think, really carries that idea further and says, "I don't think this state of nature would be a state of war." Why not? Because it's not a one-off situation. We're going to interact again and again. So if every time we fight, I realize that's actually not a very good thing; you're a constant threat to me. We face further interactions, further retaliations. I fight against you and take advantage of you. Maybe I beat you this time, but you're going to be out to get me next time. So if I cause harm, I'm likely to be harmed again in return.

- 6.14** That's why it matters so much whether there's a repeated interaction. If I can take advantage of you this time and I never see you again, well, then that's easy. But if, on the other hand, I'm going to see you again and again, then I can expect retribution, and that's going to change what I do from the very beginning.
- 6.15** There are a number of times when I've been to Las Vegas. It's a nice illustration of this in a sense, because if you interact with people on the strip, the assumption is, you're a tourist, I'm never going to see you again. Actually, there's a lot of defection. People are not really very nice to you. They're trying to take advantage of you, if you're there along Las Vegas Boulevard. On the other hand, in the rest of the city, people are really nice. Get away from that, and people are highly cooperative. They're really, really sweet.
- 6.16** Why is that? The same people might live here and work on the strip, but their attitudes are really different. Why? Because in their neighborhood, they assume they're going to be interacting with you again and again. That's a community where people actually interact repeatedly. But on the strip, the assumption is, "Hey, okay, you're here at my restaurant. I'm never going to see you again. You're just here visiting. You'll be gone in a few days. There's no possibility, really, of a repeated interaction."
- 6.17** That idea, that if we're going to interact repeatedly, that really changes the nature of our incentives, is very significant. If I know that we're going to be in a prisoner's dilemma, for example, again and again, then I have a much stronger reason to seek peace with you, because it's not something that just affects the outcome this time, it affects future outcomes. The same thing is going to be true if we're members of a continuing criminal gang, as opposed to people who just happen to be unrelated, not knowing each other, but happen to try to rob the same store at the same time.
- 6.18** All of a sudden then, I feel like, well, I have no reason to expect to ever interact with you again. I have no reason to try to protect you; you have no reason to protect me. We end up testifying against each other. But if, on the other hand, we're part of a gang that continues, then we realize we're going to interact more, and if I testify against you, you're going to really get me next time, so I'd better be very careful.
- 6.19** In a prisoner's dilemma taken in isolation, each person has an incentive to play their dominant strategy. Taking advantage of the other person really benefits them, but not really with any concern for the other person. They'll never see them again. But if the same people face these dilemmas repeatedly, the incentives become very different. And then I have to think, what's my strategy? It's not just that I interact with you today. We work at the same company. We work as part of the same group. We're going to be interacting all the time. So what do I do?
- 6.20** There are different strategies that people can adopt for these repeated interactions. Here I'm thinking about the whole series. I'm not just thinking, what do I do this time? I'm thinking, what's my pattern of behavior? How do I deal with these repeated

interactions, this continuing relationship? There are a number of strategies I might adopt.

- 6.21** I could be the devil. In other words, I could just always defect, always play my dominant strategy. Look, from one point of view, that's rational. If it were a one-off interaction, this would be entirely rational. I say, look, I'm better off doing this, no matter what you do, and so I do that. I am just always in it for me, purely for me. That is one strategy that some people have for these continued interactions.
- 6.22** We tend to not like these people because they really are completely untrustworthy. They're always going to do what benefits themselves, never what's better for the group, never what's better for the other person. They're totally selfish. But nevertheless, it is, from one point of view, a rational consideration. Hey, I'm in it for myself. If I can take advantage of you, I will.
- 6.23** We get at the opposite idea. We're always going to cooperate. You might think that it's actually morally better to cooperate, to always be the nice person, to treat others as you'd want to be treated, or to always seek peace, always seek to go along, no matter what the other person is doing. You might think, well— and this turns out to be a very successful strategy—you might adopt a strategy that's known as tit for tat. The idea is, you start by cooperating, but then you do whatever the other player did last time.
- 6.24** So I cooperate with you. If you cooperate with me, I cooperate again, because it looks like, good, this is a mutually cooperative relationship. We're ending up in that good square where we're actually seeking peace together. We're cooperating together, working together. So that's all good. But now, suppose you defect. Suppose you actually take advantage of me. The next time, I take advantage of you. So, I retaliate. If you don't cooperate with me, I retaliate against you. But if then you start cooperating with me, I cooperate with you in return. So I do whatever you've done to me.
- 6.25** And that, as we'll see, has some limitations. Nevertheless, it's highly successful overall. How forgiving should I be? Should I respond and retaliate for a single defection? Maybe I think, I've got to quickly lay down the law and make it clear that's unacceptable, and so retaliate against you. But on the other hand, I could think, I want to be more understanding than that, so maybe I have a slow version of tit for tat, which is to say, I start by cooperating, and I continue to cooperate until you've defected a certain number of times.
- 6.26** The first time, I think, maybe I misunderstood what you were trying to do. The second time, maybe I think, well, I'm not happy about this, but look, we've cooperated significantly in the past. I'll let it go. But at some point, I say, all right, enough. I have been defected against enough here. I'm not going to trust you in the future, and that would be a sensible response.
- 6.27** There's actually a biblical verse about this. How often should I forgive my neighbor? Should I do it seven times? And Jesus says, no, 70 times seven. If you take that

absolutely literally, which I'm sure is not what is intended, but you could say, I forgive the first 490. That's a very slow tit for tat. But then done, 491st. Anyway, that's something that you could think, I tolerate a certain amount of this, but not too much.

- 6.28** You could also say, I'm going to even be a little bit more tolerant than that. I will cooperate unless you've defected a certain number of times in a row. It's not that I'm going to delay; it's that I'm going to wait until you've done this a couple of times consecutively.
- 6.29** There could be reasons for this. Maybe the first time I misunderstood what you meant to do, or maybe you misunderstood the situation. Maybe from your point of view, it wasn't really a defection. It wasn't really taking advantage of me. You thought you were cooperating and just made a mistake. Here's a way of saying, look, I want to make sure. First time, shame on you. Second time, shame on me. I'm going to actually wait until it happens, say, the second time or the third time or something to respond.
- 6.30** There is another strategy, though, that's much less tolerant. It's developed by someone named Friedman. It's also called the grim strategy. I start by cooperating, but the first time you defect, I'm done with you. I always defect thereafter. Descartes, at one point, talking about the senses, said, "Never trust anything that's once deceived you." You might have that attitude, like, "Hey, that person tricked me once—never again. I will never again try to cooperate with them." So that's more or less a "one strike and you're out"-type of deal.
- 6.31** You could do that slowly. You could start by cooperating, and instead of one strike, you're out, three strikes, you're out. I will ignore the first defection, ignore the second. But the third, it's not just that then I'll defect against you as in a slow tit for tat, or as in tit for n-tats. It's more like, then I'm done with you forever. You cross the line too many times, I'm done. Forget it. I will never cooperate again.
- 6.32** There is a sneaky strategy developed by a theorist named Joss, which says, try to be nice, but occasionally sneak one in on the other person. Start by cooperating, and cooperate in general as long as the other person is cooperating. So act like you're playing tit for tat. Cooperate if the other person has cooperated, but occasionally sneak in a defection. So it's like, I'll play along. I'll cooperate, I'll cooperate. But I do something on the side to indicate defection. I'm trying to gain a little bit of an advantage without you noticing. I'm trying to do it enough that I gain, but not enough that you actually retaliate against me.
- 6.33** Then I could be random. I could just flip a coin. In general, in most of life, that seems really stupid. But on the other hand, if you think about mixed strategies in a competitive situation, maybe that's actually the best thing. Should we run or pass? I don't think a coach would be paid millions of dollars for being back there, saying, "I don't know which plays we're on. Flip a coin." But actually, in some settings, that might

be the most natural thing. You want to remain unpredictable, and that's really unpredictable.

- 6.34** The opposite of that is what happened a few years ago with University of Texas football, where it became so predictable that everyone in the stands was calling out what the play was going to be before it happened. It was a very simple thing. You'd ordinarily think, on first down, you run the ball, let's say, and then depending on where you are in the field, you might pass second down, third down, and so on. But it was as if the offense—and by the way, everybody knew the offensive coordinator's name. That's a bad sign. That's the kind of job where you don't want people to know your name. It's a little bit like a sound engineer. People shouldn't know who you are. If they know who you are, it's because you're doing a bad job.
- 6.35** This case was one where, oh, they're not going to be expecting us to throw a screen pass on first down. So, that's what we'll do, throw a screen pass. Then it's not going to work. It's going to be second and 10, and they're going to be expecting us to pass, but we won't. We'll just try to run it up the middle. The third down, since that's not going to go anywhere, it's going to be third and long, they'll be expecting us to pass and to try to get the first down. Either we'll run it again or we'll just throw it way down the field.
- 6.36** It was always a surprise play, given normal norms, but on the other hand, once you pick up on it, the whole crowd in the stands will be saying, "Okay, screen pass to the wide receiver. Yep, that happens. Loss of one yard. Run it up the middle. Gain of one yard. Throw it way down the field, incomplete, kick," and it just went that way most of the season.
- 6.37** Anyway, why was that so bad? Because it was so utterly predictable. It would have been much better if the coach had simply, or the offensive coordinator, had simply been flipping a coin. Sometimes, this is actually a good strategy. There's a strategy that really hasn't been discussed in the literature because it seems counterintuitive, but I've seen certain people actually adopt this strategy. And if you think of it in terms of a biological hierarchy, it makes more sense.
- 6.38** I call it the bully strategy, and so the idea is this. I start by cooperating, and then I do the opposite of what you did last time. Why would I do that? You cooperated. I think, "What a patsy. I can take advantage of you." Or you defected and I think, "Uh-oh, you're tough. I'd better yield to you." This is like being in a dominance hierarchy. Let's say I'm a dog, and I see another dog, and they go belly up in front of me, and I think, "Okay, I can dominate you. Good." I'm higher on the pecking order than you are.
- 6.39** But if, on the other hand, a big dog comes in and growls, then I think, "You're stronger than I am; I will go along with you." So if somebody cooperates, I see that as a sign of weakness. If somebody defects, I see that as a sign of strength, and I yield to them. I think, "Okay, I'm establishing my place in the dominance hierarchy."

- 6.40 I've been talking about the animal realm. That's one way in which this might happen, but it can happen in international relations. I think, "Is this country going to be one who cooperates with me?" Ideally, you might think in a good strategy, a tit for tat or something else, we seek cooperation with other countries. But let's say I'm a country that is not really like that. I think instead, "How strong are we? That country is weaker, we can dominate them. On the other hand, that country is stronger; we'd better yield to them." That would be an example of playing this bully strategy.
- 6.41 There's an interesting strategy that's been developed a little bit later, which is called Pavlov. It's a variant of tit for tat, but it varies in a really interesting way. If you think back on what Pavlov did in his research in psychology, he gave dogs and pigeons and other kinds of animals positive reinforcement for doing certain behaviors and negative reinforcement for doing others, the idea being positive reinforcement makes you more likely to do certain things, and negative reinforcement makes you less likely. You, in effect, get rewarded or punished for various behaviors, so they become more prevalent or less prevalent.
- 6.42 That becomes the foundation of behaviorism. You could think, even if I'm not a behaviorist, there is a kind of inspiration we could get from that. I start by cooperating, and then I see what you did last time. I count your cooperation as basically positive reinforcement for what I'm doing. That's a reward I got. But if you defect, that's like negative reinforcement. That's a punishment for what I've done. So if you've cooperated, I should do it again. If you have defected, then I should do the opposite.
- 6.43 How does that work in practice? Let's say you cooperated. Then great. That's a reward. I cooperated also, so I cooperate again. Suppose you defected and I had cooperated, then I'd see that as a punishment for cooperating, so I switch. I defect. So far, it looks like tit for tat. I'm doing what you did last time. But then it varies. Then I say, Okay, let's say that you defected last time and I defected too. Tit for tat would say, then defect again. But in this case, I defected and I got punished for it. I got a negative reinforcement because of your defection. So maybe I should change and instead cooperate.
- 6.44 So it's an interesting strategy. It's going to agree with tit for tat about cases where we both cooperated. If we both cooperated last time, TFT and Pavlov are both going to say, well, then cooperate again. From tit for tat's point of view, "Hey, you've got this ongoing cooperation going. That's a good thing. Keep it going." From Pavlov's point of view, it's like, "You cooperated, you got rewarded for that. Do it again." Suppose you cooperated, the other person defected. Tit for tat is going to say, well, retaliate, defect. Pavlov is going to say, "You cooperated and got negative reinforcement, so switch to defection." They're both going to say the same thing, but for slightly different reasons.
- 6.45 Here's where the difference emerges. You defected, they cooperated. Then tit for tat says, "Well, cooperate." They're trying to reestablish cooperation with you. That's a good thing, so cooperate in return. Show them that this is mutual. Whereas Pavlov

says, "You defected, and then you got positive reinforcement for it. You actually were able to take advantage of them. They let you. So do it again." You've got positive reinforcement for actually, let's say, dominating this other person; dominate them again. Maybe they're a patsy and you can continue dominating them.

- 6.46** On the other hand, suppose we both defected, tit for tat says, "Well, defect again. You're in this pattern of defection. They won't cooperate with you, so don't try to cooperate. It's a failure." But Pavlov says, "Wait, I defected and got punished for that. I got negative reinforcement, so I switched to cooperation." There's an interesting variation. In those cases where I've defected last time, they can differ. If the opponent has actually cooperated, Pavlov says, "Hey, sucker, I'll get you again," whereas TFT says, "Well, okay then. I retaliated against you. You're back to cooperation. Good, you learned your lesson."
- 6.47** Then what if we both defected last time? From tit for tat's point of view, it's like, "Well, all right, this is war. I defect again," whereas Pavlov is in effect saying, "Let's see if we can get back on track. You catch more flies with honey than with vinegar. Let me see if I can reestablish some cooperation. After all, I don't want to be in this war of all against all. Let's see if we can get out of this."
- 6.48** Which of these strategies, from devil to cooperate, to tit for tat, to variants of it, to Joss, to Pavlov and so on, which of these is most successful? It really depends on the payoffs. For example, we could have this pattern of payoffs in a prisoner's dilemma. We could each get 10 points if we managed to defect when the other cooperates. We could each get one if we mutually defect, two if we mutually cooperate.
- 6.49** You can just look at those numbers and think, that's a prisoner's dilemma, but how much does it pay off to cooperate? Actually, not very much. We go from one to two. But if I defect successfully and take advantage of you, I get 10 points. Actually, cooperation doesn't pay very well in that scenario. That's going to mean the cooperative strategies aren't going to do very well. Those that defect, those that take advantage of others, are going to do really well in that setting.
- 6.50** By the way, this is important for thinking about context in a society or in an organization, because some organizations are ones that really reward cooperation. Others don't. You've got to know what setting you're in. If cooperation isn't really rewarded much and defection is, then in a way, cooperating is a loser's game. It might be no reflection on general strategies about life or something, just that, hey, in this company, trying to get along with other people is hopeless. They're all ruthlessly competitive. They're going to take advantage of you every chance they get, so you've got to fight and forget about trying to seek peace with others.
- 6.51** There are probably business settings that are a bit like that. In such a setting, defection is rewarded. There's little benefit to cooperation, but you could have a different payoff matrix. Maybe mutual cooperation is worth five, mutual defection only two. Then it's

not just two times, it's two and a half times. Maybe it could even be more times. In that sort of case, you get a greater payoff for mutual cooperation, but it's still not great. In fact, if you took turns defecting, you'd do as well as cooperating. Really, that isn't very good benefit for cooperation either. You could just do as well saying, "You defect this time, I'll defect that time."

- 6.52** However, suppose we increase that. Suppose it's worth six to mutually cooperate, where defection gains you 10. You're no longer in a prisoner's dilemma, by the way, if that reward for cooperation becomes more than the 10. But if you have it be more, significantly more, than mutual defection, but on the other hand, well, less than defecting and getting away with it, you can have a more successful idea of, here it may be worth cooperating. We can mutually do better than cooperating than even if we take turns defecting.
- 6.53** So those become the really interesting cases. I'm going to look at all three of those to give us an indication of what happens when we have some of these strategies competing against other strategies. People have run examples like this on computer tournaments. Here, I'm going to just imagine 10 rounds so we can see the pattern. But you could do this over hundreds of rounds, and people have done that to try to indicate, which of these strategies actually generates more value than other strategies in this sort of setting?
- 6.54** Suppose two people playing tit for tat actually get involved in this and say, "We're going to be playing repeatedly against each other." Then they just keep cooperating. They start by cooperating, and each cooperates if the other cooperated last time, so they just cooperate, and on into infinity, wherever the game eventually stops. What points do they get? It depends on which of those payoff matrices we're thinking about. If cooperation really doesn't pay much, they each average two; if, on the other hand, it pays better, then they average six. Just keep that in mind. Mutual cooperation gives us, in the case we're most interested in, six points each.
- 6.55** But now, what if tit for tat plays against Joss? This is the person who seems to be playing the same thing but sneaks in a defection. Here, I've imagined that they just randomly sneak in a defection in round four. We've been cooperating, but then Joss says, "Let me see if I can get away with this." And what happens then? Because of the defection, tit for tat responds by defecting. On the other hand, the time before that, tit for tat had cooperated, so Joss goes back to cooperation. But now you see this alternating pattern. Wait, you defected last time; I'll defect. You cooperated last time; I'll cooperate.
- 6.56** They just keep switching back and forth between cooperation and defection. In effect, they're taking turns, defecting against the other person. If you've got a payoff matrix like the one that just rewards two points for cooperation, then that's actually pretty successful, because you get more points by this taking turns, defecting. That ends up

looking pretty good. It's way better than the two. One gets 3.6 on average, the other 4.6, and so you might say, well, look, this is great in that kind of hostile setting.

- 6.57** On the other hand, if you're in a highly cooperative setting where you get six points for mutual cooperation, here you do worse. Tit for tat only averages 4.8, Joss only 5.8. Both end up doing worse than if they simply kept cooperating. In any situation where there's any kind of reward that's significant for cooperation, it's not worth trying to sneak in the random defection. You're going to get caught. If you get caught, then that's bad. You get this cycle of mutual defection. Things get nasty, and the result of that is, you don't trust each other anymore. You both end up suffering.
- 6.58** What happens if tit for tat interacts with a bully? So tit for tat is saying, "I'm going to do whatever you did last time." If you've defected, I'm going to retaliate. But if you've cooperated, I'll cooperate with you. The bully says just the reverse. "If you cooperated, I think I can take advantage of you. But if you defected, I'd better go back to cooperation."
- 6.59** In that case, as you can see, they don't do well at all. Tit for tat starts out cooperating, but a bully immediately says, "Weak!" and defects. But then tit for tat responds with defection. And the bully says, "Well, I'd better go back to cooperating." So we get a cycle this time, but in this case, slightly offset. Each cooperates twice and then defects twice and so on. It's a bit different from the Joss type of cycle. But nevertheless, we get this repeated cycle where we get, you cooperate, then we mutually defect, then I cooperate, then we mutually cooperate, and we repeat that same cycle again and again.
- 6.60** We do better than mutual cooperation, if cooperation is not rewarded very much and defection is. But on the other hand, in a scenario where cooperation is rewarded even very significantly at all, better, even slightly, than mutual taking turns at defecting, we end up doing significantly worse, 4.2 and 5.2 instead of 6. So we both suffer from that.
- 6.61** What happens if tit for tat plays the devil? This person is just totally selfish, out to get you, and you're going in saying, "I'm a nice person. I'll retaliate, but otherwise, I want to cooperate with you." The devil just defects all the time. That's what it does. And tit for tat cooperates the first time and then realizes, oh, and so it keeps defecting. That ends up punishing both of them.
- 6.62** And it doesn't matter, actually, how much cooperation actually benefits people. It turns out, that just turns into a string of those ones. In other words, mutual defection all the way down the line. They both end up suffering. Tit for tat suffers a bit more because the devil gets in that first punch. But after that, they just both punch each other, a war of all against all.
- 6.63** What happens if tit for tat is playing somebody who's random? That's actually hard because, in fact, you don't know what to do. You don't know what to expect the next move to be. It almost doesn't matter what I do. If you figure that out, both end up

suffering again. If you were in a competitive situation instead of a cooperative situation, again, that random strategy might make sense. But if there's any possibility of cooperation and reward for it, then being random doesn't make much sense at all.

- 6.64** If we're being led off into different rooms, and the other prisoner says, "Shh, don't talk," and I say, "I'm going to flip a coin," that's not going to yield a very good outcome overall for us. What happens if the devil plays the devil? Just constant defection, very low point scores. They don't do very well at all. If you've got an organization with a bunch of selfish people who are out to get other people, it's actually not a very productive situation. None of the benefits of cooperation.
- 6.65** What if you have two bullies facing off against each other? That's a kind of interesting one. I don't know if you've noticed this, but dogs, some of them are very placid. They see another dog, they think playmate. Other dogs see another dog and are like [growling]. Interestingly, it's often the littlest dogs that are most prone to going [growling], and the biggest dogs that are like, "Hey, friend." But in any case, and then there are other dogs like my daughter's pit bull that look incredibly frightening and imposing, but actually are total sweethearts. Just like, "Oh, love me, love me." But anyway, appearances can be deceiving in that way.
- 6.66** But what happens now if we've got a bully against a bully? They start out cooperating. That's what bullies do. But then they say, "Are you weak?" They defect. But then they realize, "Oh, you're actually tougher than I thought." So they cooperate. But then they both say, "Cooperation," so they defect. They go back between mutual cooperation and mutual defection. That's not a very good strategy either. You get rewarded for the mutual cooperation, but that only happens every other time. The other times, you're mutually defecting. So the payoff is four, instead of six. Not really a great strategy. Playing with other bullies, it's not good for either side.
- 6.67** What about the bully versus the devil? That's the most extreme of all, because the devil says, "I'm just going to defect no matter what." The bully says, "Are you a tough guy or not?" You are, so the bully just cooperates. The devil just keeps defecting. So the devil just gets the full 10 points. The bully gets nothing. Being a bully and recognizing, "Oh, here's a guy much stronger than I am," the other person wins everything, and you lose completely.
- 6.68** All of those strategies are ones that I think people really do adopt, and you can think of an organizational type that goes along with them. Some people are just so nice, they become pushovers. Some really are mean and selfish, never cooperating. Some will cooperate but will retaliate. Some are pretty forgiving but eventually will retaliate. Some are really like, "Hey, I don't trust you anymore. I'm done with you." All of these sort of represent real personality types, I think.
- 6.69** But there are other strategies that may be a bit less natural, but are also known as kingmakers. They don't do well in these tournaments. They don't do well against other

strategies, but they weed out certain strategies because they are kind of a test. Can you deal with this kind of person? Often the answer turns out to be no. They're kingmakers in the sense that they don't become successful, they don't win, but they play an outsize role in determining who wins.

- 6.70** The idea is, they defect first. They see you as a potential victim of theirs, and they say, "Let me try defecting." What do you do? That does, in a sense, mimic what people do in real life. In particular, there's a strategy called Downing that tries to do this. It looks at people and it says, "How do you really interact with other people?" Here's the thought. You want to know, first of all, if the other person even responds to what you do, because if they're the sort of person who cooperates no matter what, or they defect no matter what, it doesn't even matter what you do. You know exactly what they're going to do.
- 6.71** First of all, you want to find out, do they actually respond to what I'm doing at all? But another thing is, I think, how trustworthy are you? So I look at the pattern of your defections and cooperations, and I think, "I want you to start. I want you to cooperate enough of the time." So I think, "How likely is it that you're going to cooperate with me?" I look at the past. How often have you cooperated with me? That's not an unreasonable thing.
- 6.72** When this was first proposed, people didn't propose it as a kingmaker. They thought of it as an entirely reasonable thing based on how people really make these decisions. We say, "How trustworthy is this person?" We base that on their past behavior. Seems very reasonable. But here's what makes it do poorly. It defects on the first two moves. It wants to see if you respond. So it tries defecting, and then it tries cooperating. But if somebody defects at the very beginning, immediately, you tend to get into these cycles of negativity.
- 6.73** Imagine this in real life. The first time you meet somebody, they're totally nasty to you. They make it clear they're not going to help you. And that happens the second time, too. How do you respond after that? Do you even try to cooperate with that person? Often you don't. You figure, well, that person's a nasty person. That happens with Downing.
- 6.74** After that, it's actually going to behave like a fairly normal human being might behave, except it starts saying, "Are you even going to respond? Let me see if I can take advantage of you." If it finds out, "Oh, I can't, you actually do respond, you actually retaliate," now it tries to get back in your good graces, but that turns out to be very hard. It helps to get rid of the strategies that are too cooperative. But on the other hand, it's not very successful because it starts out defecting.
- 6.75** There's one that's known as tester, and it defects on the first move. Then if the opponent ever defects in response, it cooperates once, and then it just goes to tit for tat. It has a more complicated thing, too. It cooperates on moves two and three, and

then it might defect if it looks like you're not doing that. So it's a complex thing, but the key thing is, it defects on the first move. Really, that's enough to generate a cycle of negativity with a lot of other strategies.

- 6.76** Then there is tranquilizer. It cooperates if the opponent does for a while, the first two dozen moves, in fact. You think you're playing somebody nice, like 24 times. "Yeah, I'll cooperate with it. I'll work with you." Then it starts sneaking in defections, like Joss, but it starts sneaking them in more frequently if it gets away with it, until it's reached about 25%. So what it does is basically defect a little bit, but try to keep the frequency low enough that you don't notice and that you forgive it.
- 6.77** None of those really turn out to be very good strategies. And in fact, if you do these computer tournaments, generally, tit for tat emerges as the best strategy. It's the most successful one. There's a weird kind of dilemma that arises out of this, because you look at how it does compared to every other strategy. If there's just constant mutual cooperation, then they do equally well. But if there's any kind of defection that takes place along the way, tit for tat actually loses. It never beats another strategy one on one. It either ties it or loses to it.
- 6.78** Nevertheless, it wins the tournament. You might think, well, wait, how is that possible? It's like a football team that loses, or at most ties, every single game, but somehow wins the championship. What do you mean? How is that possible? The answer is because it scores a lot of points. So imagine that in the NFL, we said, look, what's going to determine the teams that go to the Super Bowl is not a question of who wins the most games. It's who scores the most points.
- 6.79** Then you might think, oh, a team could actually lose every game, and yet get into the Super Bowl. Why? Because they scored lots of points. They had a terrible defense. In every case, the other team scored more. Nevertheless, each team played them, let's say, only once. Actually, those guys got the advantage of that only once, whereas this team was scoring lots of points every single game.
- 6.80** Then, actually, you could see that's how it could happen, that that team actually ends up better than all the others. It scored more points. That's what tit for tat is like. It encourages cooperation. It cooperates with you, it cooperates with you. Sometimes it gets taken advantage of. That's not so good. It doesn't beat anybody who's willing to take advantage of it, one on one. Nevertheless, it engages in so many productive cooperative relationships that it actually ends up looking really good. It never emerges as the best in that particular pairing, but nevertheless, it ends up looking like the best overall because it's constantly involved in these good pairings.
- 6.81** In some later tournaments, Pavlov has ended up winning. It really depends on what happens once defection occurs and what the other strategies are, how big the payoffs are for cooperation and so on. Pavlov is also a serious strategy that sometimes manages to win. But overall, as we've seen, there are only subtle differences. If people

have cooperated in the past, then actually they don't disagree. They only disagree when the person has defected in the past. And basically, Pavlov says, "Were you rewarded for defecting? Keep going. If you are punished for it, switch to cooperation." Whereas tit for tat says, "Did they cooperate? Get back to cooperation. On the other hand, if they defected, also keep defecting."

- 6.82** There is a difference when you've defected. If you stop and think about it, you can think, well, what strategies would be better for one or the other? In other words, Pavlov might do better against some strategies. Which ones? For one thing, if you get trapped in a cycle of mutual defection, maybe actually Pavlov can help you by getting you out of this constant defect, defect, defect.
- 6.83** On the other hand, it can also be taken advantage of because, let's say, against the devil, it doesn't matter what you try to do. You're not going to get me back on track. I just defect no matter what. And tit for tat might do better if, actually, it's important to recognize this is a hopeless cause. I mean, if I keep trying to respond to somebody who's not going to respond to me by cooperating, I'm going to get clobbered. So it really depends on the mix of other strategies.
- 6.84** If we start looking at some of these, what about that random defection strategy, Joss? Pavlov and tit for tat are going to look a bit different. Tit for tat will do better. It's going to end up with 4.8 points instead of 4.2 on average for Pavlov. And why? Because it actually does get trapped in a cycle, but it's a less destructive cycle than the one Pavlov gets trapped in. What about Pavlov versus the bully, or tit for tat versus the bully? Actually, it's going to turn out that neither one does terribly well against the bully, but tit for tat does better.
- 6.85** The Pavlov strategy is going to just keep fruitlessly trying to cooperate again, and the bully isn't having that. The bully interprets that as a reward for their defection. But now, what about the devil? Tit for tat recognizes very quickly, "I'm playing the devil. I just keep defecting." Pavlov keeps trying to cooperate, and so really gets punished heavily for that.
- 6.86** But that can make you think, well, does Pavlov ever do better? For example, against a random strategy, actually, it does do better, because it's capable of reestablishing some kind of reward, but it's also capable of really taking advantage of repeated cooperation. As long as the cooperation randomly appears often enough, it can actually do very, very well. What about against these kingmakers, about tester, for example, that defects the very first two times? Pavlov does better against that. It can actually recover from that sort of defection, whereas tit for tat just gets trapped into mutual defection.
- 6.87** It's somewhat difficult. There are variants of all of these that have been studied. For example, we can say, what if we're doing something like tit for tat, except if we both defected last time, I don't know whether it's better to defect the way tit for tat recommends, or cooperate the way that Pavlov recommends. What if we just choose

randomly? Say, "Yeah, I don't know. Sometimes we'll be TFT, sometimes we'll be Pavlov. We'll flip a coin." That does give a way of escaping mutually destructive defection, without getting trapped into being a victim of, say, the devil or other things. So that has some promise. It does a bit better in some contexts than one or the other, but it depends again on the context.

- 6.88** Thinking about these repeated interactions has led game theorists to a very important notion, one that's really useful in biology, which is evolutionary stability. What happens to these strategies and the communities where these strategies are being played over time? How easy is a strategy to invade? In other words, suppose we've got an organization where we're all pretty cooperative, we're all playing tit for tat, or maybe something even more cooperative than that, but we hire some people we shouldn't have hired; we hire some devils, or we hire some bullies. How much can that disrupt our organization?
- 6.89** Let's suppose we start out with 10% of these invaders, whatever this other strategy is, and 90% of us still playing tit for tat or something cooperative. Then however the scores go, we think, well, that affects the proportions of these strategies in the next round. For example, suppose those invaders do relatively well, then we end up with more of them. In biology, yes, this is a successful strategy, so those people are better at propagating, so this population grows, this one might shrink.
- 6.90** In an organizational setting, you can say, yeah, these nasty people are here, and now they're hiring more nasty people, and so on. It can go in that direction, or it could be, this is actually a pretty negative environment filled with defection, but they hire a few nice people. Can the nice people actually manage to take over the organization, or are they going to be driven out? What happens then?
- 6.91** So we can think about these things, and there are online sites where you can go and actually play some of these games and try out different strategies against each other, and say, "How evolutionarily stable are they? Can this strategy invade this other strategy?" It's well worth checking those out. It's really actually fun to play. There are ways—and here, it's just this question of, how likely is it that this resident, this existing person who's actually playing the dominant strategy, how likely are they to interact with these new invaders?
- 6.92** Those sites don't really allow for you to control that, but there are ways of controlling for that, and it matters a lot. For example, you've got this negative environment, and some nice people come in— are they together? So they interact with each other, or are they spread out? In which case, they're just constantly interacting with other people who defect. It matters a lot. Here, I'm ignoring that aspect, but it does make a difference. It matters how often they're interacting with each other, how often they're interacting with everybody else.

- 6.93** In any case, we can ask some questions then. How stable is this devil strategy? We've got an organization filled with completely competitive, completely selfish people playing their dominant strategy, defecting constantly. Can that be invaded by other people who are playing more cooperative strategies? Actually, not very easily. It is highly stable except against tit for tat. Tit for tat actually can do pretty well. So here is a run. This is assuming 1,000 meetings between 90% initially who are devils, 10% playing tit for tat.
- 6.94** This will come out differently, depending again on who's interacting with whom. You can run this repeatedly and find out what happens. But this is pretty typical. Notice what occurs. After a considerable number, the nice people are still pretty much in a minority, but then they actually start interacting with each other enough that they begin to become more numerous, the devils become less numerous. Eventually, you end up with more people playing tit for tat than you have playing this devil, completely selfish strategy.
- 6.95** In a way, that's a really optimistic thing. You can have an organization that's overwhelmingly selfish and negative, highly competitive, and you hire some people who are cooperative. As long as there are enough of them and they interact enough with each other, they can end up actually dominating the organization. For example, Axelrod says, this is how, in certain kinds of populations, among various kinds of organisms, ones playing a more tit-for-tat strategy end up taking over from those who are purely selfish. He also uses it as a historical explanation of the rise of Christianity, where basically the Roman Empire is this highly noncooperative setting, but you've got Christians who start interacting enough with each other that they actually become productive, and so, end up growing as a community.
- 6.96** But you could think about this in another kind of organization. Actually, when I became a professor at my university, I entered a department that was known to be one of the nastiest in the country. People really hated each other. It was sharply divided. There were cases where people would shout at each other and almost get into fistfights during talks and so on. It was pretty wild. But they hired nice people like me. Eventually, we managed to take over to the point where now, faculty meetings are just like, "That's a great idea. Maybe we should do that. Hey, what about this idea? Maybe it's even—yeah, you're right. Maybe that's even—"
- 6.97** I mean, it's sort of, A, we hardly ever meet because we all agree about everything, and B, when we do, it's just like, "Yeah, your candidate is fantastic. I sort of like this candidate. I think that candidate is fantastic, too." We agree so much, it's actually kind of hard to make decisions because we are all just too busy patting each other on the back and cooperating. But anyway, it's an illustration of this, that eventually this can happen.
- 6.98** How stable is a cooperate strategy, where I'm going to cooperate with you no matter what? I will do what I wish you were doing to me. I will not fight with you. I will not

retaliate. You might expect, it's highly vulnerable. As long as it's interacting with other cooperative people, it does great. But now let's imagine 10% devils enter this community of cooperators. Eventually, and again, it's going to vary time to time depending upon who interacts with whom and how frequently and so on, but still, they're going to end up being at least half the group. They're going to be able to take over eventually. But it takes a long time, on the other hand.

- 6.99** Here, after a thousand meetings, a thousand of these repeated interactions, they're only at about 50%. So the devils are going to continually gain ground, but very slowly, which is a nice thing. On the other hand, not always. Sometimes, they're going to take over pretty completely. And why? Because they start interacting with each other enough, and with the cooperators enough. The cooperators are interacting with too many devils, they don't get enough benefit from their willingness to cooperate.
- 6.100** What about bullies entering this picture? We're all cooperative, and then some bullies come in. They can stay alive. They can continue. We're not going to wipe them out because cooperators never retaliate, and so they're not really going to push out any invaders. On the other hand, they also don't allow them to grow. The bully strategy just isn't successful enough to get beyond that very low level.
- 6.101** Suppose we have a community that's full of bullies. We're 90% bullies. Now, some other invaders enter in. How stable is this bully strategy? We can say, well, what happens if, let's say, a bunch of cooperators come in? It's just the inverse of what we saw. They remain at a very low level. They aren't wiped out, but on the other hand, they don't really increase either, at least not generally and not very much.
- 6.102** Occasionally, you get a result like this. If the cooperators can interact with one another frequently enough, they can end up going up to about 50%. Sometimes they just stay at that low level of 10%, sometimes they increase to maybe about 50%. The cooperators don't end up taking over, usually, but on the other hand, they can increase their numbers. It again depends on how often they're interacting with bullies, how often they're interacting with one another.
- 6.103** What about tit for tat? Can it invade bully? It's about the same story. It's trying to be cooperative with people who aren't going to cooperate with it, really. They see that as a sign of weakness, so most of the time it stays at a low level. Pavlov does even worse. It stays at a lower level. It doesn't get wiped out, but doesn't increase. But suppose bullies face off against the devil. The devil clobbers the bullies. Bullies do not try to interact with the devil. You will lose. Basically, they reward people for being nasty to them, and so the people who are nasty take over.
- 6.104** What about tit for tat? After all, I've said that's in general the best strategy. It's the one that seems to be most successful in these tournaments. So how stable is it? In general, pretty stable. Introduce devils into a bunch of tit-for-tat players, and they tend to not be wiped out by tit for tat, but on the other hand, they stay at a low level most of the

time. What if you get some bullies enter in? They do a little bit better because they sometimes engage in mutual cooperation, but overall, they don't become dominant. They become a significant minority.

- 6.105** What about Pavlov itself? Does that do better in terms of evolutionary stability? Put a bunch of bullies in there, and they actually, typically, stay at a low level. If they're dominant, they will destroy Pavlov. But on the other hand, if they're just a minority, they can't do all that well. It will depend, of course, run to run. Sometimes they do better than other times. They might even increase to a significant minority, but it's going to depend on how often these interactions take place among the different groups. Put the devil in, and well, the devil can get about parity with Pavlov, so it can invade it fairly successfully.
- 6.106** I've been talking about evolutionary stability—which ones are more stable? Which one's less stable? And against which possible invaders? It's hard to generalize about that. It's complicated. But now, what, in general, makes something a good strategy? What makes something succeed if we're engaged in these repeated interactions?
- 6.107** The first rule is, be nice. I said I was going to tell you why niceness is such an important virtue. Here's why it is. The strategies that start off defecting once or more than once do terribly. They immediately get into the cycle of negativity. They are not successful overall. It's always good to begin by cooperating. So never defect first. Always be the one who, at most, defects in response to somebody else's defection. Never do it first.
- 6.108** Tester, tranquilizer, and things like that are going to be bad for this reason. Joss will be bad. It won't defect in the first move, but it will start defection. Those are very negative things, and that strategy ends up being punished significantly. Bully is like this. The bully will be the first to defect. That's going to lead to cycles of negativity.
- 6.109** You've got to be optimistic about the possibility of cooperation. Always begin by being cooperative. The nice rules do really well with each other. They're hard to invade, and as long as there are enough of them, they tend to win. Even if they start out as a minority, if they interact with each other enough, those interactions are so positive, generate so much value, they tend to win. Being nice is a really important virtue.
- 6.110** The second characteristic of good strategies is that they are clear. Other strategies can understand what they're doing. If the others don't understand what you're doing, it's going to look random to them and they can't respond appropriately. So too much complexity looks like chaos. Back when I was the chairman of a philosophy department, if I had a fault, it was this fault. I was doing things that were too complicated, and I was playing a certain game, and other people, some of them knew what the game was, a lot of them didn't know what the game was.
- 6.111** Sometimes I intentionally thought, "I don't want them to know what the game is. Here's what I want, and if they know what I want, they're going to really work to defeat me. So I'm going to keep them confused about what I want." But a lot of times, it really

wasn't due to that; it was because I didn't understand this rule. Other people have to know what you're going for, what you're trying to do. They have to understand your strategy. If they don't, it's just going to look random, like there is no strategy at all.

- 6.112** A third feature is, you've got to be retaliatory. You have to be willing to punish bad behavior. If you don't retaliate, you're easily exploited. So think about that cooperate strategy. It's great until you encounter the devil or the bully or somebody else who is going to take advantage of your willingness to cooperate. You are easily exploited. You've got to be willing to hit back.
- 6.113** How often should you hit back? How sternly? That really depends on the benefits of cooperation, on how often you're interacting with other cooperators and so on. There's no general rule about that, whether it's tit for tat, or slow tit for tat, or Pavlov, or something in that ballpark that does better in this particular environment. But you have to be able to retaliate.
- 6.114** You also have to be forgiving. That is to say, the strategy that says, "You defect once, I'm done with you forever," that is not very successful either. You've got to avoid these cycles of mutual retaliation, mutual defection. How forgiving, again, is going to depend on the environment, but you've got to somehow get back to cooperation. That's one way in which Pavlov tries to do something different and actually says, "Mutual defection, let's see if we can break the cycle." The problem is that if it fails at breaking the cycle, it doesn't learn its lesson, and it just keeps trying; it keeps failing.
- 6.115** Here is, I think, a very important rule. Don't be envious. Envy is self-destructive. Think about, actually, both Pavlov and tit for tat. They're very successful strategies, but they at most tie every single interaction. They never end up with more points than the opposition. Nevertheless, they win the tournament because they generate more points overall. There are lots of positive interactions. If you start thinking, "I don't look as good as the other person in this interaction," overall, that's a negative thought that is going to be self-destructive. Don't care about that.
- 6.116** In fact, to the extent that you can make the other person look good, you're going to look good, too. The result of that is, overall, you're going to be more successful. So don't compare yourself to others. Just think, "Am I doing as well as possible, given what they're doing, in these repeated interactions?" Tit for tat does not beat the other player; it just elicits behavior that serves both really well, so it comes out doing very well overall.
- 6.117** I don't want to conclude by saying, so just follow tit for tat, or be Pavlovian or whatever, and that's the simple thing. There are problems with tit for tat. There are certain circumstances where it doesn't work very well. For example, power relations might make retaliation impossible. The boss defects, suddenly says, "Hey, I know it's last minute, but I need you to work all night doing this." You may not be in a position to

retaliate, right? Ideally, you could retaliate, but the fact is, you'll lose your job if you retaliate, so don't do that.

- 6.118** It's also maybe incompatible with friendship and love. I actually knew a woman who lived in my neighborhood who was dating someone, and any time she said, "No, I can't do that," he would, the next time, say, "Well, no, I can't do that." It was very much a tit-for-tat type of thing where, hey, you said no, then I say no to you next time, and so on. You can imagine, they broke up shortly after that once she realized the pattern. In certain kinds of relationships, you're supposed to be more forgiving and more cooperative than that.
- 6.119** There is a possibility of misperception. I think you've defected, but maybe that's not what you meant to do at all. Those mistakes can just keep echoing back and forth. That's one motivation for thinking, be more forgiving. Wait until somebody's defected twice or three times in a row, or do this slowly because there's too great a chance of miscommunication.
- 6.120** You could think, start by cooperating and just keep doing it until that level of apparent defection becomes unacceptable. Maybe it's, I'll tolerate this three times and then I'm done. I'll go back to tit for tat. Or maybe it's simply a matter of, as long as it happens less than 10% of the time, I'll ignore it. But whatever it is, then basically try to be a bit more forgiving than tit for tat is.
- 6.121** That can indeed really help. Something that's a random defection or even a misinterpretation of what somebody's done can keep echoing, can be negative. But on the other hand, if we're slower to respond, we keep cooperating, we restore it much more quickly, and so we're much better off. Those cycles of retaliation are a problem. We can end up in a Hatfields & McCoys situation, where we just keep trading back and forth, defection and cooperation. There's no way out.
- 6.122** Finally, there is actually a way of beating tit for tat. This is a surprising way. It requires having a confederate. You've got to recognize a partner. Then you always cooperate, they always defect. You can do great because you always win. They know, anytime they interact with you, it's like, "Okay, buddy, you get to beat me up." When that happens, you're, let's say, playing tit for tat otherwise with everybody else, so you're doing pretty well.
- 6.123** Then you recognize your partner, and your partner agrees to just like, "Okay, you win, you win." That allows you to accumulate extra points. It's just like the movie Talladega Nights. I think of it as the shake-and-bake strategy, where Cal basically gets into the lead with Ricky Bobby right behind him. They won't let anybody pass them. Then, at the last minute, Cal lets Ricky Bobby win. That's exactly this sort of strategy that does beat tit for tat.
- 6.124** On the other hand, it requires that you have this kind of partner who's willing to be dominated in this way. It also requires that you recognize who that partner is. That may

or may not work in an ordinary business setting, but maybe it does. Maybe you've got somebody who you're really good friends with, and that friend is willing to let you take the credit for all sorts of things, then you can actually do better than the average tit-for-tat player.

- 6.125** How do we encourage cooperation? Think about the prisoner's dilemma. We can have people communicate, talk to one another, actually try to cooperate, try to coordinate, and give yourself incentives to actually do this. We can change the payoffs. We can find ways of rewarding cooperation and punishing defection. We can do that sometimes through financial means, or we could do it in other ways. But in general, we have to change those payoffs.
- 6.126** We could get rid of defectors. We could just fire the people who are not cooperating. We could encourage them to leave, make it difficult and unpleasant for them. We could disperse them. After all, if people, whatever their strategy, are interacting with each other, they end up growing much more quickly than if they're interacting with the majority. Disperse them, spread them around, and they're going to have a harder time interacting with each other. They'll have to assimilate to the mainstream.
- 6.127** But the most important bit of advice is to enlarge the footprint of the future. In other words, if I think this is a one-off interaction, I have every incentive to defect. But if I think we're going to be interacting, especially frequently in the future, then I've got an incentive to cooperate with you. I've got to see this as part of a repeated interaction, and so I can make that future seem more significant.
- 6.128** I can also make our choices dependent, make what I'm doing depend on what you do. So I need you to succeed. That encourages us to cooperate. In all of those ways, we can actually try to build these relationships of trust. I'll come back to my theme of social capital. We can actually build social capital in these ways by encouraging people to realize, "I'm going to be interacting with you a lot in the future."
- 6.129** One thing that Coleman has stressed is that the more dimensions in which I interact with you, the better. Am I cooperating with you purely on this project or that project? It helps if there are lots of projects. In fact, it helps if we're both on the softball team or in some other way interacting; maybe we go to the same church. We interact in different dimensions. So I realize, I might be nasty to you at work, but I'm going to see you Sunday morning, or I'm going to see you Wednesday night when we show up for softball practice or something like that.
- 6.130** And then I realize, oh, those interactions, recognizing that they are continuing and that maybe they're occurring in different contexts with different incentives, all of that gives me very powerful reasons to be trustworthy myself, to seek cooperation with you, and in return, for you to cooperate with me. It builds up those norms and those relationships. That ends up being vital, which is why I saved this answer for the end of

this one. It really requires repeated interactions and both sides to say, yes, we're part of a continuing enterprise here. We're going to be interacting again and again.

- 6.131** I can't help but think of, I brought this up yesterday, win-win or no deal as part of that, and how that motivates people to speak the truth kindly upfront, but if it doesn't work out, then it doesn't work out. How do you see that in game theory terms?
- 6.132** That's a good question. Can you give me a practical scenario that will make this a little more concrete?
- 6.133** Sure, marriage.
- 6.134** Okay, yes. Yeah, good. Actually, marriage is a great example, because if I think, look, we are in this till death do us part, I've got huge incentives. I mean, the footprint of the future is very large for me, and the same thing is going to be true for you. The result of that is going to be, we have a huge incentive to be cooperative, to be forgiving, to essentially be nice to one another. Maybe not "Gift of the Magi"-nice, but nevertheless, really nice to the other person. The more we think about this as a relationship with the future, the stronger the incentives to cooperate, the greater these norms of cooperation and coordination and so on.
- 6.135** If, on the other hand, I think of this as a short-term relationship, then it becomes a very different thing. Let's say, I don't know, I'm a college student who thinks, "Well, this is fine, but pretty soon I'm going to graduate and I'm going to go off in this direction. She's going to go off in this direction. It's a short-term thing," then the incentives for this are much less, because there isn't really that much of a future. There's some future like, hey, we have until May, but then it's going to change. At that point, there's just much less commitment.
- 6.136** So, there's this weird sense in which the strength of the relationship is going to depend on the level of commitment. You might think the level of commitment is going to depend on the strength, but to some extent, it goes the other way around. I think there's been a huge change, for example, since divorce laws were liberalized. If you really think "till death do us part," then there is an immensely large future. Its footprint is very big, so you've got huge incentives.
- 6.137** But if you think, "Well, I can get out of this if it doesn't work out," then it's more tentative, it's less. You might think, well, it's not just that divorce got easier, so you've got more divorced couples. It's that now the footprint of the future has been shrunk. It's no longer necessarily the next 40 years of my life. It's the next, well, N years of my life until I get tired of it or something happens, and that changes the incentives. There's still a large enough future that you would think it's still more or less continuing, but it's a bit different.
- 6.138** Let me put it in a different context. When people got jobs at companies and thought, "This is a job for life," I mean, my grandfather worked for the railroad for 50 years.

Actually, I've worked for the university now for 46 years, so I'm getting close to that. But there's a strong footprint of the future there. You think this is a continuing relationship, and that changes it. Whereas if I think, "Well, I'm doing this for a few years, but then I'm planning to do that, and then I'm hoping to do that," that's really a different attitude about work, and it makes you much less cooperative.

- 6.139** You're there to see how this can help you, but you're not really that interested in a long-term relationship with this particular company. I think a lot of people have written about the fact that that's changed the employment relationship. It's changed cooperation, it's changed people's commitments to the job, and so on. All of this, I think, can be explained in exactly these game theoretic terms. It's, you have an incentive to play your dominant strategy, if you think you're just there for a short time.
- 6.140** In fact, you can really see this, actually, in both kinds of contexts. Let's say you submit your resignation, but you're not leaving for the next month. How much work are you going to get done in that next month? Or somebody says, "I'm going to retire at the end of June." You think, well, are they going to produce much between now and the end of June? Their incentive is not to do very much.
- 6.141** Similarly, actually, I think in relationships. If a couple gets into trouble, it becomes dangerous, because they start realizing, wait, maybe there isn't a future. That gives each less commitment and less willingness to actually engage in cooperative behavior. It can be a kind of thing in the corporate world and in human relationships, where once things start going wrong, you start thinking, "Maybe there's not the future I thought there was." You become less cooperative, and things get worse and worse quickly.
- 6.142** It's a way in which, actually, good things can accelerate, but bad things can accelerate, too. As I see, "Hey, this is good, I want to continue with this," great, I've got greater incentives. That increases the likelihood that we'll keep doing this. But if on the other hand, I start thinking, "Oh, maybe I need to leave the company, or maybe this relationship is doomed," then my incentives change, the other person's incentives change, and it goes downhill quickly.
- 6.143** All that's predicted by this sort of theory as well. The footprint of the future matters a lot. The more I think we're going to interact in the future, the more cooperative I become. But the more I question that, the less cooperative I become. That can ruin relationships at an accelerating rate.

Lecture 7: Sequential Games

At a glance — Sequential games.

- 7.1 Aristotle talks about the fact that, in some societies, people work together, and in other societies, people are like an isolated piece in a board game. I want to use that as our inspiration today, to think from another angle about the ideas of trust, of collective decision-making, of people working together, and how all that works in terms of the games being played, the goals that people have, and the assumptions they have about the world.
- 7.2 So, in today's lecture, I want to talk about sequential games. So far, we've mostly been dealing with simultaneous games, ones where we both move before the other knows what the other is doing. And lots of games are like that. We've been discussing many of those, but many games are sequential. One player moves, then the next moves, and so on. People take turns, typically. This might be a matter of chess. White moves first, then black moves, and so on and so forth, or maybe it's checkers, or maybe it's Monopoly, or whatever it is.
- 7.3 But it's not as if you both, in Monopoly, roll the dice simultaneously, or you close your eyes in chess and say, "I'm moving here." "Oh, I am, too." Who got there first? Or something. Instead, you take turns. And so, sequential games are really most of the games that we're used to playing in real life. So, they're an important class, and they reveal different things.
- 7.4 We've been talking about Nash equilibria. We're going to talk about a new kind of Nash equilibrium this time. It's going to be like the other sort, which is very useful in certain contexts, but also lands you in a place you really don't want to be in other contexts. So, I want to begin by saying, there are a lot of games that we've been talking about so far, that if we turned them into sequential games, would be completely ruined.
- 7.5 Consider rock, paper, scissors. Okay, you go first. You say, scissors. Then I say, rock. The second player would always win. Similarly, with our simple game of two-finger morra, if I call even and I see you put down one, then I know to put down one. And so, the game would be utterly trivialized if it were turned into a sequential game. It has to be simultaneous for it to work. The traveler's game that we talked about last time is also like that, because then you and I are both putting down a value for this piece that's somewhere between \$180 and \$300. And if I know what you're putting down, then I can put down \$1 less. I get rewarded for making the lower bid and you get penalized.
- 7.6 That's why, by the way, the minimum is the Nash equilibrium there, even though it's a stupid Nash equilibrium, because that way you can't undercut me. If it were a sequential game and you had to move first, then I could always get more than you. Conversely, if I had to move first, then you would always get more than me, so making it sequential would ruin it.

- 7.7 On the other hand, some games wouldn't make any difference at all. We talked about Gresham's law, where bad money drives out good. The buyer and the seller, both independently, have some reason to use bad money in their transactions and to hoard the good money. Well, that's not affected by who goes first. Even if the buyer offers the money to the seller, and the seller recognizes this is the crummy money, this is bad money, it doesn't change things. His incentive was to give bad money in change, anyway.
- 7.8 So, some games are utterly ruined if you make them sequential. Some games, on the other hand, aren't affected at all, like that one. In fact, whenever you see people with dominant strategies, you might think the sequential aspect of this doesn't really make any difference. Even in the classic prisoner's dilemma, I find out what the guy in the other room did. I find out he's remained silent. I can still benefit by giving testimony against him. If I find out that he has ratted me out, then I have even more incentive to rat him out. In a way, it doesn't change the circumstance at all, if one of us goes first and the other knows what the move was.
- 7.9 But if we think about typical board games or something like chess or checkers, you realize, wait a minute, they have to be sequential. It doesn't even make any sense, really, to have these moves be simultaneous. So, when we think about a sequential game, things become a little bit more complicated. Yes, we're going to have a set of players in the game, our rational decision-makers, and we're going to have possible actions available to each player. We're going to have payoffs associated with those. So those are the kinds of things we've been talking about all the way along.
- 7.10 But in this case, we're also going to have to have rules determining the order in which the players make their moves. We have to know who goes first, and then how do we go? Maybe we go clockwise around the group of players, maybe counterclockwise. Maybe it alternates between two players. Maybe there's some more complicated rules about who plays. Think about volleyball, where the serving, at least, is sequential, and the hitting of the ball is sequential. I do get a chance to find out what you've done in hitting the ball, in order to make my defensive move. I don't have to just wait and be ignorant of that. I see the ball coming over.
- 7.11 And so, there are going to be rules like that. There will be rules that determine when the game ends. A simultaneous game, we both play, and well, that's that. But in this case, we're going to have a sequence of moves, and at some point it stops. Then we'll have to know, what does that ending mean? Okay, the game's over, but who won? What does it mean? And so, we'll have individual payoffs, but there's usually some benefit to winning the game as well.
- 7.12 Well, let's think about a simple game. Actually, when I've tried this with my students, I find out it's not so simple. They get very frustrated because I say, "All right, you can play against me," and they always lose, which is kind of fun. I enjoy that. But here's the idea. It's very simple. You just start with the numbers, 1 to 21, in a row. Then you can

have people play against each other individually or as a group against the professor or whatever you do.

7.13 But anyway, here's the thing. On your turn, you get to remove either 1 or 2 or 3 of the consecutive lowest numbers. Let's say you go first. You can either say, I cross out 1, or I cross out 1 and 2, or I cross out 1 and 2 and 3. And in the end, the player who takes 21, who crosses out the 21, wins. Okay. Now, in a certain sense, it's a very simple game. On the other hand, it takes students a while to catch on to actually what's going on here. In fact, some of them never get it.

7.14 But you can simplify it. You can help them, like, say it's just up to 17. Well, then it's a little easier to see what's going on. If I lower that to 13, it gets a bit easier than that. But most people don't actually get what's happening until I lower it to a game of seven. Then you can pretty quickly start seeing what happens. In this case, the player who takes seven wins. If you're playing, you realize, wait. Okay, you can take up to three numbers.

7.15 So, if I hand this to the other player with just seven left, or with six and seven left, or with five and six and seven, that player wins. And so, I don't want to do that. I want to make sure that, let's say, four, five, six, and seven are handed over. So suppose I'm playing first in this game, then I can say, I take one, two, and three. Then the most that the next player can take is to take through six, and I win because I get seven. Or suppose I take one, then the other player can say, "I take two and three," and now I'm the one who's stuck.

7.16 So, you basically always want to do it in such a way that the next number after your turn is going to be a multiple of four. If you play that way, you will win. If you play first, you actually have an advantage. But in fact, you will win, really, if you play as well as possible, because you can put the other person constantly behind and you can guarantee that you can always get it to that multiple of four. In that case, you'll win.

7.17 So that's why it's easy for me to play against them and win, because usually they don't figure it out. No matter, even if they go first, I realize, "Oh, I know exactly what to do." Now, if they're clever, they can actually guarantee that I can't win. But it's not until the game of seven that some people start figuring it out. This is an example of a sequential game that, in a certain sense, is very simple. Once you see how to play it, you realize, oh, this is kind of trivial.

7.18 It's a bit like tic-tac-toe. I remember as a little kid playing tic-tac-toe. Then you realize there's an algorithm for this. I'm guaranteed not to lose if I play in the right way. With my little brother, at that point, I realized, my first move is always going to be stupid, so that then I see if I can recover from the stupid move, and then it got interesting again. But this is a game where, in a sense, you could do that. You could make a foolish play, even if you're going first, and then see if you can recover and outsmart the next person.

- 7.19** But the point of this is, sometimes there is a huge advantage in a sequential game to going first. Sometimes there's a huge advantage to going second. We're going to have to pay attention to that kind of thing, because sometimes the first move, if it's clever, can actually guarantee the outcome of the game. Well, how do we do this? Notice what happened when I took you through how I figured out the game of seven, and then figured out, oh, it's the same for all those other games.
- 7.20** It would be different, by the way, if it ended on an even number. But if it ends on an odd number, that's just something like 7 or 13 or 17 or 21, that strategy will work. But here's what we did. We looked ahead to the ending. We said, I want to be in a position to get seven, and so I'm going to be in that position, being able to take one, two, or three of these numbers, as long as we're up to there, at least. Then I think, let's work backwards. How do I get there?
- 7.21** That is going to be our general strategy in thinking about these sequential games. It's called backward induction. I think, how do I win in the end? What would the ending of the game look like if I win? Then I work backwards. How do I get to that position? I take that step by step. Now, in some games, like 21, it's really remarkably easy to do that. You can figure out, I want to be here. How do I get there? Once you see the pattern, you realize, oh, that's how I do it. But if I'm thinking of it at the start, the way most of my students do, they don't get it. They're just taking some number more or less at random.
- 7.22** Whereas if you work backwards from the ending, you can actually, as it were, solve the game. Now, not all games are like that. It's not as if you can easily say, what would checkmate look like? There are many things checkmate would look like. That's part of the reason chess is an interesting game. It's immensely complicated, and it requires supercomputers to even play it as well as a really good human. But typical games aren't like that. We think, let's think about the ending. How do I win? How do I put myself in this position? And then work backwards, step by step, to see how I get there.
- 7.23** Here's another example of such a game. This is called the game of Nim. There are two piles of matches and two players, and a player in her turn can remove any number of matches from either pile, but only from one pile at a time. Now, that's an odd rule. What's the point of that? Well, each player is required to remove some matches if either pile has any matches in it, and the player who removes the last match wins the game.
- 7.24** Suppose each pile, to make it simple, contains just three matches. Does either player have an advantage in playing this game? You think about what winning would look like, and you realize, wait, I can take any number of matches from a pile? Why don't I just grab all the ones from the first pile? Well, because then the second player grabs all the ones from the other pile and wins the game. And so, if I'm starting out, I realize, I don't want to do that. What do I want to have happen? I want to have some left in just one pile, so that I can take them.

- 7.25 But now, how do I get myself into that position? Let's say each one does have three. I want there to be some matches left here. And so, I have to think, how do I get you to leave me just one pile with some matches? It's actually a really tricky thing to do. It's a kind of thinking we're not used to. Even though it's an incredibly simple game, you might think, "I'm not used to thinking this way." I want to put you in that position.
- 7.26 Now, if I just say, hey, look, I'll just take one from this and all of these are gone, and then you take one and then I get the third, why would you take just one? You'll take all the rest. So I need to actually have matches in both, but one match in both, and hand it to you in that way. Then you can only take from one pile, so you'll take one. Then I'm left taking the other. So I have to somehow get to that one-in-one situation. Well, how do I do that? That's the tricky part. Okay, that's hard.
- 7.27 For the moment, I'm going to leave that as an exercise to the viewer because you think, wait, I've got to get you to leave that situation. But why would you do that? Well, because you don't want to take all the ones from one pile, right? Because after all, if you do, then I take the last one and win. So, actually, here's a way you might start thinking about it. This is the way a student will typically approach it and say, what if I take one of them? So, I've got now—let me pretend my fingers are matches. Okay, here are the piles.
- 7.28 I think, if I'm going first, I don't know what to do, so I'll take one here. Then you think, what? What if I take—well, wait a minute. I can't take all of these, for example, or you win; you get the last match. I've got to leave things in both piles, so I'll just take one here. You take one there, I take one here, and now you take one. I win. One way to do it is to start thinking, all right, that worked out for me. But on the other hand, what if I take one and then you take two? Now, what do I do? Maybe I take one, you take one, I win again.
- 7.29 You might start thinking, look, there is an advantage in this case to playing first, partly because there's an even number. I'm always leaving an odd number. That's a key to how I end up playing this game successfully. But notice again what we've done. We've said, all right, partly we figure it out just by trial and error; what if we did this and see if that works out? But partly, we think in the end, I want to be in this type of situation. How do I get there? And that helps me a great deal.
- 7.30 Now, one way we're going to draw these, instead of just boxes with columns and rows as we did with simultaneous games, we're going to need sequential games to have a kind of structure that's more complicated. We'll represent them in terms of game trees. In a simultaneous game, the moves are hidden from each player as they're moving, and so the order doesn't matter, but here the moves are ordered. In a game of perfect information, which we'll restrict ourselves to for the moment, the moves are ordered and they're known, and we need to know what the payoffs are in advance.

- 7.31** So we need to specify decision nodes, like, whose turn is it? Who's deciding now? Then we talk about the actions, the various possibilities they have for making a decision. Then once all the players have moved and we reach the end of the game, what are the payoffs? What are the results there? I've drawn one here that's very simple. Player one gets to move left or gets to move right. Then player two, in either case, no matter what they've done, gets to move up or gets to move down.
- 7.32** Then there are different players. If it's been left and up, the first player gets one, the second player gets three. If it's up and, or left and down, then three and two are the payoffs. If it's right and up, four and two are the payoffs. Then if it's right and then down, three and one are the payoffs. Now I'm thinking, as player one, and I think, my payoffs are the first number in those pairs, so one, three, four, or three. Well, I want to end up with the four. That's the best I can do. So I'm thinking, yeah, I want to go right and have the other player go up.
- 7.33** But now I realize, so I move right. That turns out to be a good move in this game for me. That'll be the right move. But I don't really know that yet. How would I figure that out? I've got to think, all right, I get there. But before I get there, player two is going to have a move. And so, I need to think ahead, to think, what is player two going to do? Well, if they have that choice, I've gone right, and now they can either go up, payoffs four and two, or they can go down, payoffs three and one. So what are they going to do?
- 7.34** Their choice is to get two units of value or one unit. They'd prefer the two. So they will go up in that circumstance. But that's great for me. So if I go right, they're faced with that choice, but they have an incentive to do exactly what pays off for, as it happens, both of us. We're both better off. But that player is going to be motivated to get the two instead of the one, and that gives me four instead of three. Yay. So, by going right, player one can actually win this game.
- 7.35** Now, suppose I am player two, then I'm thinking, let's see. My payoffs are going from right to left, one, two, two, and three. I'd like to get that three. But that requires, not only my decision to go up in that context, but player one to go left. But player one has the first move. I have no control over that, and so I can't get there, because I look at this and I think, what's player one going to do? They're going to go right. So I don't really have any way of getting my maximum value. This game is going to end up at four, two, at right and up, if each player plays rationally.
- 7.36** So that's how we're going to think about these games. We'll call these games in extensive form, where we represent a sequence in terms of a tree. It's a directed graph, and each of the vertices is going to be a position in the game. All right, here's where we've reached; now your turn. There will be a root that represents the starting position of the game. Who goes first? There will be an initial decision by some player. Then wherever it ends, that's a terminal position. That will be a way the game can end.

- 7.37 And so, it's really intuitive, in a sense. We start out with this. This player goes first. Here are their possible choices. We draw those. What happens in those cases? Well, it goes to the other player's turn, let's say. Then they have these choices. Then maybe it goes back to player one. They have these choices. Finally, we get to places where the game just ends. So, here would be an example of a game like that. We start out with player number one having decision one, their first decision. And so, it might be to go this way or that way.
- 7.38 Then at each of those, player two gets a turn, so player two now has a decision to make. I can go this way or that way. They might go this way, make choice one, or over here, they make choice two. Then in this simple game, it never goes back to one. It's just, we're done. The payoffs are, for players one and two, it could be one and five, or one and seven, or 10 and two, or negative one and zero. But in any case, if I'm player one, I'm interested in maximizing my value. If I'm player two, I'm interested in maximizing my value.
- 7.39 Now, this game turns out to be, even though it's incredibly simple, somewhat interesting from our point of view. Here's a game history. That is to say, the first player says, "Well, I'm going this way." Then the second player says, "Well, I'm going that way." That will be a possible history of the game. In fact, if you think about this, you realize, there are four possible histories here. We could end up here or here or here or here. We can go different ways depending on what moves the players make.
- 7.40 Now, this is an incredibly simple game, not only because each just plays once, but also because each just has two choices at each node. If you think, for example, I'm white in a chess match and I am faced with the opening move, there are a lot of opening moves I can make. It's not like I go, "Well, this pawn or that pawn, I'm done." There are many things I can do, and that's true of a lot of games.
- 7.41 So here is a subhistory where player one has made that choice, but that's only part of this game. That's a subhistory; it's not a full history, because the game didn't end. A history will take us all the way to the end of a game. This is a subhistory. It's just one of the moves in the game, one of the parts of the game. We can actually come up with official definitions here, but writing out the definitions may help you or it may hinder your understanding, to be frank, because the intuitive idea here is really clear. It's like, here's the first move. I have these choices. Then another player moves, they have these choices, and we just keep going down until the game is over.
- 7.42 What happens here is a strategy. Now, this is why it's worth maybe thinking about the definition. What does a strategy look like for me? Well, it's not just, here's what I'll do in this first move. I have to think about all the subsequent moves where it gets to be my turn and say what I'm going to do in those cases. If I'm going to at least move again, it's not just a matter of, my first choice, I do this. That's a strategy. It'll be, now what subsequent choices might I make and what will I do then?

- 7.43 Now, most of us in playing a game have not thought through all of those strategies, and for good reason. What are all the possible chess positions where I might have to make a move? It's a vast number. I would think even a chess master has not just thought them all through and said, "Well, oh, yes, that's position number 357, 609. Here's what I do in that case." So, the more complicated the game, the less likely it is you'll have thought through that particular situation.
- 7.44 But a strategy would actually specify that. In our simple games here, it's not hard to see what these strategies might be. Then an action is just a move. You're at a decision node, it's your turn, in other words, and you do something. You make a choice, you go this way or that way, you either spin the spinner or you roll the dice or whatever it is in that particular game.
- 7.45 Now, let's go back to that game we were talking about a moment ago. Here is an action for player two. Add a decision node. In this case, I've designated as 2.1, to indicate it's now the second player's turn. This is option one that they might choose from. They play A, the thing that I've labeled as A. In that case, we end up with a payoff of player one getting one unit and player two getting five. Well, that's one option.
- 7.46 Here's another strategy for player two. The player two could say, if player one has gone this way, then I will choose this. If player one has gone this way, then I'll choose that. And so, because player two is playing second, they can't just say, "I'll do this." They have to say, "Well, if the other player—if I'm in this position, I'll do that. If I'm in this position because of the other player's move, then I'll do that."
- 7.47 And so, what does the strategy look like? Well, it's telling me what I'll do at every time it's my turn. So, player one might say, I can go left or right. I can choose A or B. I'll choose A. But then if I end up in a situation where my choice is between C and D, given what the other player did, I choose D. And so A, D would be one strategy. I'll move here if I'm in that situation; I'll move there if I'm in that situation.
- 7.48 Now, what is a strategy profile? Well, that's just the set of strategies for all of the players. That's going to determine a path through the tree, because player one might say, "Well, if it's my move first, I do this." Player two will say, "If you do that, then I'm going to do that." That'll be part of their strategy. Player one, "Well, if you do that, then I'm going to do this," and so on. We can actually have these strategies determine a path through the whole tree. It will determine an entire history.
- 7.49 Now, let's think about what a Nash equilibrium would be for such a game. It's going to be a little more complicated and, frankly, a little bit uglier than what we've been talking about up to now. Because suppose we tried to convert this into normal form; we realize, now each player doesn't just have two choices. It's more complicated. I have to think, what are the possibilities in this game? And I realize it might be that a strategy here, in a sense for player one, is very simple. Look at the tree again. Player one just faces a choice between X and Y there.

- 7.50** But then what player two does depends on the situation they're in, given player one's choice. So, the choice might be between A and B, or it might be between C and D. That tells us, good, the players are in different positions. Player one can choose X or Y. Player two can say, here's one strategy for me, A and then C. Here's another, A and then D. One is B and then C. One is B and then D. And so, it's a bit more complicated. The result of all that is that we have to think through the best response in each of those cases. It's a little harder.
- 7.51** If I'm the player who has the choice between X and Y, I'm going to think, suppose you play A, C. That's your strategy. Well, then I should play Y. I get 10 units of value out of that. Suppose you're playing A and D. Well, then if I play Y, I actually lose points. That's not good for me. I'd better play X. Suppose you're playing B and C. Well, then I get 10 if I play Y, so good. Y looks really good. But what if you're playing B and D? Then actually, again, I'm going to lose one, so I'd better pick X.
- 7.52** What are the Nash equilibria here? Well, the ones where both players are playing their best responses, and so I can think that through from the other player's point of view and think, if you play X, then I'm better off playing either B, C, or D. If you play Y, I'm better off playing A, C, or B. And so, we end up saying the Nash equilibria are, Y together with A and C, Y together with B and C, and then X together with B and D.
- 7.53** Now, all of that's kind of ugly, right? Kind of complicated. You might think, then, yeah, there are different outcomes of the game that are possible here, so I guess it all just depends. In fact, if you're a player and you're analyzing it this way, you might think, so I'm the player who has this first choice between X and Y. What should I do? Well, I guess, I don't know. A couple of Nash equilibria are good for me, some are bad for me. It's kind of 50-50, but I get a much bigger payoff for Y, so I'll choose Y.
- 7.54** That's, I mean, that's not unreasonable here, but on the other hand, it's a pretty sloppy way of thinking, and it's not always going to work out. So, how should we really think about it? Well, I go back to this tree, and I think, suppose I choose X. Then I leave player two in a position where they have to choose A or B. What are they going to choose? Now, if they choose A, they get a payoff of five. If they choose B, they get a payoff of seven. They're going to choose B.
- 7.55** And so, if I take X, it's not like I really have to think through, "Oh, they might take A or they might take B." I can look at that and say, "If they're rational, they'll choose B." Similarly, suppose I choose Y. That's what I really want to do because that gives me a payoff of 10. The thinking a moment ago said, "Yeah, might as well do that. It's kind of 50-50, big payoff. Hey, I'll do it." But wait a minute. They get a choice here. They're thinking, "Yeah, I can pick C or D." Now, what are they going to do?
- 7.56** Well, actually, it works out well for me in this case, because I can say, would player two play D? They've got a choice. Once I've made choice Y, they can either get two units of

value or none. They're going to take two. And so, in that case, they're not going to pick the one that yields a bad result for me. I'm going to win.

- 7.57** Now, if I analyze it in terms of, "Here's what I'd really like to do. I'd like to end up with 10 points there, but how do I get there?", player two has a choice, and they get either two units of value or zero. Well, great, because they're going to pick the two, and that gives me the 10. So, actually, Y is the right choice here. But before, I was just guessing. Here, I can say, it's rational for them.
- 7.58** Now, suppose that last node were not minus one and zero, but were minus one and five, let's say, then I'd know, faced with that choice, they'd choose D, and I end up losing, in which case, I'd better not choose Y, because that player is then going to be putting me in a difficult position.
- 7.59** So, one reason I'm a horrible chess player is that I find this really hard to do. I just do that sloppy thinking and think, who knows what that guy is going to do? I basically assume that my opponent is as dumb as I am about playing chess. And so, I say, yeah, who knows? It's like there's a chance he's going to do this, chance he's going to do that. And so, I often make stupid moves.
- 7.60** In fact, once playing Hans Kamp, I actually lost in the fewest possible moves ever. It was like that game that people talk about, like, "Oh, but that actually happened to me." Because I was really stupid and just thinking, "Who knows what Hans is going to do?" And I'm like, "Oh, that's what Hans is going to do." He actually understands how to play this game.
- 7.61** Anyway, that's what happens when we do this kind of thinking. We can actually say, wait a minute. What is the next player going to do? And then I can anticipate that and I can realize, oh, then I'm safe in making the choice that looks really attractive here. Or if the game were a little different, I'd say, it's really attractive, but it's a trap. It's a trap. If I do that, the other player is going to make this move, and I'm in big trouble.
- 7.62** Well, if we adopt this, we can think that game I mentioned earlier through, and now we have the tools to fully understand what we were doing. We can say, yes, I want to choose the best action for myself as player one, and ideally for me, I end up with that four-point prize. How do I get there? Well, I go right. The second player has that choice. They're going to get either two or one. And so, wonderful, they're going to choose two; they're going to go up, and I'm going to end up with my four points. So again, I'm not guessing here. I can actually say, yes, I can actually achieve that. It will be rational for them to make a move that benefits me as well.
- 7.63** But not all games are going to be like that. Here is a classic example of this backward induction applied to something that happens in economics all the time. We have a challenger who might decide to enter a market. Say you're a company, and there's another company that's been succeeding well in this new market they developed with

a product. You think, we ought to have a competitor to that. But you're thinking, do we go into the market or not?

7.64 I mean, this is risky. If we spend a lot to develop our own product here and compete against them, they've got the advantage of already having a position in that market. So, can we challenge them successfully or not? We've got to invest quite a bit, so it's a risky move. So, what should we do? Well, we can either go into this market or we can stay out and just let them have, as it were, a monopoly. Then we have to think through, "But they're going to respond in some way." Suppose we enter this market. Now, if we stay out, they're just going to do their thing. Game ends. But if we decide to go into this market, they're going to typically respond.

7.65 Now, what might they do? They might accommodate. That is to say, they might say, "Okay, great. We'll try to outcompete you." But they aren't going to do anything to try to undercut us. Or maybe they'll fight. Maybe they'll drop the price of their product to try to undercut us, so that we're losing money, even if we successfully take over part of that market. If that happens, then we have to decide what to do in turn and so on, but we're going to end up in a price war, and it'll be damaging to our company. So what should we do?

7.66 Well, suppose the payoffs are like this. If we stay out, we gain no profits here. Let's say the other company ends up with two units of whatever this is. Maybe it's \$2 million profits. Let's just say these are millions. If we enter this, then we can actually take over a significant part of the market. In fact, we think we can make the product better. We'll have an improved version. And so, that's great. We'll have now the \$2 million, and they'll just have \$1 million. Their profits will be cut in half from this particular product line.

7.67 But then what if they fight? What if we're ending up in a price war? Well, that's going to be costly for us and costly for them. Maybe we lose \$3 million. We lose what we actually spent to develop this product line and all the marketing and all of that. They will also suffer. They'll end up with a loss of a million dollars because they are actually now selling their product at a loss, just to drive us out of the market.

7.68 So we're trying to decide whether to enter this market or not. We think there's a real risk of a price war. But on the other hand, there's a possibility of real profits. Now, if so far we're just thinking of it in those terms, and thinking about it in terms of, let's say, expected utility and all of that, we think, "Well, what's the other player going to do?" I don't know. Now, maybe it's like, hey, how risk-averse am I? Or how tolerant of risk am I? It'll look like that kind of decision.

7.69 But once we analyze it in terms of a sequential game, we realize it's not really that kind of decision. We have to say, suppose we stay out; well, then that's the end of the game. But suppose we enter in. Now it is their choice. Do they fight a price war against us, or do they accommodate and just try to compete normally? And then we think, now from

their point of view, what are they going to do? Well, if they accommodate, they still have a million dollars. Their profits are cut in half, but they're still making a million dollars.

- 7.70** On the other hand, if they engage in this price war, they're losing a million dollars. So what choice are they going to make? Well, if they're rational and think we can make a million dollars or lose a million dollars, they're going to make a million dollars. And so, we say, so let's get into the market, because they are going to, if they're rational, accommodate us, and then we'll be making \$2 million. We'll do well.
- 7.71** So, now it's not a question of how risk-averse am I; it's really just a matter of, put yourself in their shoes. What are they going to do? It's going to be really costly for them to actually fight a price war against us, so they're not going to do it, and so we can actually enter the market. Now, that looks reassuring, like, great. But the moment you start thinking about this seriously, you realize it's a little simple, right? Why is it a little simple?
- 7.72** Well, first of all, we're assuming that those are the outcomes, and maybe that won't be the outcome. Maybe our product is not going to be as popular with the public as their product. And so, I'm saying, "Oh, two and one. We'll make \$2 million. They'll have their profits cut to \$1 million." Maybe, but maybe not. Maybe, actually, our marketing won't do well. Or maybe consumers will try our product out and find out it breaks down pretty quickly, and they'll go back to that other product and so on. Maybe that won't work. There's a risk there.
- 7.73** A second risk, though, is, here I'm thinking, they're going to make a million dollars a year or lose a million dollars in this coming year; what are they going to choose? But from their point of view, I'm not just thinking about this year. I'm thinking about the future. If I drive you out, then I keep my \$2 million after that. Yeah, I lose a million this year instead of gaining a million. But on the other hand, I'm then, once you're out of this market and I've driven you out, then I'm back to my \$2 million for an indefinite future.
- 7.74** You are stuck now out of the market, whereas if I accommodate you, then I'm making a million dollars a year. So I'm thinking in terms of the future here. The game doesn't end there. In other words, it keeps going. When I think about the keep going part, I realize, yeah, what's the next move? If you are then, as a result of the price war, out of this market, then I go back to making \$2 million a year.
- 7.75** So, in short, this looks like it's a great simplification, but actually, that's a bit misleading, because if I'm the incumbent who's got the control of this market, I'm not going to just say, "Oh, yeah, competing on price, that'd be pretty rough. We're going to lose money, so let's don't do it." That's way too simple. I'm going to think, "If I can successfully drive you out and keep my monopoly, then I've got a future of big profits," and here, I've got

a future of much less profit, so maybe it's worth losing that money for a year or two in order to drive you out.

7.76 Anyway, this is a kind of danger to thinking about this. We tend to simplify games in our heads, especially, but also sometimes in our economic analysis, in such a way that looks like, oh, we can predict what the other player is going to do. But they won't always do that, because they may think, "That's not the end of the game." In general, we have to be cautious about that then.

7.77 In some games, you know what the end is. Checkmate, game over. If it's a video game, you directly get a game over. You know what the end of the game is. But in real human relationships, in real economic analysis and so on, you may or may not be right about where the game ends. There may be further moves to be made.

7.78 So, in this case, it would look, if we put it in one of those tables, as if there are multiple Nash equilibria, one where we enter the market and then they accommodate, and one where we stay out. But on the other hand—oh, sorry, the other where we enter the market and they fight. But there's a simple idea here. Look, one of those Nash equilibria is really not very credible, at least as this game stands. If we just thought about it as representing either the whole future or just assume that, after that, that they're very short-sighted, just thinking about that. Then we'd say, look, one of them requires them to actually carry out a non-credible threat.

7.79 And so, we're going to refer to the Nash equilibrium that survives that test. But would they rationally make that choice? That is going to be called a subgame perfect Nash equilibrium. That becomes one of the central concepts in sequential game theory. It's not just a Nash equilibrium now, it's a subgame perfect Nash equilibrium. Now, what does that mean? Well, it's an equilibrium we can achieve if all the players are behaving rationally. It is one that we can achieve in a way that doesn't involve any incredible threats.

7.80 That is to say, I may say, "Look, if you do that, I'm going to— I don't know. If you do that, I'm going to cut off my nose." Well, you're going to think, "You're not going to cut off your nose. That's not a credible threat." If I tell my little kid, "Come on, study for that test, or I'll cut off my own nose," she's not going to believe me, and rightly so, I'm not going to do it. There are incredible threats. There are credible threats. In this case, fighting the price war, is that credible or incredible?

7.81 Well, I have to think, what does it look like from their point of view? Even if you think this diagram is way too simple, there can be situations where it would simply be too costly in order for them to actually engage in that price war. We realize, look, you're not in a position to do that. You can't afford that. You may threaten, if you enter this market, "We'll drop our prices," and blah, blah, blah. I think, yeah, I'm looking at your financial statements. You're not going to do that. It would put you out of business. And so, in that case, it really would be an incredible threat. At that point, we say, yeah,

there is an equilibrium here where we make those moves, but you're not going to make that move. I can realize that's not going to happen.

- 7.82** What we do then is start at the last decision nodes. We work backwards through backwards induction. We think, "What is the best action for that player at that node?" We find out, given that everybody's making their best choice along the way, where do we end up? That is going to be a subgame perfect Nash equilibrium.
- 7.83** In that simple game we started with, it'll be going right and then going up. So player one gets four points, player two gets two points. But sometimes, it's very useful in that respect. We can say, "Here's where we're going to end up." And so, it's really helpful in playing games. You think, "If I do that," not just, "Hey, let's roll the dice." There are times when decision-makers do that.
- 7.84** We now have access to the decision-making, if I can call it that, of the Japanese high council before Pearl Harbor, when they made the decision to attack Pearl Harbor. You might expect deep strategic analysis, lots of sequential game theory. No, instead, you find somebody saying, "Maybe the Americans will attack us in a few years." And the naval minister saying, "Well, will they attack us or won't they?" Nobody responds. He asks two more times. Nobody responds. He finally says, "I don't think they'll do it." And then another person says, "Well, maybe they will, maybe they won't. There are times when you just have to jump off the veranda of the temple." And so, they do. That is the level of strategic analysis that went on here?
- 7.85** I remember reading the White House transcripts from the Nixon era, and it's not like I was astounded, not because of the Watergate-related stuff. It seemed to me kind of thin on all of that. Instead, it was just like, wait, this is the level of analysis going on in the White House? Nixon was a highly intelligent man, but still—and it was better than the, "Let's just jump off the temple," but not a lot better. Okay, so you might think, wait a minute. We'd be a lot better off, generally, if people were making decisions according to thinking about backwards induction and how they actually get, given that other players are going to be involved, to where they really want to go.
- 7.86** But sometimes, subgame perfect Nash equilibria aren't so great. Let's talk about a few cases like that where it may not really give us something that's very useful, or maybe it will. Here's a simple ultimatum game. Alf is going to choose. Remember the ultimatum game, the idea is, you get \$100 and then you can share it with another person who can accept their share or can reject it, in which case you get nothing. Alf, let's say, goes first, and he can either make a fair offer to Betty, let's say it's somewhere around half, would be a fair offer. He can also make an unfair offer.
- 7.87** As we saw, one Nash equilibrium here is simply, "Oh, I'll give you \$1. That'll make you better off. I'll keep \$99." But that's a pretty unfair offer. And then Betty has a choice. She can accept or reject in each case. We think through, I make a fair offer. I say, "Let's split it 50-50, Betty." She can accept or reject. Well, what's she going to do? I think,

where is she better off? If she rejects it, we both get zero. But if she accepts, we both get half the value. But then, what if I make an unfair offer? I keep \$99. Here, Betty, here's \$1. Well, she can accept that or she can reject.

7.88 Now, the classic way of thinking about this is to say, she's a dollar better off if she accepts, so she's going to accept. And so, what should I, as Alf, do? Make an unfair offer. Because, actually, look, as long as I offer her anything at all, she's going to accept. It's rational for her to accept. She's better off getting something than getting nothing. And so, great, I'll make an unfair offer.

7.89 And that is one way to think about this. One equilibrium here is where Alf plays an unfair offer, and Betty accepts, no matter what he plays. Maybe he just offers \$1. She said, "Okay, I guess I'm better off with even \$1." But another equilibrium is where he plays a fair offer, and then she rejects it if it's unfair, but she accepts it if it's fair. In this case, what's going to happen is that Alf makes a fair offer. "Hey, let's split it 50-50," and Betty accepts. So both of those are Nash equilibria.

7.90 Now, you might think, what's the subgame perfect one? If Alf looks ahead and says, "Hm, she's going to accept no matter what," well, I might as well choose what advantages me, and make the unfair offer. But on the other hand, if you look at what people really do, that's not what they do. Most people propose somewhere between 40% and 50%, and most people who are responding in Betty's position actually will reject anything under about a quarter. So I keep \$75; I'll give you \$25—it's kind of iffy whether somebody will accept that. But they probably will. But if I say, I'll give you \$15, and I'll keep \$85, it turns out most people will reject that offer. Just say, "That's not fair."

7.91 Now, that's interesting because it challenges this idea of pure rationality. I should accept even if you've offered me \$1 because, hey, I'm a dollar better off. But most people won't do it. They think, "That stinks. I don't like you. I don't want you to have \$99, and keeping you from having \$99 is worth it to me. All I have to do is lose \$1, so I'll do it."

7.92 Now, that suggests that rationality maybe isn't purely self-interested, or maybe it's just, I think, a different thing going on, which is more, going back to Hume on conventions, it might be like, look, there are certain conventions about how we treat each other in this society, and we don't take advantage of people like that. So, I agree, I'm a little better off, but you're way better off. That's not fair. That's the kind of division that, if I reward that, is going to lead to social chaos, so I won't reward it.

7.93 But in any case, whatever is going on here, it suggests that only one, as it happens, of those Nash equilibria is really, given human psychology, likely to happen. It may be that subgame perfect equilibrium is a useful concept, but it's actually a bit too weak to capture what really happens. We need to think about the things other than just pure rationality—am I any better off?—that in fact go on in people's heads.

- 7.94** And one of them might be, as we saw in that simple "are you in or are you out" game, it might be, hey, look, that's not the whole future. There are further decisions to be made. But in some cases like this, it may be, there are larger social issues at stake. I don't want to be treated that way. It's important to uphold those norms for general purposes that have nothing to do with this particular situation. So, I think there's an interesting issue here about what other factors might be involved.
- 7.95** But in any case, if we want to find these subgame perfect equilibria, what we want to do is work backwards, start with the endpoints, and then think, what decisions are people going to be making if they're rational at each of those subgames, and work back toward the beginning. If we've got people making rational decisions all the way through, then we end up in a position that can be a subgame perfect Nash equilibrium. There may be other equilibria, but we're not going to get there, because people would not choose that.
- 7.96** Okay, so this is going to refine the idea of a Nash equilibrium. Lots of Nash equilibria, we can now toss that aside. There are going to often be a lot of them in these sequential games, but most of them are uninteresting. We only care about the ones that we could get to, given rational decisions. I don't want to think about equilibria where, yeah, I mean, if I did that, and then he did that, and then I did this, and then—but we'd all be crazy to get there. That's a Nash equilibrium, but it's not going to happen.
- 7.97** And so, we can think through officially how this happens. Now, here's a little game that I think will give us the feeling of this concept more than actual looking at formal definitions. Player one can choose A or B, and if player one chooses B, they get two, and the other player just gets nothing. But now the second player gets to choose, and if they make a certain choice, then it goes back to player one. But they can also end the game. They get one—or, sorry, they get five, player one gets one. Then it goes back to player one, and depending on the choice they make, they can get either 10 or nine, but the other player gets two or 10.
- 7.98** I think, how's this likely to go? I think this is a kind of interesting one. If I think backwards, I'm player one, let's say, so I think, "Well, that 10-point prize looks awfully juicy. I'd like to get there." So can I do it? Well, if I choose little A, then it goes to player two. They're looking at this, and they're thinking, "Okay, I can, given my choices now, get two points or 10 points or five points. I'd love that 10. Can I get there? Suppose I choose big A."
- 7.99** Well, then it goes back to player one, and they're going to get 10 or nine. They have an incentive to choose the 10. They're going to choose that, and I'm left with only two, whereas if I choose capital letter B, I get five. And so, I'm going to say, "Oh, I'd better choose B and end the game right there," which means player one only gets one point. So, if I'm player one, I think, "Oh, if I take A in quest of the nine or the 10, I'm going to

end up with one, so I'm going to choose B." And so, the other guy gets nothing, but I at least get two.

- 7.100** What's the subgame perfect Nash equilibrium here? It's actually that the first player ends the game promptly by just saying, "I get two points, you get nothing." That's weird. You'd think, but look, we can both be better off down there. But yeah, you're not going to let me get there, because you're going to take the five, and then the game on next turn where I'm even worse off. Okay.
- 7.101** So, there's a general point we've been talking about here concerning credible and non-credible threats. Ordinarily, if you're going to threaten anybody, you want it to be a credible threat, a threat that you might actually carry out, not one that it would be completely irrational for you to carry out. You might think, yeah, a rational person, like a rational politician, a rational statesman, would never make a non-credible threat, would they? Oh, I don't know. I see lots of non-credible threats being made and so on. It's a good thing they aren't carried out.
- 7.102** But in any case, if we think through the way that game is going to be played, player one is going to say, "I end the game right here. I take my two and call it done." And so, player two is never even going to get to make a play. They could both do way better, but on the other hand, the incentive is to end the game early.
- 7.103** All right, let's try another one. Here's a simple game, actually, one that we've talked about before, so I think we've already actually gone through this one, where if I'm player one, I say, "I could get 10 points. Can I get there?" Well, if I make the right choice, then player two gets a choice, but then they end up with a choice between two and zero, they will choose the two. So yay, I can get to the 10 points. I just have to make the right choices. Now, that's only one of three Nash equilibria here, but it's the one that's subgame perfect. It's the one that results from rational choices from each player.
- 7.104** Okay, and there are other games like this. Here's a more complicated one, where player one says, "Yeah, I can choose T or D," and then it goes to player two, and they've got choices in each case. Now, if I'm player one, I'm thinking through, and I think, "Hm, okay, my best payoff is five. So, I would really like to go this way, then have them choose if they are going to go left or right. I'd like to choose— I hope they choose right. Then I can choose X, and I get five points, but that leaves them with zero." And so, I think, what are they going to do if I go that way?
- 7.105** Actually, then they look at it and they say, I could get a payoff of 10, but you would face a choice between five points and four points. You're going to choose the five, and I'll be left with nothing. So, I don't want to do that. It's not just, oh, it's risky to do that. I know it's rational for you to do something that will actually lead to my defeat if I do that, so I don't want to do that. I'll go the other way. In that case, you get two, I get five. And so, that's how the game is going to end up, if I go that way. If I go the other way and leave

it in player two's hands, then they've got a choice between zero and minus six. They're going to take the zero rather than lose the six, and so I get one point.

7.106 Actually, I realize, oh, at first this game looks pretty good for me. I could get five, I could get four. But now I realize, no, realistically, my choices are between one and two, because given the way the other player is going to play the game, that's what's going to end up. So it's rational for me to seek the two. So, in short, player one here should go up, and then player two should go left, and player two gets five points, player one only two. It turns out this is actually a surprisingly simple game once you think, what are people really going to do?

7.107 What are the morals here? Some Nash equilibria are really unrealistic in sequential play. Thinking about just what's a Nash equilibrium and what's not is, first of all, complicated, and secondly, not very helpful. We have to think, what is a subgame perfect Nash equilibrium that would actually result from rational choices? Because yeah, we could end up there, but we're not going to, because if you actually have a turn, you're not going to choose that, unless you are irrational in some way at that point, and I don't want to count on that. So, it's going to turn out that, if we analyze the game, there are going to be lots of paths that are just not going to be chosen, and we can ignore them.

7.108 Now, let's think about a stag hunt in this kind of context. In this case, we're a highly sophisticated hunter-gatherer tribe, and I've got cameras around this forest, so I know whether you're hunting rabbits, or whether you're staying at your post and helping me hunt the stag, and then I get to respond appropriately. Before, we said, there are two Nash equilibria here—one where we both hunt rabbits, one that's better where we both hunt the stag, and it becomes a difficult question of trust. Can I trust you to stay at your post, or are you going to go off hunting a rabbit?

7.109 Well, that is different if we think of it as sequential, where, I think it's still a question of trust. You can say, "Oh, yeah, I'll help you hunt the stag," but I still can't trust you to do it. But suppose I can monitor that and then respond. So I've got my little camera at your post, and I can watch and I can check. Yep, still there, so this is a very strange hunter-gatherer society, but maybe it's our society after the holocaust, when the nuclear weapons have destroyed things, but miraculously left our electronic devices alone. This is getting more implausible. I'm not helping myself.

7.110 But anyway, so I think through, good. How do I think about this? Now, suppose you are—I think, I'm going to go hunt the stag or hunt the hare, but then you've got this choice. That's one way of thinking about it. But another way of thinking about it is, I'm monitoring what you're actually doing. So, actually, I am watching. You're player one now. I'm responding to you. And so, I'm seeing, is he there helping me hunt the stag or not? Is he chasing rabbits?

- 7.111** I think, what am I going to do if he's there at his post, hunting the stag? Well, I'm clearly better off hunting the stag, and so I'm going to stay at my post. Now, what if he's hunting rabbits? Well, then I'm going to go hunt rabbits. It looks like we have the same Nash equilibria. But now, I think, from his point of view, what's he actually going to do? Forget the cameras for a moment and think of it as sequential. I think, what would it be rational for him to do?
- 7.112** Now, here, since, in effect, I could turn it around and think, let's say he's player two; then it's going to be rational for him, if I'm hunting the stag, to actually hunt the stag with me. If, on the other hand, I'm off hunting rabbits, then it's going to be rational for him to hunt rabbits. So, if I can monitor him, then I can find out what he's doing and respond appropriately, making myself player two.
- 7.113** But on the other hand, if I think he can monitor me as well, suppose we have this mutually going thing, where we can see what the other player is doing; well, then we both have a very strong incentive to hunt the stag, because we realize, "Hey, look, as long as I can make sure he's doing it with me, this is the rational thing." So we're going to get a different outcome. It'll turn out that we both hunt the stag, because we can be aware of what the other player would do, and it would be rational for them to hunt the stag if we're hunting the stag. So it becomes rational for us to hunt the stag.
- 7.114** Now, here's a case. In that case, we can think this analysis is actually good. The better option ends up being subgame perfect. But now, what about this one, a centipede game? This is a game that challenges the utility of this idea. It's kind of like the sequential version of a prisoner's dilemma, if you want to think of it that way, which shows that Nash equilibria aren't always good. In this case, that even subgame perfect Nash equilibria aren't always good.
- 7.115** Why? Well, it's a little bit of a complicated game, but at each node, each player takes pass, in which case the pot grows, or takes whatever's in the pot and ends the game. At the first position, you can either take two points and the other person gets zero, or you can pass it to the other player. Now, if they take it, you get one and they get three. But if they pass it, then it goes back to you. If you take it, you get four. They only get two. If you pass it, and so on, it keeps going like this.
- 7.116** The whole idea is, in each case, if you pass, the pot grows, but your part gets a little less. Okay? In a sense, you have an incentive to say, "I should stop it here, because if they stop it at the next turn, I'm worse off." And yet, since it keeps growing, there's an incentive to keep this game going as long as possible. I've actually had my students play this game in class, and it turns out that it lasts quite a while, because I'm not going to tell them when the timer is going to go off.
- 7.117** So, it's just a question of, how long are we going to keep this going? They usually manage to keep it going so that the payoffs end up being seven or eight or nine or something, and then they start getting nervous. Like, the pot is getting big enough, and

at some point, the game is going to end here, so maybe I should take it. But what is the subgame perfect Nash equilibrium here? If we think about that and use backward induction, we end up saying, there are lots of Nash equilibria. But on the other hand, in every case, I think, oh, should I take it or should I pass it along? I could be worse off if I pass it along and then it ends at the next turn. I'd better take it.

- 7.118** And so, actually, the subgame perfect Nash equilibrium is that I just take at the very first opportunity. I take the two points, the other person gets zero, game over. Uninteresting. But in fact, nobody does that because everybody realizes, "But that would be a terrible outcome." And so, it requires trust. You've got to trust that the other player is also going to pass it along or else you'll be worse off. People are willing to do that for a while. But then the pot becomes significant enough that people start worrying.
- 7.119** Think about a startup company where everybody agrees to just share all the profits equally until, at some point, the profits start growing significantly. Then all of a sudden, it's like, I don't know. You know what? We really need a financial chief officer here. I'll take over that role. And all of a sudden—I mean, this happens very frequently. How did Microsoft start, and how did one person end being the one who's fabulously wealthy out of all that? You might think, yeah, at some point, somebody, as it were, takes and doesn't do it because the pot has grown big enough. And so, the incentives change.
- 7.120** Now, I think this is pretty interesting. People do tend to pass along. We can think about this in terms of Hobbes and the state of nature. People, in fact, don't fight the war of all against all, typically. They will at some point decide it's worth it to trust the person, or, things are good enough now that I can actually take my share. In politics, yes, there are arms races, and so on. But on the other hand, people don't just build weapons systems without end. They place some kind of limit on it.
- 7.121** In economics, firms will cooperate with each other, at least short term for a while, and people cooperate with each other much more than the theory predicts. Why is that true? I think it's because we're able to recognize that some of these subgame perfect Nash equilibria are nevertheless not very good. And so, we're willing to trust, realizing there's a risk here. This might put me in a worse situation, but it might put me in a much better situation if you turn out to be trustworthy.
- 7.122** And so, in that respect, it is very much like the stag hunt, very much like the prisoner's dilemma, where we can be a lot better off, both of us, if we trust each other, at least in the short term. At some point, it may be that we fear, oh, we're getting close to the end of these interactions, or we start thinking, the payoffs are big enough now, I can't resist. But that can happen.
- 7.123** And so, to close, we can think about that farmer's game, adapted from David Hume. We agree to help each other with our crops. That goes on for years. Why does it go on? In each case, I could say, it's best for me this year to have you help me with mine, and

then I don't help you with yours. But I realize that cuts off all future interactions. There's a lot of payoff to those future interactions, just as in the centipede game. In a way, this is like a centipede game. But if I know that this is the last time—I'm retiring and moving to Florida—should I bother to help you? That becomes the puzzle.

7.124 And so, we saw it's something like a prisoner's dilemma if we think about this as a simultaneous game. If we think of it as a sequential game, the neighbor can help or not, then I get the choice of helping or not, and the neighbor may well say that last year, "Look, if I help you, what are you going to actually do?", your first choice would be that you don't bother to help me. And so, if that's true, that's my worst-case scenario. So, will I help you, knowing that it's rational for you not to help me in return? Maybe I won't.

7.125 And so, it looks like, actually, the subgame perfect Nash equilibrium here, if I think, "You're not going to help me, no matter what, then I won't help you." But on the other hand, all too often, even if it's the last time, people actually do this. The employee who is retiring or is leaving the company will—and they have an incentive not to do much—but they sometimes continue to work hard up until the very last day. And why?

7.126 Well, Hume says, it's because we realize, just as in other kinds of games, we realize, like in the ultimatum game, "I'm better off with the dollar, but I don't want to reinforce a norm that says you can treat other people that way." So similarly here, I may say it's to my advantage not to help you in return, but nevertheless, I want to uphold a norm that we can trust each other, that we keep our promises, and that we can have these kinds of relationships. I don't want to undercut that.

7.127 And so, I may see that whatever small effect I can have on the larger social picture as more significant than whether I actually gain the dollar, or whether, in this case, I actually exert the energy to help you.

Lecture 8: Social Choice Theory

At a glance — Social choice theory, the final lecture.

- 8.1** In this final lecture, I'm going to do something insane. I am going to try to motivate and then prove Arrow's impossibility theorem for you, and have time left over to talk about the Gibbard-Satterthwaite theorem, and thereby introduce this whole field that really is, from one point of view, an offshoot of game theory. It's called social choice theory.
- 8.2** Now, most people would think this is a crazy thing to try to do in one hour, and maybe it is crazy, but I've got to give credit to my colleague, Johan Gustafsson, who wrote a paper on Arrow's theorem, proving Arrow's theorem, without any tears. It's the cleanest proof I've ever seen of this theorem, and it's what made me think I can actually do this. Now, I did it at my own university for my students, and they at least seemed to understand it. Maybe they understood it. Maybe they just nodded in ways that were encouraging, just to make me feel better. But in any case, that's what I'm going to be trying to do.
- 8.3** So, Aristotle noticed long ago that people want to own their own lives. They want to choose their goals and projects for themselves. So far, we've been assuming in game theory that people are making decisions in terms of their own rational self-interest, that they are exercising a certain view of rationality and actually pursuing whatever it is; whether it's a good thing that they're pursuing or a bad thing, they're doing it rationally, and they're making choices in terms of what is better for them, what they prefer, or what their own higher utility is, whatever that happens to be.
- 8.4** But a lot of decisions, we've started to notice, can't fully be explained in that way. People care more about each other than that would seem to indicate. We aren't entirely selfish in the way we make decisions. Moreover, we are often faced with decisions where, okay, there's a cost to me, but I think there's a broader social payoff. And so, we've seen people are more trusting, are more altruistic than game theory would actually predict. And so, we can think about this in a more systematic way.
- 8.5** We can think, yes, sometimes I'm making decisions in terms of the benefit to me. Maybe if I'm investing money in various things, I'm thinking, okay, I'm interested in the payoff to me of that investment. But in a lot of other kinds of situations in society, we're making choices that involve the welfare of other people as well as ourselves, maybe people we love or our friends or our associates or people just nearby us, even if they're strangers.
- 8.6** But sometimes we're making decisions together as part of a group, for example, when we vote in some sort of election. Then we realize, "I'm deciding for the group here. It's not just, what's better for me? What do I think is better for us?" Often we make those kinds of decisions. So yes, I want to live my own life. I want to own my own life. I want to choose my goals and projects for myself. But some of those involve other people,

and some of them involve all of us, the whole group that I'm a part of. That's what Arrow's theorem is about.

- 8.7** In general, that's really what social choice theory is about. How do we go about making decisions together? Not just me as an individual playing a game, but us collectively. So, it's also part of the theory of rational decision-making, but in this case, it's about collective decision-making, collective choices. How do we do things as a group?
- 8.8** Now, the moment you start thinking about this, you realize this is more complicated than what we've been talking about up to now, because it simplifies things a lot, if I think, "Hey, I'm just in it for me." I'm looking at my own utilities, my own preferences. But if I'm choosing for the group or just part of a group of people that's making a choice together, I realize, "Oh, this gets more complicated." Even if I and all other members of the group are rational, in the sense that our preferences are rational, they're transitive, they don't lead to any cycles or anything like that.
- 8.9** Nevertheless, the preferences of the group can lead to cycles. This is something noticed by Condorcet several hundred years ago. Every time there's more than two options, this can happen. So, here is the classic case. We have three people, A, B, and C, and they rank three options, X, Y, and Z, as their first choice, second choice, third choice. A prefers X to Y and Y to Z. B prefers Y to Z and Z to X. Then C prefers Z to X and X to Y.
- 8.10** Now, so far, there's nothing that's strange about that. That can easily happen. In fact, in groups, that kind of thing often happens. But now, suppose we think in terms of just comparing X and Y here. Both A and C say X is better than Y. B says, no, I think Y is quite a bit better than X. But still, it's two to one. If we adopt majority voting, for example, X is going to beat Y. Now, what about Y and Z? A likes Y better than Z. B likes Y better than Z. C disagrees, but again, it's two against one. What about X and Z? Gosh, two out of the three people prefer Z to X. So, again, it's two against one.
- 8.11** But put all those together. Suppose we just say, let's let the majority decide. That's what we do in democratic countries, right? That's what we do in republics. We have a majority of the people or a majority of the legislators making decisions, even a majority of the members of the Supreme Court or what have you. We adopt majority voting. We think that's a rational way to make choices.
- 8.12** But wait a minute. The majority says X is better than Y, that Y is better than Z, and that Z is better than X. And we have a cycle. If we have majority voting, we say, oh. Even though each of those individual voters is completely rational—there's nothing weird about those choices—nevertheless, the group, if we vote that way, ends up being led in a cycle. It looks like the group has no rational preference ordering, even though the individuals in it do. That's Condorcet's paradox. There is just no rational social ordering of these alternatives, given those preferences.
- 8.13** Starting from any alternative, moving to a socially preferred one, you can end up back where you started. This could actually lead to what is known in the literature as a kind

of money pump, where you say, "Oh, you prefer Z to X? Hey, pay me a buck, and I'll give you Z instead." You say, "You prefer Y to Z? Give me another dollar, I'll give you Y." "Okay, you prefer X to Y. Give me another dollar, I'll give you X." "Okay." Then you're going to say, "Wait, I'm \$3 poorer, and I'm back with X where I started." And so, this is not a good thing. This is irrational.

- 8.14** Now, in that situation, there's not what we call a Condorcet winner. In other words, there's nothing that could win a majority vote against every other option. In that kind of situation, we often get led into a group irrationality. So, people have studied this quite a lot. It's a real danger if you have a small number of people voting or if you have a large number of options. On the other hand, in large populations where there are relatively few options, it's really highly unlikely. It can happen, it's not very likely. It's not a big practical problem in a large democracy.
- 8.15** It is a big practical problem, however, in small groups. Your group at work is trying to decide what to do in a decision like this, or you're in some small—you're on a board of directors and you're trying to make a decision, then actually this is a significant danger. In fact, I've seen it happen in a board meeting like that, where people started out with option one. Somebody suggested improving it to option two. Then somebody said, well, how about option three? We said, yeah, that's better. How about option four? That's even better.
- 8.16** Then somebody said, well, what if we changed it just this way? We got led back to option one. I was thinking, this is a Condorcet paradox. This shows this group is completely irrational on this issue. Instead, everybody else in the group said, oh, so we're back to where we started. So actually, that was the right thing all along. I thought, no, no matter where we start in that cycle, we'd be back to the same place. But they took it as a great confirmation of where we had begun. I still haven't quite figured that one out. I'm not part of that group anymore.
- 8.17** But in any event, this has a real practical significance because suppose you want a certain outcome and you're in a meeting like this. It's following Robert's Rules of Order, and you think, well, what motion do I want to make? I want option X, but I happen to know that this is the preference of the various members of the group. So what motion should I make? Now, the most intuitive answer is to say, well, you want option X, move that X, right? No. Do not do that, because remember that a majority does prefer X to Y and Y to Z, but a majority prefers Z to X.
- 8.18** So if you move X, what's going to happen? B says, "That's my worst-case scenario. No, I vote against this." C says, "Well, I really prefer Z." So C votes against it, you lose, and your favorite option is off the table. So what do you want to do? You want to say, wait a minute, what can beat me? Option Z. I'm going to move Z. Or better yet, since I'm for X, I'll try to trick somebody else into going first and actually moving that Z.

- 8.19** Then B is going to say, "Well, actually, I prefer Y, I vote no." I'm going to say, "No, I don't like Z at all. I vote no." So Z will lose. If Z is out of the picture, X wins. If you're somebody who is in one of these meetings, think about this thing very carefully. There are certain motions you want on the floor first, not because they're good and they'll pass, because you want them to lose, because they're the threat.
- 8.20** I was able to use this a lot back in the days when I was a department chair. Once, some fool made a motion that was out of the preferred order. I had a plan. We were going to move this first. It was going to lose. Then I was going to do this. Instead, somebody said, "I move the—." You just ruined the plan. I didn't get my way. I'm still annoyed. But what you want to do is think, what's the threat to my plan? Is it something that is supported by a majority? If it's not, get that on the table first so it can lose, so that then the threat is eliminated.
- 8.21** All right, well, let's come to the general issue then of voting. There are all sorts of different methods of voting. You might say, yeah, you were saying this about majority voting, but maybe that's a bad idea. Maybe there are better voting systems. But plurality, in other words, whoever gets the most votes, or a majority is whoever gets the most—whatever gets a majority of the votes, yeah, that's a very common system.
- 8.22** But there are other systems. You could have a majority vote with a runoff in case nobody gets 50%. Instead of just who got the most votes, you could say, take the top two among those if nobody got more than 50%, and that's something that's often used. There could be the Condorcet method. Is there anybody who defeats all the other candidates, one-on-one? Sometimes there will be, but often there won't.
- 8.23** You could use what's called the Borda rule. You could say, give me a preference ordering of the candidates, and then we'll number them. We'll either assign one to your least preferred candidate and go up, and then we'll add up all those numbers and find out who gets the most points. That would be a way of doing things. The Academy Awards are determined in that kind of way.
- 8.24** You could use the Hare method, also known as the ranked choice voting method. It's used in Australia, Ireland, and now Alaska, where you count the number of first-place votes, and whoever gets the least first-place votes has the ballots redistributed to whoever their second choice is, and then you do it again and again until there's only one candidate left.
- 8.25** You could have an approval voting, where you can vote for as many candidates or as many options as you want, just all those you like. If you vote for everybody, you have no effect, really, but you vote for any of the ones that you find acceptable. Or you could say, each person gets 10 votes, and you get to distribute them. So maybe say, "Well, I love this candidate. I don't like anybody else. They get all 10." But maybe say, "Well, I really like these two people. I'll give them each five," and so on.

- 8.26** There are lots of different options. There have been many times in American history, where, actually, it might well have mattered which of these we were using to determine the outcome. There have been a lot of very close elections. Maybe the most outrageous is the 1824 presidential election. Andrew Jackson, John Quincy Adams, Henry Clay, and William Crawford were the candidates, and they were running against each other. Nobody got a majority of the votes. Nobody got a majority, even in the Electoral College, so it was decided in the House of Representatives.
- 8.27** It was quite controversial because the most votes went actually to Andrew Jackson. He got over 40% of the vote. But nevertheless, John Quincy Adams ended up winning the vote in the House of Representatives, even though only 32% of the people actually had voted for Adams, and he had had fewer electoral votes than Jackson had. This is a case where it was so close in so many places that whether we think of it in terms of total vote totals, or we think of it in terms of the Electoral College, a different voting method may well have affected the outcome.
- 8.28** The 1860 presidential election was like this. Abraham Lincoln won, hence the Civil War that resulted in part from the reaction to that. But he got fewer than 40% of the votes. He did have a majority in the Electoral College, but there were three other candidates who split the other parts of the vote, and so it ended up being very, very complicated. All of those candidates ended up getting significant support in a wide variety of states. If you look at this in terms of a county-by-county picture, you realize it's even more complicated than just looking at it state by state.
- 8.29** In 1912, there was a very close election. Woodrow Wilson was elected with less than 42% of the vote. Roosevelt and Taft split the Republican vote. Taft was the Republican candidate. Theodore Roosevelt was furious about not being picked, and so ran as the Bull Moose candidate, a third-party candidate, got a lot of support, but that split the vote, hence Wilson became president, and mostly from very strong support in the South.
- 8.30** In 1992, that election was actually remarkably close. Clinton was elected with only 43% of the vote. 2000, it was incredibly close. Both came close to 50%, but neither one ended up above it. If there had been a Hare voting system, for example, and the votes that went to Ralph Nader had been redistributed, it might have come out differently. In any case, it makes a huge practical difference. Sometimes it's not just, what decision should we make, but how do we make the decision? That turns out to be crucial to determining which decision actually is made.
- 8.31** The person who's really investigated this most deeply is Kenneth Arrow, an economist at Harvard, who developed the idea of a social welfare function, and in general, deals with the problem of preference aggregation—how we take the preferences of the group and convert them into, well, one group preference out of all the individual members' preferences.

- 8.32** So we're thinking, either we want a social preference ordering—that's what a social welfare function does, takes all these preference orderings and combines them into a group preference—or a collective choice rule that says, I don't have to know all the elements of the preference ranking, but I at least have to know what's chosen. I have to know what's best, what the choice is out of this, even if I don't really care which one is fourth place and which one is fifth place.
- 8.33** Now, Arrow says, if this is going to be rational, we want certain conditions to be satisfied. So one of them is the unrestricted domain condition. We want to say, I've got to have an answer. I've got to have a social welfare function that produces a social preference ordering for every possible set of individual preference orderings. I've got to be willing to deal with this Condorcet paradox-type situation. I can't say that you guys are just weirdos. Your preferences don't count. It's just a tilt. We don't get anything. Like, who won the presidential election? "Nobody. The preferences are just irrational. Can't do it, sorry." No, we don't want that to happen. We want this to work, no matter what people's preferences are.
- 8.34** Secondly, we want a weak Pareto condition to be satisfied. This is really just a unanimity condition. If everybody strictly prefers X to Y, then the group preference should be for X over Y. Let's say we have a presidential election and it's just unanimous. Hard to imagine. I'm not even going to offend people by picking possible candidates for whom this could happen. But every single person in the country votes for this person over that person. Then the group selection should be that person.
- 8.35** There should be a non-dictatorship condition. There's no one person whose preferences determine the entire preference ordering. It'd be weird if we say, "Oh, yeah, we have a democracy, but that guy is the only one who gets to vote." That's a dictatorship; that's not a voting method.
- 8.36** Then finally, the most controversial of these is independence of irrelevant alternatives. If I'm trying to determine the social preference between X and Y, it should depend only on the individual preferences between X and Y. In other words, what else is on the ballot and so on shouldn't really make any difference.
- 8.37** If I change people's preferences about other things, it shouldn't affect the ranking here. If I add an option, it shouldn't affect how these two are ranked. And if I subtract an option, it shouldn't affect how those two are ranked. So let's say I prefer a banana to an apple. You offer me a piece of fruit, say you want an apple or a banana. I say, "Oh, I want the banana." Say, "Oh, we have oranges, too. You can have an orange." Then it would be bizarre if I say, "Oh, well, now I prefer the apple." No, I shouldn't. That would be crazy.
- 8.38** Or here's a classic case. Supposedly, this really happened. Sidney Morgenbesser was quite a joker. He walks into a restaurant, has dinner, then asks the waitress what they have for dessert. She says apple pie and blueberry pie. He says he'll have the apple pie.

She comes back in a moment and says, "Oh, we also have cherry." Then he said, "Oh, in that case, I'll have blueberry." That's supposed to be funny. My students don't laugh at it, either.

- 8.39** But actually, it's something that's absurd, right? "Apple or blueberry?" "Oh, I take the apple." "Oh, there's also cherry." "Well, then I'll have blueberry." That seems bizarre, right? It doesn't make any sense. If I subtract an option, it's the same type of thing. If I'm deciding between these two which I like better, why would it matter what else you have, if I'm just thinking, which of those two do I like better?
- 8.40** This has a close relationship to Samuelson's idea of revelation that we've already talked about. If you choose X when Y is also available, you shouldn't be in the position of choosing Y when X is also available. So, I realize in ordinary choices, sometimes you do, depending on the past, like, "Oh, I just had spaghetti yesterday. I'm not going to choose it again today." But nevertheless, we can think of that as a different situation.
- 8.41** The whole idea is, if I like this better than that in this situation, given that there are other options, too, but just focusing on those, I'd rather have this than that, then I don't later, given the choice of other options or the change in any other options, say, "Oh, now I prefer that one to this." It's independent of what other options there are available.
- 8.42** So far, so good. What is the theorem? There is no collective choice rule. There is no social welfare function that actually satisfies those four constraints. We don't want a dictator. We don't want independent alternatives to mess up preference rankings. We don't want some things to just tilt the system and say there's no outcome. We want unanimous votes to count. They all seem pretty weak and pretty plausible, but put them together, nothing satisfies them. That's a shocking result.
- 8.43** So, how do we get there? Let's start with the concept of a decisive coalition. A set of people is a coalition that is decisive for an option against another option. Why? If and only if that option is socially preferred whenever anybody in that group prefers it. So, in other words, if we all prefer something, then the whole group prefers it. We, together, determine the social preference.
- 8.44** If at first you think that's a weird idea, think about majority voting. Any majority is like that. A majority of people is decisive, because if a majority says X is better than Y, then, well, they can choose that. The rule says, whenever the majority prefers X to Y, we all then, as a group, prefer X to Y. So, it's actually a much more intuitive idea than it at first sounds.
- 8.45** If you think about a majority, it's not just decisive on this issue as opposed to this issue. It's actually, in a majority system, decisive on any issue. You put things up for a majority vote, and the majority gets to vote about that, and says that becomes the social preference. Now, that would be true, at least if we had a pure democracy where majorities determined everything. Our system is carefully set up so that does not

happen. But in any case, you might think, look, if this group gets to decide that issue, it gets to decide other issues, too. It's large enough.

- 8.46** There's always such a decisive set because we have a rule of unanimity. If everybody agrees that X is better than Y, that carries the day. There's always at least one decisive set, which is the set of everybody. And so, that's pretty simple. We've got one; it's the whole group.
- 8.47** But now, here's our strategy for proving the theorem. First of all, decisiveness spreads. If a set of people is decisive for some option against another option, then it's globally decisive. It's like a majority in the sense that, if it's good enough, if it's strong enough to actually determine the outcome of this issue, it determines the outcome of every issue. It's globally decisive if it's decisive anywhere.
- 8.48** Then we're going to have a strategy of contraction. We're going to say, if a set like the unanimous set is actually strong enough to be decisive, then a smaller subset is also decisive. If we keep doing that, we get down to one person. We start with everybody being globally decisive; we end up with one person being globally decisive, a dictator. That's our strategy.
- 8.49** First of all, spread of decisiveness. We're going to show that if a group is decisive with respect to this issue, it's decisive with respect to any issue. So, say that X and Y are distinct options, and say that G is a group that's decisive for X against Y. It decides X is better than Y. The people within it say that, and so, great, the group decides that. Then it's a globally decisive group. That's what we're going to show.
- 8.50** Let's start about the against side. In other words, this group is big enough to defeat Y. If it's decisive for X against Y, that set is also decisive for X against any other option. In other words, this group is strong enough to decide that this one's better than that one. If so, it's strong enough to show that this one's better than anything else. That's kind of surprising. But suppose X, Y, and Z are distinct options. All of this is going to work only if we have at least three voters on at least three options, as in the Condorcet problem.
- 8.51** We assume that this set G is decisive for X against Y. Then it's going to be decisive for X against Z. It can defeat Z too, even though Z is just some other arbitrary thing. First, why would that follow? But here's the idea. Let's assume that it's decisive for X against Y, but now we've got that unrestricted domain condition. We can let everybody in G prefer X to Y and Y to Z and X to Z. We can let all the individuals outside of G, if there are any, prefer Y to Z. Then we don't have to really specify anything else.
- 8.52** So we've got the one group, the decisive group, saying X is better than Y, Y is better than Z, so X is, of course, better than Z. We have the other people on the outside; it doesn't really matter what their preferences are, so just assume they prefer Y to Z. Now, we think, how does this work then? In G, they have those preferences. The others at least prefer Y to Z. So G, everybody prefers Y to Z then—the ones in our group and the ones outside. So by unanimity, yay, the society prefers Y to Z.

- 8.53** But G is decisive for X over Y, so our society does that, too. It prefers X to Y, and it prefers Y to Z, so of course it prefers X to Z by transitivity. So, guess what? Our group actually is strong enough to decide that X is better than Z. Wow. Now, the social preference ranking of X and Z only depends on individuals' preference rankings of X and Z. But the only ones we've specified are those in our group G, so G is indeed decisive. We don't have to worry that, in that other group, something might happen that defeats this. No, it won't, because G is decisive for X against Z. So that shows that if we can defeat one thing, we can defeat anything against X.
- 8.54** Now, what about the for side? So let's assume we've got this set that is decisive for X against Y. Then if it can show that X is better than Y, we're going to show that set is also decisive for any other option against Y. In other words, if it's big enough to show that X is better than Y, it can show that any other thing is better than Y. So, how does that work?
- 8.55** If X, Y, and Z are, again, distinct options, and the set of individuals, G, is decisive for X against Y, we're going to show that it's decisive for some Z against Y. That turns out to be similar. We divide the society into these two groups, G, our decisive group, and then the rest. We say, well, in G, Z is preferred to X, X to Y, and Z to Y. Let's let the others, since we have an unrestricted domain condition, let's assume they prefer Z to X. Then we've got Z preferred to X by unanimity. We've got X preferred to Y, because G is decisive that X is beating Y. So Z beats Y by transitivity.
- 8.56** In other words, just the same thing. But instead of saying, "If we can beat this, we can beat anything," if we can show that this loses to this thing, we can show it loses to anything. So, in short, we get it both directions. Because of independence of irrelevant alternatives, we don't have to worry that some other preference ranking in our outgroup is going to matter. They won't matter. And so, we've got G being decisive for Z against Y.
- 8.57** Now, it's simple to show then that there's a kind of symmetry here. If a set of people is decisive for X against Y, it's also decisive for Y against X. In other words, if we can decide the preference ordering is going like this, we could also decide that it's like that. It's not like it's a one-sided thing. This group is big enough now to actually show that there's a preference in this order. We could just flip things around and show that it's strong enough in that order. We could go through that in detail, but it's kind of obvious. You just flip it around and replace all the variables. It's the same strategy.
- 8.58** Here's the idea then. We have global decisiveness. For any distinct options X and Y, G, let's say, is decisive for X against Y, then G is globally decisive. It can decide that Z is better than W because it's large enough to do that. So it can decide things; whatever it is above, whatever it is below, it's globally decisive. All we had to use to get there, really, was independence of irrelevant alternatives, transitivity, unanimity, and our unrestricted domain condition. So it was actually fairly straightforward.

- 8.59** Now, it's weird, it's surprising, it's a bit shocking. But again, think about a majority. If you've got a pure democracy and a majority, if it's a large enough majority to decide this issue, it's a large enough majority to decide any issue. That's the whole idea. So that part, once you think of it that way, is not so implausible. In other words, what it takes to decide an issue doesn't depend on the issue; it just depends on the size.
- 8.60** Now, we can start contracting, and this is where it gets weird. This is where it's not just that what I'm trying to do today seems insane. It's where this whole thing seems insane. Wait a minute. How can this be? Something's going wrong. And as we'll see, almost everybody thinks, at this stage, something really weird happens, and nobody can quite agree what. So, there's a sense that, look, Arrow's theorem holds, the logic holds up; and yet, one of our assumptions has to be bad, because somehow this result is strange. But on the other hand, you could look at it and say, "But they all seem reasonable." So what goes wrong?
- 8.61** Here's the controversial part, contraction of decisive sets. We've got a decisive set that is globally decisive now. It can decide any issues, and it contains at least two people. We know there is one because the set of all of us turns out to be big enough to do this. Then a smaller subset of G is globally decisive. If this group is big enough, we can subtract some people, and it's still strong enough.
- 8.62** Here's where the majority voting thing starts to break down, because you can say, wait a minute, 51%, that's enough to decide the issues in this pure democracy, so 50% is. I don't know. Then 49% is. Then 48% is. Then 1% is. Then this guy is. That's the strategy of our proof. We want to say, no, no, something weird happens. But what?
- 8.63** All right, here's the way the proof goes. Let G be a globally decisive set, at least two people in it, and we're going to divide it into two subsets. So, we've got our majority, let's say, this globally decisive set, and now split it into two parts.
- 8.64** Let X , Y , and Z , again, be these distinct options. Let's suppose everybody in G_1 prefers X to Y and X to Z . It's not yet specified how they rank Y and Z . Everybody in our second group prefers X to Z and Y to Z . But what they think about X and Y is so far undetermined. So we've got that set up. Group one, X is better than Y and Z . Group two, X and Y are both better than Z . Now we say, all right, what then is true of the others? Who cares, really?
- 8.65** But now, so far, we can't come to any quick conclusion, but we can say this. Either those preference rankings are enough to figure out that X is preferred to Y or they're not. So, we have to consider two cases. Suppose they are. Suppose the social preference ranking of X and Y only depend on the individual preference rankings of X and Y . That's what independence of irrelevant alternatives says, so we'd better assume that. Then the only thing we've specified about X and Y are actually those of the individuals in G_1 .

- 8.66 And so, outside that, well, they haven't been specified at all. We haven't said anything about how X and Y are related. But we've assumed that, let's say in G2, they all prefer X to Z and Y to Z, but that's compatible with any pattern of preference rankings they might have between X and Y. And so, whatever that pattern might be, X is preferred to Y when all the individuals in G prefer X to Y, but the only ones who were specified are in G1. So, in other words, it all depends on that preference ranking within G1. So, that's itself decisive for X against Y.
- 8.67 But notice that's a subset of the original set. By spread of decisiveness, G1 is globally decisive. It's decisive for X against Y, so it's decisive, period. You might think, well, wait, wait. But yes, it turns out, look, we haven't said anything about the ordering of X and Y among this group, so they don't matter. This group was enough. Now, that's the first part.
- 8.68 But now, suppose that they're not sufficient. You might say, well, so far, yeah, that's a weird result, but you've assumed that this is enough to decide the preference ranking between X and Y, so no surprise that that group then is enough to determine the preference ranking between X and Y.
- 8.69 What if they're not? What if we've gone now from just above a majority to below a majority? Well, there's got to be some extension then where X is not preferred to Y. And so, according to independence of irrelevant alternatives, the only part of the extension that's needed to derive that X isn't socially preferred to Y are the preference rankings between X and Y. So we've got G1, where X is preferred to Y and to Z, and G2, where X is preferred to Z and so is Y.
- 8.70 And then we say, all right, there's some preference ordering then, some preference ranking between X and Y, in whatever that extension is; that's enough to say, well, look, it is possible that we don't have X being preferred to Y here. Presumably, within G2, they have some view about this. I don't know what it is, so let's just call it $X \succ Y$. So, here's what we've assumed. In G1, X is preferred to both Y and Z. In G2, X is preferred to Z, and so is Y. Then X and Y are compared somehow, but we haven't figured out how yet.
- 8.71 Then what about the people outside this set? By unrestricted domain, we can assume anything we want about that. Let's assume they also have whatever this unspecified relation is between X and Y. They also $X \succ Y$, whatever that turns out to be. Now, we put all that together. And by global decisiveness, we get that X is preferred to Z. Wait, why? Because G1 and G2 agree about that. X is preferred to Z. They're globally decisive. So that's enough to show that X is preferred to Z.
- 8.72 And then, what about Y and X? By completeness, Y has to be at least as preferred as X. Now, why? Actually, "why?" and variable Y sound alike, so I've got to be a little careful in how I express this, I realize here. But by completeness, Y has to be at least as preferred as X, because we're assuming that this was not enough to actually determine

the preference ordering between X and Y. And so, if it's not true that X is preferred to Y in this extension we're considering, then Y must be at least as good as X in that extension. That's just the negation of X is preferred to Y.

8.73 Then here's the trick. By transitivity, Y is preferred to Z. Now, why is that the trick? Y is at least as good as X, and X is preferred to Z, so that means Y is either equal to X or it's better than it, so it's got to be better than Z, since X is. And so, we can say the society then prefers Y to Z. But again, the preference ordering of Y and Z depends only on individuals' preference rankings between Y and Z. It shouldn't depend on where X is. The only ones we've specified are those in group 2, and so G2 is decisive for Y against Z. So, by spread of decisiveness, G2 is globally decisive.

8.74 So, what did we do? We said in the first scenario, where what we've said is enough to actually show that X is preferred to Y, then we said, so then G1, that first subgroup, that's decisive. And then we said, well, suppose it's not. Then it turns out the other group, G2, the other part of it was decisive. So in either case, there is a subset that is decisive if the whole thing is.

8.75 Now we can easily prove the theorem itself. By unanimity, for all options X and Y, whenever everyone in G prefers X to Y, then X is socially preferred to Y. So G is globally decisive, let's say, and this could be the whole set of voters if you want, or it could be some subset. But now we just keep applying contraction of decisive sets. We keep saying, well, if this set is actually decisive, a smaller one is, too. But if that is, a smaller one is, too. We keep shrinking it down until, since this is a finite group, we assume we do not have an infinite electorate here. But as long as we've got a finite electorate, a finite group of people, we finally do this a finite number of times until we get just one person left in that group, a dictator.

8.76 That's a sort of startling reaction. If there is a group, like the group of all individuals that's powerful enough to decide even one question, they can decide any question, and then there's a dictator as a result, because we can keep shrinking it down and finding a subset that's decisive. That's pretty surprising. If X and Y are distinct options, and G is decisive for X against Y, G is globally decisive. So, if Group G gets to decide something, G gets to decide everything.

8.77 Now, in the kind of voting system we're considering, where there is one system, well, yes. Now, in society, really, we might say, oh, for example, as John Stuart Mill said, wait a minute, wait a minute. There should be self-regarding actions. Do we all get to vote on what color socks that guy wears? No, we leave it up to him, right. And so, we end up saying, yeah, wait a minute. We don't really do that in society. But notice what we're doing in doing that. We're saying certain decisions are not to be included in this. This is about how we make group decisions.

8.78 We may say there's a certain set of options where the group doesn't get to vote. So those are on the side. But anything we consider within the field of things that can be

collectively decided is going to be affected by this. So, in a certain sense, you could say this doesn't apply to every single decision, every single option. I think that's fair. But on the other hand, within the sphere of things we consider political matters, this holds. And so, yes, we can say there are some things that are off to the side here. We're not going to let this affect them, but it doesn't affect the larger point, really.

- 8.79** Now, we could affect it in a deeper way by having a principle of subsidiarity, saying, we're going to allow the larger group to decide only certain kinds of questions; whenever possible, we're going to let a subgroup decide. But if you start thinking about it that way, you realize, is that an exception to this? Or is that really just a way of building this in, saying, you know what, this should really be a local government matter, so we're going to let the city council decide this, or the voters of this city decide this. That then ends up being a socially decisive subset. And so, you might think that's just a way of, as it were, not shrinking it little by little, but shrinking it all at once, to say, now, within the city, isn't the same thing going to happen?
- 8.80** Then the libertarian paradox. We've talked about this a little bit, not a whole lot, but the whole idea here is that, if each person gets to make a decision about certain matters, and unanimity actually carries the day, we can get problems generated. We've already talked about this example without calling it this. One version of it is a person who loves blue and is a conformist, against the person who is a nonconformist, who doesn't like blue. What colors should the houses be painted?
- 8.81** That's our case where, really, it looks like each person should have the decision to paint her house whatever she wants, and yet they're both going to prefer certain things to certain other things. If you put all that together, you get led into that cycle of repainting that we talked about. That is one example of what's called the libertarian paradox. It will turn out, well, that's closely related, actually, to this theorem. It's not so much an objection to it; it's instead a way of saying, you might try to say you get to control certain things, but if other people have preferences about them, then actually you can get led right back into this puzzle.
- 8.82** That part is a bit surprising, but maybe stronger than you might think initially. What about the contraction of decisive sets? It feels like that's where things go wrong. Wait a minute, we could all make this decision. Yes, that seems right, at least within the sphere of things we allow to be political decisions. But wait, that means a subset does in every single case, so we can finally get it down to one person? I thought this was a democracy. It turns out, now we've turned into a monarchy. In effect, this is, hey, look, every pure democracy would be a monarchy. There's always one dictator. It seems like that's just not true.
- 8.83** What's going on here? We've shown that if a set of individuals is globally decisive, and if it contains at least two people, there's a subset, a smaller subset, that is actually globally decisive. If this group gets to decide, some small subgroup within it gets to decide. As it were, every group has a cabal within it that is run by one leader.

- 8.84** You might think, wow, okay. It seems to clearly fail for majority voting. Or a super-majority voting, let's say, no, this requires a two-thirds vote or a three-quarters vote. Then when you fall below two-thirds or three-quarters, it doesn't work, right?
- 8.85** Or suppose we have unanimity, like the Polish Parliament of the 17th century. Everybody has to agree. It has to be unanimous. They really did have such a rule. One person dissents, it defeats it. All that had to happen in Poland at that time was one person who said, "I object." And not only that law didn't pass, but actually, everything that parliament had decided was immediately thrown out.
- 8.86** And for about 100 years or more, that actually worked. And then other neighboring countries realized, "If I bribe just one guy, I can pull him from doing anything." So it all fell apart. But in any case, yeah, I mean, you'd say, wait, all it takes is one person to defeat this. What do you mean, a smaller subset can do it? How is this possible? So, think about majority rule, for example. What would go wrong here with majority rule?
- 8.87** There's going to be an assumption of ours that does not apply in majority rule. We don't get all of those things happening. And what is it? It's still true that unanimity works, so that's not a problem. If a majority prefer this candidate to that candidate, it's not like then that candidate somehow wins because of people's preferences about other people in the ballot, as long as the votes between those two don't remain unchanged. What goes wrong here?
- 8.88** It's really revealed in the Condorcet paradox. That is to say, sometimes this is actually going to lead us into a cycle, and so it's going to lead us into irrational decisions. So, it's not going to be the case that the contraction always works. But why? Because it's not going to satisfy our other constraints. Because sometimes it's going to lead us into irrational cycles.
- 8.89** What about a triumvirate, let's say, with unanimity? All three of our leaders have to agree. That was tried in Rome. How does that work? You might say, but look, two can't do it. It requires the three to be unanimous. If you've got Pompey and Caesar together, that's not enough. Crassus has to agree, too. So we don't get a subset, do we? No, we don't.
- 8.90** But on the other hand, what goes wrong here? Often they're not going to agree. So, there are going to be lots of cases where we don't get a social preference ranking. Come on, triumvirate, tell us what to do. Do you prefer X to Y or Y to X? And it will turn out, some of them prefer X to Y, some Y to X, there won't be a decision. And so, that's going to fail that criterion. You might say, maybe they're the problem here. Or maybe it's unrestricted domain.
- 8.91** Maybe sometimes, if people have preferences like those in the Condorcet paradox, we should just say the group's preferences are just together weird. They're just irrational. No individual is irrational, but as a group, this combination just doesn't work. We don't know how to deal with things. It's not going to give an answer. Just tilt. It's like the

pinball machine where you whack it too hard, and it just says tilt. Maybe sometimes democracy is like that. You whack it too hard, and it just says tilt.

- 8.92** Or what about a unanimity rule? Maybe something goes wrong there, but that's bizarre. I mean, it might be true that everybody together makes a mistake. Indeed, there have been systems where unanimous votes are actually subject to reversal or, for that matter, a great suspicion, just because the thought is, this could be groupthink. There's some kind of hidden assumption. If nobody disagrees, then maybe there's something weird going on. But still, it seems really plausible. Look, if we all agree that X is better than Y, then as a group, we should say X is better than Y.
- 8.93** What about a dictatorship? I mean, is it Okay sometimes to have a dictator? If we didn't have spread of decisiveness, it wouldn't be so bad, because what color socks you're wearing or what you're having for dinner, we could say, okay, you get to be dictator with respect to those decisions. But we don't want to say, you get to choose your own sock color, so you're the dictator about everything. You get to choose everything. No. And so, a dictatorship, if it were just on this issue or that issue, wouldn't be so bad.
- 8.94** That's kind of what the libertarian thinks. Hey, we each have a set of decisions we should be able to make and, in effect, be our own dictator with respect to those. Yeah, but it's the spread of decisiveness that becomes a problem. So we don't really want to give that up. What about independence of irrelevant alternatives? Now, lots of people have said that's the problem; that's the condition that gives us difficulty. And yet, it is closely related to Samuelson's idea of revelation.
- 8.95** It's very weird if I say I like X better than Y, but then you say, well, there's also this menu option Z. I say, oh, well, now I like Y better than X. It would make perfect sense to say I don't want Z, or to say, oh, I like Z better than X even. But to say it affects my decision between X and Y seems really bizarre. So it's hard to see how it could be rational to abandon this.
- 8.96** What about transitivity? Maybe we should just live with cycles, but that seems pretty destructive. I mean, it might be a good thing if you're the leader and can say, "I know what motion to make first," and so on; it might let you get your way. But it doesn't seem in general like a very rational way of producing group decisions. What about connectedness, though? What about this idea that, in the end, we want an answer and we want to be able to assume that X is preferred to Y, socially, or Y is preferred to X, or they're ranked the same?
- 8.97** Maybe that's too strong. Maybe sometimes we end up saying X and Y are incomparable, and yet that seems bad. That seems like this tilt example again. And so, it's kind of hard to give that up—though the way this proof went, actually, one of the things I like about Gustafsson's version of the proof is that it really points out, that's way more important than people noticed. That's how we got from, X is not preferred to

Y in our extension, to Y is at least as good as X in the extension, and by transitivity, that then it's better than Z.

- 8.98** It actually played a really vital role in the most strange part of the proof. Maybe it's really the problem in the first place. Maybe sometimes a voting method is just going to misfire and not really going to give us any preference ordering. In fact, this is one way in which people have said, maybe the most we can get is some collective choice rule. Just tell me what number one is. I don't need the whole preference ordering. But that's not always going to really solve the problem either. Think about that Condorcet paradox.
- 8.99** So that's Arrow's theorem, and we've done it in less than an hour. So that's amazing. Thank you, Johan. But the idea here really is that we have said, look, there are very plausible constraints to put on elections or any kind of plausible collective decision thing, where we're deciding together for the group. It's not me saying what's best for me, you saying what's best for you, and so on; it's rather we saying what's best for us, and we find out it's really hard to do that in a way that counts as rational all of the time.
- 8.100** Now, that is one kind of, you might say, collective irrationality problem. It looks like I have to violate at least one of those plausible conditions of rationality. But Allan Gibbard and Satterthwaite, independently, developed a theorem that is closely related to Arrow's theorem, that says, these voting systems are actually irrational and dangerous in another way, too. So, this has become known as the Gibbard-Satterthwaite theorem.
- 8.101** And the basic idea is, any decent voting system can be strategically manipulated in some way or other. So, consider any deterministic voting rule. This could actually be extended to one involving chance. Like, hey, if it's a tie, we just flip a coin. I went through a graduate seminar where Gibbard was proving that extension of this, and it took us weeks. I'm not really going to talk about that, but something very close to this holds, even if we bring in random elements.
- 8.102** Consider a deterministic voting rule that takes, as its input, preference rankings among the various people. Let's assume there are no ties. This is going to be true. It's going to hold for any number of voters and any set of alternatives. Let's assume there are at least three voters and three alternatives again. Here's what he shows. Any non-dictatorial voting rule with at least three alternatives is manipulable. People can change their preference ordering in such a way that they do better.
- 8.103** So, I may say, I like X better than Y, but in this case, I'm going to actually put on my ballot, I like Y better than X, because it yields a better outcome for me if I do that. Now, that's weird, but it turns out, it's always possible to do that in any plausible voting scheme. What are we going to say about this, really? Here's the import of the theorem.
- 8.104** If you've got at least three possible outcomes, and hence, you might say, I see some of the advantages of a two-party system or of a run-off system where we have only two

candidates, then this doesn't apply. There's no reason why, if my choices are just A and B, I really like A a lot better, I said, "But I'm going to vote B," because somehow I think that will improve my odds, no, it won't if they're only two. But if there are three, that always is a possibility.

- 8.105** Now, why? Here are our options. Essentially, we have something that is a dictatorial system. As in Arrow's theorem, we just say you're a dictator. That is one possibility. One is to say, maybe there are alternatives that can never win. Here are the candidates, A, B, and C. C is a fake. He'll never be elected. He can't win. I don't care if everybody votes for him; he can't win. He's going to be disallowed by the courts or whatever. Because why does that change things? It's because then it shrinks it to two alternatives. There have to be three real alternatives.
- 8.106** Then it's manipulable. So anything that is not a dictatorship, and actually really has three real options or more, is manipulable. There are situations where you can improve the outcome for yourself by actually misrepresenting your preferences. So, our assumptions are, the domain is unrestricted. Voters do have these strict preferences. The rule's deterministic; there's no randomness, though something close holds even if there is. It always picks exactly one winner. Here we don't even need the full preference ranking. We just need to know who wins. The voting system is ordinal or ranked voting. It's not approval voting or something like that. That has a related result, but not quite the same.
- 8.107** So, there are a variety of these real-world manipulable systems. For example, a plurality system, first past the post. Whoever gets the most votes wins. How could it be that that could be manipulated? Actually, it's super easy to imagine, right? You really prefer the third-party candidate. You are really a fan of Ralph Nader or Ross Perot or Henry Clay or whoever it happens to be in that particular election. It doesn't really matter.
- 8.108** The point is, are you going to vote for that person? Are you going to think, "But he's not going to win. So I'll vote for the candidate and say I prefer that candidate, even though, really, I would rather have my third-party person"? Tons of people who prefer the third party actually do misrepresent their preferences, because they think, "Yeah, I'd like to vote for this guy, but he's not going to win, so I'll vote for the one who has the best chance, who's closest to me." So that's a routine kind of manipulation of votes.
- 8.109** What about an instant runoff or a ranked choice voting? It applies to either one of them. It can reward burying your preferences. So, you say, Okay, once again, I prefer A to B and B to C, but actually, it looks to me like B has a pretty good chance of winning. Do I want to risk the fact that my vote might end up actually going to B? No, I don't. I think B is better than this insane person, C. But I'm going to put C above B, just because I don't want my vote to count for B, because I really want A. Even in a ranked choice system, I can have an incentive to misrepresent my vote.

- 8.110** What about a Borda count or some other point-based system? Then definitely, I can say, "Oh, I'm putting you way down here, buddy. You're the close contender to my candidate. I realize you might win, but I'll put you at the bottom, so you only get one point out of my ballot. And so, this other crazy person, well, I'll put that person high, so they get more points than you do." In short, no matter what the voting system is like this, I can misrepresent things. And this has really happened.
- 8.111** For example, Alaska, I mentioned, adopted this sort of ranked choice voting, and the result of that was that Murkowski beat Tshibaka. If it had been a straight sort of system, it probably wouldn't have happened. But there's another sort of manipulation going on here as well. Chesbro, the Democratic candidate, had up to 17% support in polls before the election, but in the actual first round of the election, only got 10%. So already, about 7% of Alaskans were misrepresenting their preferences. They really wanted Chesbro, but they thought, "Chesbro is not going to win. So I'm going to vote Murkowski over Tshibaka."
- 8.112** And so, already, Murkowski, in that first round, ended up with 43% as opposed to 42.6% for the second-place finisher. But on the other hand, that was already given 7% of people who were misrepresenting their preferences. And then, since Chesbro was eliminated, those votes ended up going, well, in this case, mostly to Murkowski. And so, that tells us, yeah, here was a case where that actually affected the outcome of the election.
- 8.113** Now, another sort of abstract case is where we take something like the Condorcet paradox. We've already seen that, in majority voting, that's just X beats Y, Y beats Z, Z beats X. We get this weird cycle. But let's assume that there are different proportions supporting it. Let's say 44% say X is better than Y, which is better than Z. 33% say Y is better than Z, which is better than X. Then 23% say, actually, Z is the best, then X, then Y.
- 8.114** What happens as a result? X wins [67%] to 33%, when we think what actually happens in terms of a ranked choice system. But on the other hand, we think, well, wait a minute. Suppose I am in this group that's going to lose. I look at this and I think, X beating Y. If I look at that group, it's 67% to 33%. If I'm in Group B here that prefers Y, I can think, wait a minute, wait a minute. I don't like that outcome at all. I lost, and the person who won is the candidate I like least. Suppose I misrepresent my vote.
- 8.115** Suppose, instead of saying, "I like Y better than Z," I say, "I like Z better than Y." Then what happens? Z wins. Z wins in this case, 56% to 44%. And so, I've gone from getting my worst-case scenario to at least getting my second choice. I've improved my situation by misrepresenting my preferences. Even if I've got a ranked choice system—and often people advocate that because they say, "Oh, it gives a better chance to these—you can vote your honest ballot, even if you prefer a third-party candidate," and so on. The fact is, it's also vulnerable to manipulation, because in this case, you can do better by misrepresenting, and that sort of thing happens all the time.

- 8.116** What about approval voting? I've mentioned this doesn't apply to it. Suppose I just say, how many candidates do you approve of? Still, even though it's not manipulable, I never have a reason to say I disapprove of somebody I do approve of or vice versa. But on the other hand, well, I should even be careful about saying that, because it can still be strategic. Because I'm giving you two bits of information—which candidates I like better than which others. But I'm also saying, where do I draw the line?
- 8.117** Let's say I like X better than Y, Y better than Z. I'm not going to approve everybody, or my ballot won't have any effect. So I'm going to disapprove Z. But what about X and Y? Should I just approve X, or should I approve X and Y, too? I've got a strategic choice. If I think Y is a serious threat to X, then I'm going to say I disapprove of Y. It turns out, this is also manipulable, but in a different way.
- 8.118** Let me close by going back to 1824 and that presidential election. Now, it's very hard to say, what would various voters' ballots be, if instead of just voting for one candidate, they were voting a preference ranking, and then we were using approval voting or the Borda count or some other rule? But I am guessing. I went through and looked at the results, state by state, and the percentages cast, and basically the states fell into eight different groups in terms of their preferences. They had eight different preference rankings that seemed to be the dominant preference ranking in that state.
- 8.119** And so, I said, well, how would it work out? Plurality voting, well, Jackson would have won, either by the Electoral College or by popular vote. He got not the majority of the popular vote, but he got more than anybody else. If there were a runoff, it would have been between Jackson and Adams, and Jackson, I think, would have narrowly won. If it was the Borda rule, where we give people points according to preference orderings, then it seems like Adams would have won, relatively narrowly, but nevertheless, more of the people who preferred the others would have ranked those other candidates over Jackson.
- 8.120** Then what about the Hare system? Here it would have been very, very close, and I'm kind of guessing, but probably Jackson, very, very narrowly. What if it were just approval voting? Then actually, given those preference orderings, I think it's plausible to think that Adams would have won. But on the other hand, just slightly different assumptions, it would have been Jackson. So, it turns out that—I'm sorry, Henry Clay, I probably would have voted for you, but nevertheless, I don't think you really had a chance. The same thing is true for Crawford. But who won between Jackson and Adams really came down to what voting method are we using.
- 8.121** Now, as it happens, we have a majority sort of thing, but in the Electoral College, nobody got the majority, so then it was simply a state by state, each state casting one vote in the Electoral College; there was a huge advantage to having one more of the small states as opposed to fewer larger states. So, as it happens, Adams ended up beating Jackson. And so, that was yet another way of voting that led to Adams' victory.

8.122 But the point of that little historical exercise is simply to say, there are occasions where our choice of those methods really makes a huge difference, like a difference in some cases of whether or not there's a civil war. So, what's the overall moral of what I've been talking about for the last hour? There is no perfect voter, but also there is no perfect voting system. We have to decide among the various things that we would value, and we really cannot have it all.